

Recent advancements in latent heat phase change materials and their applications for thermal energy storage and buildings: A state of the art review

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ABSTRACT

Phase change materials (PCMs) have received substantial interest for their ability to store and release latent heat for energy conservation and thermal control purposes. PCMs are available in a variety of latent heat and melting points but their performance is low due to low value of thermal conductivity which limits its usage. The addition of highly thermal conductivity nanoparticles, porous metal foams and encapsulation methods have been used to address this issue and try to fix the low thermal conductivity of PCMs which is broadly discussed in this manuscript. The ability of PCM to store and release the thermal energy prompted the researchers to use it in potential applications. The energy retained and emitted by PCMs may be used for a variety of purposes, such as in photovoltaic (PV) panels, thermoelectric generators, building air-conditioning, air and water heating systems, heat exchangers, desalination solar stills, textiles, thermal management of electronic equipment and batteries and food packaging. Therefore, in this review, first the advancements in thermal properties of PCMs are thoroughly discussed in terms of enhancement in melting and solidification rates. After that, the use of PCMs in various applications is then explored, and conclusions are drawn accordingly. Based on analysis of recent literature, it was discovered that the phase transition temperature, phase transition enthalpy and thermal conductivity are three important parameters for the selection of an appropriate PCM for use in various applications. The current status of these advanced energy storage materials is also presented in this review. Lastly, some challenges and future recommendation are also proposed for future researchers which will bring a revolution in thermal management field.

Introduction

The expression “energy crisis” refers to ever-increasing energy demand and the depletion of traditional resources. Conventional resources are commonly used around the world because this is a low-cost method

to meet the energy demands but along aside, these have negative consequences such as air and water pollution, ozone layer depletion, habitat degradation, and climate change [1]. As a result, research into green energies and renewable energy options such as wind, solar, biomass, geothermal, and nuclear energy is a necessity [2]. Because of the rapid growth of industries and cities, there has been an increase in the use of

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Nomenclature		SEBS	Styrene-b-(ethylene-co-butylene)-b-styrene
Abbreviations		TES	Thermal Energy Storage
Ag	Silver	TEG	Thermoelectric Generator
ANN	Artificial Neural Network	TEM	Thermoelectric Module
Al ₂ O ₃	Aluminium Oxide	TiO ₂	Titanium Dioxide
CPC	Compound Parabolic Collector	TEOS	Tetraethyl Ortho Silicate
CNTs	Carbon Nanotubes	USD	US Dollar
CNFsv	Carbon Nanofibers	xGNPs	Halogenated Graphene Nanoplatelets
Cu	Copper	ZnO	Zinc Oxide
CuO	Copper Oxide	Symbols	
EG	Expanded Graphite	A	Surface Area (m ²)
Fe ₂ O ₃	Iron Oxide	E	Energy (J)
GNPs	Graphene Nanoplatelets	G	Solar radiation intensity (W/m ²)
FESEM	Field Emission Scanning Electron Microscope	h	Heat transfer coefficient (W/m ² .K)
GO	Graphene Oxide	k	Thermal conductivity (W/m.K)
GCP	Grainy Carbon Powder	L/day	Liter/day
HE	Heat Exchanger	L/h	Liter per hour
HT	Heat Transfer	L/m	Liter per minute
HVACv	Heating, Ventilation and Air-conditioning	L/m ²	Liter/meter square
HTF	Heat Transfer Fluid	M	Mass (kg)
LHSv	Latent Heat Storage	$\dot{m}$	Flow rate (kg/s)
MWCNTs	Multiwall Carbon Nanotubes	Nu = hl/k	Nusselt Number
MEPCMs	Microencapsulated Phase Change Materials	P	Power (W)
NPs	Nanoparticles	Q	Heat transfer (W)
NEPCMs	Nano enhanced PCMs	Re = $\rho vl/\mu$	Reynold Number
PCM	Phase Change Material	$\dot{q}$	Heat Flux (W/m ²)
PEG	Polyethylene Glycol	t	Time (s)
PPI	Pores Per Inch	T	Temperature (K)
SAT	Sodium Acetate Trihydrate	v	velocity (m/s)
SiC	Silicon Carbide	V	Voltage (V)
SiO ₂	Silicon Dioxide	μ	Dynamic Viscosity (Ns/m ²)
SEM	Scanning Electron Microscope		

fossil fuels, resulting in increased pollution. The world is reducing its reliance on fossil fuels by adopting green energy sources and using creative approaches to boost energy system efficiency and increase the use of innovative and sustainable energy sources. One of the innovative methods is to use latent heat Thermal energy storage (TES) using PCMs. TES systems can help save energy and reduce the harmful effects of energy usage on the climate.

Phase change materials (PCMs) are a cost-effective energy-saving materials and can be classified as clean energy sources [3]. Because of promising properties, PCMs are regarded as decent choice for TES because they can retain and release large amount of latent heat during the phase change process. PCMs are being studied for a variety of applications due to their unique characteristics such as low cost, high heat capacity, high density, reliability, thermal stability, capacity to consistently melt, non-segregation, non-corrosiveness, toxic less, and low or no super-cooling. Also, after thousands of phase change cycles, the PCMs can maintain latent thermal energy without changing the properties [4,5].

PCMs are classified into three categories based on their chemical composition such as organic, inorganic, and eutectic, as illustrated in the Fig. 1. However, organic PCMs based on paraffin are the most extensively used in various applications. In contrast to organic PCMs, inorganic PCMs have a high latent heat per unit mass and volume, highly cost-effective, inexpensive, and non-flammable. These materials, however, have some drawbacks such as subcooling and isolation, but they are being studied and utilized because of their high thermal conductivity [6,7].

Even though PCMs have better thermal properties than water, the mechanism of thermal enhancement is still unknown due to inaccuracy

in research findings. Nanoparticles [9–12], expanded graphite [13,14], micro-encapsulation [15], macro-encapsulation [16], extended surfaces [17], multiple or cascaded PCMs [18,19], eutectic PCMs [20], porous metal foam [21–23], geometric modifications [24,25], heat pipe [26], and different finned tube structures [27,28] have been used in PCM research to improve thermal stability, thermal conductivity, and heat transfer rate for TES applications.

Safaei et al. [29] utilized graphene oxide with various concentrations

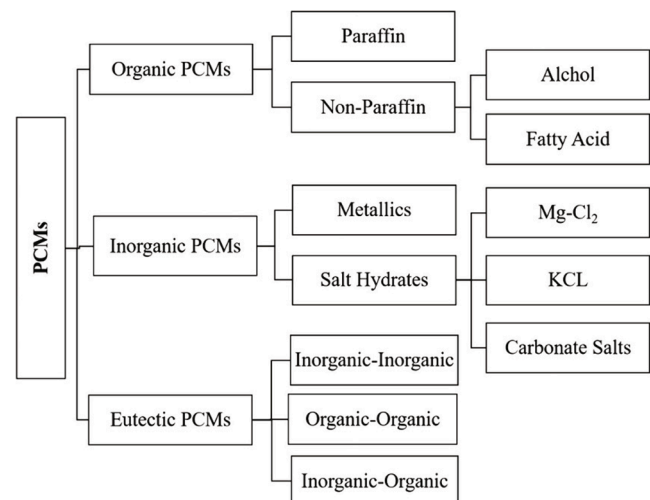


Fig. 1. Classification of phase change materials [8].

dispersed in Paraffin PCM for solar still efficiency improvement. Solar still productivity enhanced by 25% by incorporating GO/Paraffin as compared to simple PCM case. By increasing concentrations of nano particles, efficiency of solar still improved considerably. Sarafranz et al. [30] experimentally tested the thermal performance of heat pipe filled with binary mixtures of n-pentane-methanol and n-pentane-acetone together with pristine acetone, n-pentane and methanol. According to findings the n-pentane-acetone mixture exhibited highest thermal efficiency while the value of filling ratio and tilt angle was small for binary mixture compared to pure liquids. Ahmadi et al. [31] compared several machine learning techniques for modelling the dynamic viscosity of CuO/water nanofluid. Variables such as concentration, temperature and size of nanoparticles were explored in terms of their influence on dynamic viscosity. By relative importance of input variables, size of nanoparticles displayed lowest influence whereas concentration of nanoparticles showed highest impact on dynamic viscosity. Bahiraei et al. [32] explored the graphene-silver based hybrid nanofluid in a microchannel heat sink along with secondary channels and the ribs for improving cooling performance. The results depicted that with the nanoparticles concentration from 0 to 0.1% and Reynolds number of 100, the convective heat transfer coefficient enhanced by 17% and the bottom surface temperature of heat sink reduced by 13.88 K by increasing Reynolds number from 100 to 500 at 0.1% nanoparticles concentration.

Bahmani et al. [33] investigated numerically the thermal characteristics of $\text{Al}_2\text{O}_3/\text{H}_2\text{O}$ nanofluid in parallel and counter flow double pipe heat exchangers. The effect of nanoparticles concentration, flow direction and Reynolds number were explored for thermal characteristics evaluation. According to results, the increase in nanoparticles concentration and Reynolds number gave maximum enhancement of average Nusselt number and thermal efficiency by 32.7% and 30% respectively. Optimal concentration of nanoparticles was noticed to be 5% and counter flow heat exchanger was recommended at higher Reynolds number. Giwa et al. [34] investigated the electrical conductivity and effective viscosity of hybrid nanofluids of $\text{Al}_2\text{O}_3/\text{Fe}_2\text{O}_3$ (25:75 mass %) with base fluid of deionized water and ethylene glycol (50:50 vol%). The influence of parameters like nanoparticles concentration, temperatures and base fluid was evaluated. It was revealed that effective viscosity of hybrid nanofluid with deionized water as base fluid was much lower compared to deionized water-ethylene glycol based nanofluid. While electrical efficiency enhancement was more in case of deionized water-ethylene glycol based nanofluid. Bagherzadeh et al. [35] numerically studied the flow and thermal behavior of microchannel heat sink with slip boundary condition and water/ Al_2O_3 nanofluid jet for heat transfer enhancement in the presence of homogenous magnetic field. The performance of heat sink was explored by varying parameters of Reynolds number, nanoparticles concentration and magnetic field strength. Results showed that the magnetic field lowered the vertical velocity gradient but at creeping flow the magnetic field showed no influence on Nusselt number enhancement. Using homogenous magnetic field, Reynolds number of 50 and nanoparticles of 0.06 vol% in water enhanced the Nusselt number which caused better cooling of heated wall. Bagherzadeh et al. [36] developed a novel approach of enhanced ANN for sensitive analysis of hybrid nanofluid functionalized-MWCNTs- Fe_2O_3 /Ethylene Glycol. Analysis was done by ANN analytically, gave authentic results with less computation time and cost. Akbari et al. [37] used two-phase mixture model to analyze the heat transfer and fluid flow characteristics of laminar Cu/ H_2O nanofluid in a 3-D curved microtube. The results were illustrated for coefficient of friction, Nusselt number patterns, dimensionless axial velocity and dimensionless temperature profiles. Results showed that the due to centrifugal and buoyancy forces, the axial velocity profiles showed no symmetry. As the number of nanoparticles enhanced, isotherms attained the desired temperature quickly.

Goshayeshi et al. [38] dispersed various types of Fe_2O_3 NPs into the Kerosene to investigate the closed-loop pulsating copper heat pipe

thermal behavior. The influence of magnetic field, NPs type and size were evaluated for thermal attribution of heat pipe. Results showed that the addition of NPs enhanced the thermal performance of pulsated heat pipe, magnetic field reduced the thermal resistance by 12% packed with gamma Fe_2O_3 NPs. Gamma Fe_2O_3 NPs of 20 nm size were observed to be best candidate for attaining better thermal performance.

The addition of high conductivity nanoparticles (NPs), porous metal foam, expanded graphite, and encapsulation are promising methods for improving the thermophysical, and chemical characteristics of PCMs, hence enabling wider applications of PCM at improved system performance. These advanced PCMs are researched for extensive applications such as thermal energy storage [39,40], solar heating [41,3,42], automobiles [43–45], textiles [46,47], refrigeration [48,49], waste heat recovery [50,51], air conditioning applications [52,53]. These researches are mainly concerned with the rate of energy consumption, energy storage capacity, energy savings, efficient heat charging/discharging and PCM thermal conductivity enhancement [54–56]. Similarly, PCM usage for solar cooking, drying, power generation and desalination applications has been studied. PCMs usage in textiles enhance their thermal and cooling properties. Methods to introduce PCM in textiles are covering, spraying, and encapsulation. The use of nanofiber technologies for stability of PCMs is also one of the most recent approaches to the study of these materials. Researchers have proposed and studied the use of PCMs for cooling of electronic equipment and batteries. Their findings revealed that the heat sink performs well before the PCMs fully melts. After PCM has fully melted, the heat sink's productivity begins to deteriorate rapidly. The latent heat capacity of PCMs with suitable transition temperature can be used to provide thermal comfort and minimize building loads. In recent years PCMs use in building heating-cooling systems and hot water supply has received a lot of attention due to diverse variety of melting temperatures of PCMs. Accumulating solar thermal energy during daytime and supply it during night for heating application may be a practical option for lowering the building's energy costs.

Extensive experimental and numerical studies have been conducted to improve the efficiency of energy storage materials to date, but no definitive conclusion has been reached. Fig. 2 depicts the number of publications on advanced energy storage materials from 2010 to 2020, based on "Web of Science" results. It is apparent that the number of publications is steadily growing, indicating that the world is increasingly focused on this domain.

Advanced phase change materials have been broadly used in various

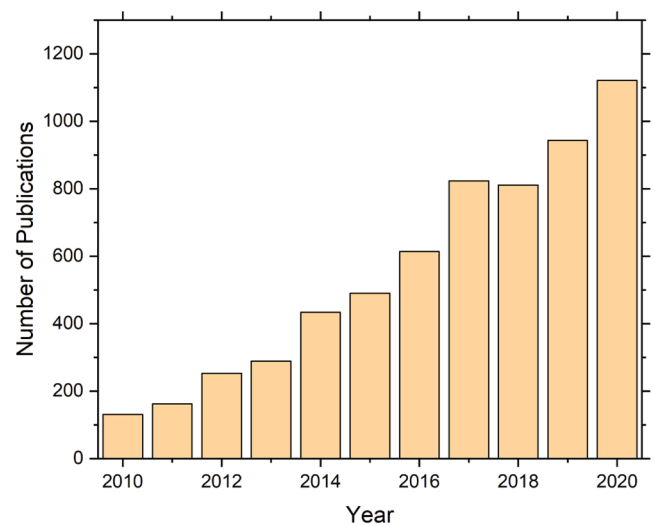


Fig. 2. Web of Science stats regarding number of publications during (2010–2020) when searched with keywords “advanced energy storage materials”.

applications. Zhu et al. [57] investigated the trombe wall packed with PCM for indoor thermal comfort in Wuhan climate conditions. The author utilized the TRANSYS heat transfer model and GenOpt optimization tool to determine the best value of critical parameters and energy consumption rate. Trombe wall optimization findings indicated that the ideal air gap thickness was 0.05 m, optimal wall thickness was 0.68 m, optimal vent area was 0.6 m², and optimal melting temperatures of lower and higher temperature PCM layers were 16.5 C and 27.75 C respectively. The annual building load was lowered by 13.52% in optimized PCM packed trombe wall as compared to conventional trombe wall. Kozelj et al. [58] analyzed the thermal storage tank with encapsulated PCM for building heating purposes. Paraffin was used as PCM and represented 15% of total volume of water thermal storage tank. Results showed that use of encapsulated PCM in water tank, the heat storage was enhanced by 70% compared to conventional case of water based thermal storage system. Further, author did numerical simulation using TRANSYS and validated the experimental results.

Qu et al. [59] conducted a multi-factor analysis with PCMs incorporated in building envelopes for attaining thermal comfort and energy savings using EnergyPlus software. The most influential parameter for energy usage and interior thermal comfort was found to be PCM envelope type, followed by PCM layer layout, PCM type, and PCM layer thickness. According to author, the appropriate selection of PCMs based on local climatic conditions could result in energy savings ranging from 4.8% to 34.8%. Guo and Zhang [60] examined the performance of PCM-packed building envelope for seasonal fluctuations. The PCM placed in the center of the envelope was evaluated using enthalpy porous model. The results revealed that the application of PCM layer reduced the energy consumption rate but exhibited varied performances for seasonal fluctuations with negative performance in the cold season. The melting point and volume fraction of PCM showed significant influence on the optimal PCM selection for different climate conditions. Yang et al. [61] prepared flexible shape-stabilized PCMs by using cross linked polymer swelling strategy for thermal management of building and electronic equipment. According to the findings, the prepared composite PCM demonstrated reliable mechanical properties, and its thermal storage and passive radiative cooling capacity could lower electronic device and building temperatures by 5.5 C and 9.3 C respectively.

Arici et al. [62] performed numerical study for thermal optimization of building wall containing PCM for heating/cooling thermal loads for three cities of Turkey. The considered parameters were PCM thickness, PCM melting temperature and PCM location. Results showed that maximum energy savings were attained for placing PCM near exterior plaster for summers and interior plaster for winters. Optimal PCM melting temperature range was 6–34 C and PCM layer thickness was 1–20 mm, subject to certain climatic conditions. Annual energy savings of 18% were achieved for Diyarbakir city and time delay of 10.3 h in maximum peak was attained in Konya city. Tuncbilek et al. [63] explored numerically the thermal behavior of office wall containing PCM layer under random cooling operation. The PCM parameters of location, thickness and phase change temperature were optimized for exploiting latent heat of PCM effectively. Findings revealed that PCM addition near interior was effective, optimum melting temperature of PCM was 25 C and 23 mm thick PCM layer gave energy saving of 12.8% compared to wall without PCM.

Tuncbilek et al. [64] investigated the PCM parameters of location, quantity and melting temperature by filling PCM into typical building brick under the ambient condition of Marmara region of Turkey. According to results, PCM filling in brick near indoor gave better energy savings, optimum PCM melting temperature ranged from 18 C for summer to 26 C for winter and annual optimum PCM melting temperature which resulted in highest energy conservation was 18 C. With the inclusion of PCM in brick, annual thermal load reduced by 17.6%, 13.2% of which was due to latent heat of PCM.

Wonorahardjo et al. [65] studied the passive cooling system comprising of coconut oil and water PCMs as internal thermal mass

coupled with stratum air ventilation under tropical climatic conditions. Author defined storage panel filled with coconut oil or water with inlet and outlet openings for indoor air and conditioned air respectively. Results showed that the coconut oil-based storage medium gave 0.5 C lower peak temperature in middle zone and more temperature reductions in upper zone. Passive cooling system led to a uniform temperature throughout the day and revealed low-cost and energy-efficient solution for ever increasing building energy demand challenge. Li et al. [66] explored the thermal and optical performance of window unit filled with NPs enhanced PCM. NPs of different types, different volume fraction and sizes were dispersed in PCM for thermal and optical characterization of window. Results revealed that the NPs volume fraction and size influence was significant during dawn and dusk times. CuO NPs of volume fraction below 1% and size below 15 nm were proposed for better thermal and optical performance.

Wang et al. [67] investigated the Al₂O₃/paraffin nanoPCM thermal radiative properties for the harvesting of solar energy. The parameters of volume concentration, NPs size, temperature and optical path were adjusted to improve the solar energy absorption ability of nanoPCM. The absorption coefficient of nanoPCM at 0.01% was found to be 15.5 times greater in relation to paraffin. With NPs size reduction from 30 to 10 nm, the reflectivity of nanofluid dropped by 37.04%. Similarly, the increase in temperature raised the Al₂O₃ nanoPCM absorption coefficient and decreased the reflectivity. Li et al. [68] performed thermo economic and environmental analysis of PCM loaded walls in rural residence of Northeast China by EnergyPlus tool. The parameters examined related to PCM were PCM layer position, PCM wall orientation and PCM melting point. Results indicated that energy saving of 12.9% was achieved by PCM filled in wall near the interior surface. Compared to baseline case, PCM filled wall in the south facade decreased 12.8% heating load. The optimum PCM melting temperature was reported as 16 C for interior design temperature of 18 C Carbon footprints were reduced by 52.7 kg/m² by utilizing the appropriate PCM wall throughout its lifecycle.

According to review of recent literature, the phase transition temperature, phase transition enthalpy, and thermal conductivity are three significant criteria for selecting a suitable PCM for use in potential applications. PCMs retain thermal energy during heating before undergoing phase transition and emit heat when cooled. The ability of PCMs to store and release thermal energy has prompted researchers to investigate PCM's thermophysical properties for possible applications. Owing to low thermal response rate of most of PCMs, various heat transfer improvement techniques are employed to increase energy storage and retrieval rates of PCMs. In this article, the studies for advancement in thermal properties of PCMs are extensively explored. Also, recently a lot of PCM based review papers have been published in which PCM specific application is discussed or specific improvement in thermal properties of PCM is discussed. Ahmed et al. [56] reviewed fatty acids and their eutectics PCMs enhanced by carbon based fillers. Authors reported that the latent heat storage capacity of PCMs reduced by increasing concentration of carbon-based fillers. Christopher et al. [18] studied the cascaded PCM arrangement for latent heat TES systems. It was shown that using multiple PCM improved the TES system in terms of energy and exergy efficiency, charging/discharging rate, energy storage, power density. Tariq et al. [69] discussed NEPCM preparation techniques and discussed different NEPCM applications related to TES and thermal management. Alehosseini and Jafari [70] reviewed the nanoencapsulated PCMs preparation techniques and discussed the applications of nano-encapsulated PCMs in various fields. Shen et al. [71] reviewed the thermal management performance enhancement techniques of lithium ion batteries and discussed the improvement in heat transfer and TES capacity of PCMs. Preet [72] reviewed the thermal management of PV panels using PCMs and NEPCMs. Author discussed the numerical models of PV thermal systems coupled with PCMs and NEPCMs and stated the PCM selection criteria and determination of phase change temperature of PCM. Mahian et al. [73] evaluated the building integrated PCM based PV systems, and systematically discussed the PCM-PV experimental,

numerical, optimization and economics studies. Farooq and Zhang [74] discussed the basics, materials, and approaches of the smart textiles and also analyzed the usage of PCMs in smart textiles for personal thermal management. Most recent reviews on PCMs have explored specific enhancement method and specific applications of PCM based systems. Therefore, a comprehensive review covering major PCM enhancements techniques and applications of PCM based systems needs to be addressed. This review presents a comprehensive overview on the advancements in the PCM and the use of advanced PCM in energy storage and thermal management applications. In first section, different thermal performance enhancement techniques of PCMs are discussed which includes addition of nanoparticles, fins, metal foams, encapsulation. In second section, various PCM applications are discussed thoroughly through recent literature. In third section major challenges and future recommendations related to PCM advancements and applications are discussed. The surveyed literature was collected from Google Scholar, Elsevier's ScienceDirect, and Web of Science with several keywords. The keywords included phase change materials, PCMs, NEPCMs, porous metal foams, fins, encapsulation, shape stable PCMs, thermal energy storage, latent heat TES, thermal management, thermal comfort, PV cooling techniques, solar energy, battery thermal management, building thermal management, solar collector, solar heating and cooling, heat exchanger, smart textiles, electronic devices, thermo electric generator, food. This is the novelty of this review paper that it encompasses major thermal performance enhancements in PCMs and extensively discusses diverse PCM applications. So, overall, the current manuscript will help researchers to provide a valuable insight into recent developments in latent heat thermal energy storage along with thermal management as well as their thermal energy storage applications.

Various advancements in PCMs

PCMs have drawbacks like low thermal conductivity, leakage, volume change during phase transition, flammability, super cooling, corrosiveness, and poor stability. Thermal conductivity is a critical property of PCM that influences heat transfer capacity and its low value reduces thermal response rate [75]. A variety of techniques have been investigated in literature to improve the performance of PCMs that include addition of NPs, adding expanded graphite [76,77], and preparing eutectic PCMs [20], porous metal foams [78,79], heat pipes [80,81] and capsulation of PCM as represented in Fig. 3. Encapsulation, or microencapsulation, is the process of completely covering and isolating the core from its surroundings. The thermal stability and leakage resistance of PCMs increased significantly after encapsulation, greatly expanding the application domains of PCMs. Microencapsulation is described as a physical or chemical reaction that produces a homogeneous and stable shell material enveloping the core material. On the basis of the formation mechanism and shell state, microencapsulation methods may be divided into three categories: physical, chemical, and physicochemical [82].

The initial stage in producing microencapsulated PCM and nano encapsulated PCMs is often the production of an emulsion (oil/water). The dispersed phase is primarily comprised of one or more PCM. The composition and synthesis technique of the coating around the PCM droplets vary depending on the synthesis methodology. The emulsion polymerization, suspension polymerization, and coacervation encapsulation techniques are primarily used to synthesize PCM encapsulated in organic polymers. Interfacial polymerization and in situ polymerization techniques are used to synthesize PCM encapsulated in organic and inorganic coatings. PCM encapsulated in inorganic coatings is usually produced by the sol-gel technique [83]. The various capsulation methods of PCM are summarized in Table 1.

Therefore, the above discussed are some ways to increase PCM thermal performance that recently gained significant attention.

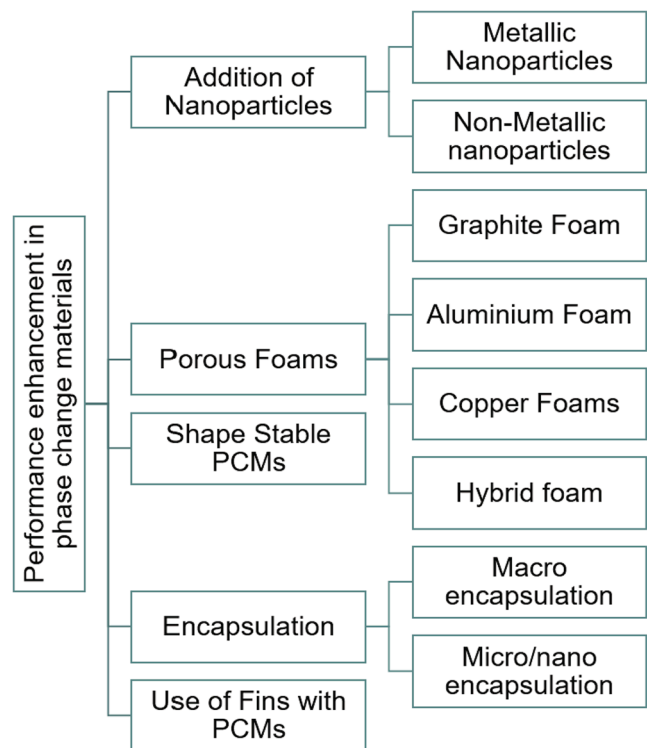


Fig. 3. Techniques for the enhancement in performance of PCMs [85].

Addition of nanoparticles in base PCM

The addition of high thermal conductivity NPs improves the heat transfer within PCMs. NPs function as a link for thermal energy transfer since they have a low thermal resistance. Nano enhanced PCMs (NEPCMs) are being studied due to high thermal conductivity of NPs and their effect of enhancement of heat storage and retrieval rate of PCMs. There are a lot of studies which explore the influence of NPs parameters like size, shape, materials and concentration on the PCM thermal properties. It is reported that the heat transfer rate and thermal reliability of PCMs have improved with the addition of NPs. The nonmetallic NPs have better influence on thermal characteristics of PCMs than metallic NPs [86]. Thermal conductivity of NEPCMs is reported to enhance by 20% to 100% as compared pristine PCM. The NPs addition in salt hydrates PCM improve their sub cooling, phase segregation and thermal conductivity properties [87].

Nizetic et al. [88] critically reviewed studies of NPs enhanced fluid in terms of preparation, applications and thermal properties. Findings showed that the use of NPs in PCMs increased the thermal conductivity but marginally lowered latent heat. In TES and cooling systems, NEPCMs are mainly used. Authors concluded that literature does not provide the optimal preparation process for nanofluids, economic and environmental assessments, safety concerns and handling guidance.

Wu et al. [89] explored the thermal conductivity of paraffin by

Table 1
Summary of various methods of capsulation of PCM [84].

Category	Various Methods
Physical method	Pan coating, air-suspension coating, centrifugal extrusion, spray drying, vibrational nozzle, solvent evaporation/ extraction, vacuum impregnation, electrostatic binding
Chemical method	Interfacial polymerization, emulsion polymerization, suspension polymerization, in situ polymerization
Physical-chemical method	Complex coacervation, Sol-gel Synthesis, ionic gelation.

dispersing graphene nanoplatelets (GNPs) and carbon nanotubes (CNTs) NPs. The dispersant WinSpense3050 was added for stable dispersion of NPs in PCM. Dispersion of 5% GNPs in PCM enhanced the thermal conductivity by 37.1% than pristine PCM. Melting/freezing rates and thermal stability of PCM were improved by the addition of CNTs and GNPs. Kalidasan et al. [90] prepared nanocomposite of polyaniline and cobalt NPs and mixed it with paraffin wax in various proportions (FESEM image shown in Fig. 4). According to the findings, enriching the PCM with polymer nanocomposite increased the latent heat of fusion and thermal conductivity by 15.9% and 20.4%, respectively and improved thermal stability for 200 thermal cycles.

Daneshazarian et al. [91] numerically inspected the thermal performance of halogen graphene nanoplatelets (xGNP) enriched PCM packed in vertical cylindrical cavity at different xGNP concentrations. Results showed that with 0.5 wt% of xGNP in the PCM, melting rate decreased by 9.7% and heat storage rate improved by 12.6%. Authors further noted that the inclusion of NPs in PCM by more than 0.5 wt% decreased melting rate and heat storage rate, as well as increased the dynamic viscosity. Liu et al. [92] prepared hybrid NPs through coating carbon layer on aluminium NPs and dispersing them in sodium sulphate decahydrate for thermal efficiency enhancement. According to the findings, the use of 3 wt% hybrid NPs increased thermal conductivity by 26.41%, decreased the freezing latent heat enthalpy by 5.13%, and strengthened the stability of PCM.

Raj et al. [93] studied thermal characteristics of shape-stabilized polyethylene glycol dispersed with 1 wt% MWCNTs and GNPs. Various thermal, structural, and chemical analysis demonstrated that the NEPCM possessed better thermal response, good chemical and thermal stability. Results concluded that addition of MWCNTs and GNPs in PCM enhanced the thermal conductivity of PCM by 61.73% and 84.48% respectively and lowered the heat sink base temperature by maximum of 9.77%. Benbrika et al. [94] numerically analyzed the melting/solidification characteristics of a latent heat TES unit of horizontal cylinders by diffusing GNPs in PCM. According to the findings, with the addition of 3 wt% GNPs in PCM, solidification time was reduced by 50%, melting time slightly reduced and energy storage capacity was reduced by 14%.

Sun et al. [95] investigated melting process of TES system utilizing paraffin PCM and graphite and nano coconut shell charcoal. The concentration of NPs was 0.02, 0.06, and 0.10 wt%. It was found that the melting time shortened by 21% with 0.06% graphite nano particles in PCM with 2 wt% oleic acid dispersant. Qu et al. [96] studied the face stabilized paraffin with the hybrid nano additives of expanded graphite (EG)/MWCNTs and EG/carbon nano fillers (CNF). The shaping material for paraffin was polyethylene and the concentration of NPs and EG was varied. According to findings, PCM thermal conductivity enhanced by 60% and 21.2% for MWCNTs/EG and CNF/EG for mass ratio of 4:1 respectively compared to case of 5% EG.

Shahsavari et al. [97] performed experimental and numerical study to evaluate the effect of temperature and concentration of NPs on viscosity and thermal conductivity of paraffin/ Fe_2O_3 PCM. For improving NPs dispersion, oleic acid was utilized. According to findings the increase in

NPs concentration augmented both “k” and viscosity of nanofluid while the increase of temperature enhanced the “k” and decreased the viscosity. Furthermore, the artificial neural network (ANN) model was utilized to simulate the “k” and viscosity of nanofluid using experimental data. It was observed that the developed ANN model provided accurate results.

Arici et al. [98] explored the melting of PCM with various arrangement of fins, various wall orientations, and CuO NPs. As compared to the no fin case, the integration of fins in the cavity shortened the melting time by maximum value of 52% and 68% based on fins length and fins position, respectively, with short fins performing better with NPs. While it was observed that the combine use of fins and NPs augmented the melting rate by 54%. Sarani et al. [99] examined the effect of discontinuous fins configuration and CuO NPs on PCM solidification rate enhancement. Findings showed that the inclusion of discontinuous fins increased the discharging rate by 89%. Maximum enhancement in discharge rate of 77.8% and 59.5% was observed for the combination of CuO NPs with copper and aluminium discontinuous fins respectively. Sheikholeslami et al. [100] explored the melting/freezing of RT35 PCM with triangular fins and CuO NPs in triple tube heat exchanger. The use of triangular fins increased freezing rate by about 40.75%, and the addition of CuO NPs lowered freezing time by 44.88%. The combine use of fins and NPs in the system lowered the solidification time from 20.2 min to 11.13 min. Alizadeh and colleagues [101] investigated numerically the solidification process of latent heat TES device with curved fins and numerous NPs. The results revealed that with 4% single walled carbon nanotubes, increased solidification rate of PCM by 40.9% and optimized fin shortened the freezing time of PCM by 61.54%. The combine use of optimized fins and NPs increased the rate of solidification by 68.81 %.

Use of fins along with base PCM

PCMs have good energy density, and constant phase change temperature but the major drawback of PCMs is weak heat transfer during melting/freezing cycles especially in case of organic PCMs. So for accelerating the PCM responsiveness for heat transfer, increasing contact area by utilization of fins is widely researched technique due to favorable outcomes compared to no fin case [102,103]. Different types of fins i.e. rectangular [104], longitudinal [105,106], radial [107,108], annular [109], tree [110,111], plate [112], spiral [113,114], triangular [115], trapezoidal [116], corrugated [117], perforated [118,119], helical [120], and some others shapes [121–124] have been studied for optimizing the geometric shapes, size, orientation and materials of fins. In recent studies, various types of fins along with NEPCMs and metal foams are utilizing for thermal performance enhancement of latent heat thermal energy storage systems [125,126–131]. By using various type of fins with base PCM, the effect of heat transfer is increased due to large area availability.

Ren et al. [132] studied the improvement in melting of PCM by incorporating triangular fins and copper nanoparticles in latent heat enclosure. The results showed that the melting rate of PCM was

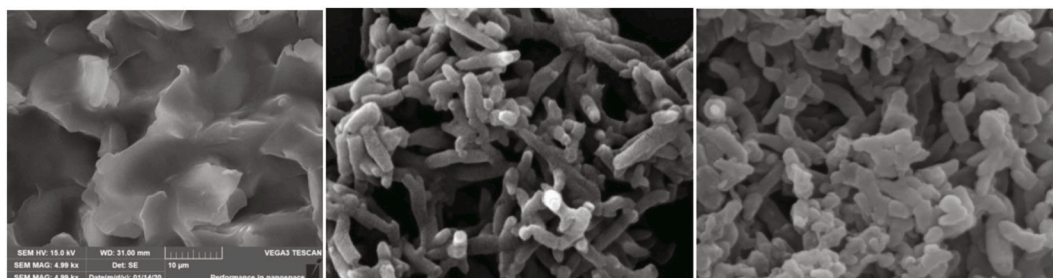


Fig. 4. FESEM images of (a) Pure paraffin (b) polyaniline and 1% cobalt (c) polyaniline and 2% cobalt [90].

improved with triangular fins as compared to rectangular fins and the addition of NPs. The long and slender triangular fins were recommended for attaining high energy storage efficiency. Hasnain et al. [133] investigated numerically the melting of PCM by utilizing branched fins and Al_2O_3 nanoparticles in horizontal latent heat TES system. It was revealed that the double branched fins lowered melting time by 45.9% and inclusion of 10% Al_2O_3 nanoparticles in PCM reduced melting time by 26.8% as compared to simple PCM case.

Zhang et al. [111] designed and investigated the PCM based TES system with novel shape fins. Topology optimization technique was used to optimize the PCM and fins layout in TES system. Results showed that the optimal volume fraction of fins to be 20%, while charging and discharging rate of PCM enhanced by 46.8% and 47.1% respectively compared to longitudinal fins case. Huang et al. [134] investigated the charging and discharging behavior of fractal tree shaped fins based latent heat TES system. Results showed that inclination angle influence melting process more than freezing process. The vertical fractal latent heat system gave 52.5% reduction in melting time while horizontal fractal latent heat system gave 94.7% and 101.5% improvement in melting and freezing rate respectively.

Yang et al. [135] studied the thermal performance of irregularly spaced annular fins based shell and tube latent heat TES system. Results indicated a 62.8% reduction in total melting duration, 34.4% temperature difference reduction in PCM region and 84.7% improvement in melting rate uniformity.

Liu et al. [136] investigated the solidification ability of latent TES system using novel triangular shaped longitudinal fins depicted in Fig. 5. Aside from fin geometry, the impact of initial temperature and fin material on solidification rate was also investigated. The results showed that the novel fin shortened solidification time by 38.30% as compared to the simple longitudinal fin case. The optimal temperature gradient between the inner wall and the PCM was over 20 K for a quick solidification rate.

Kothari et al. [137] studied the effect of varying the inclination angles of paraffin base heat sinks with plate fins. According to the findings, for critical temperature of 75 °C, three finned heat sink at a 0° inclination angle increased operating time by 74%, while a decrease in inclination angle from 90° to 0° enhanced operating time by 66%, 46%, and 43% for no fin, one fin, and two fin PCM based heat sinks, respectively. With reduction in inclination angle from 90° to 0°, the PCM melting time for unfinned and finned heat sinks was reduced by 44 % and 30 %, respectively.

PCMs saturated with porous materials

Metallic and non-metallic porous foams saturated with organic paraffin type PCMs are being explored for their thermal performance enhancement of PCMs and their possible usage in diverse applications. The integration of porous materials and PCMs results in shape-stabilized PCMs which have better thermal conductivity and no leakage issues

[40].

Yao and Wu [138] explored the thermal characteristics of composite PCM of copper metal foam and paraffin by varying the foam porosity and pore density. The low porosity and high pores density were observed as ideal for the uniform distribution of temperature and low temperature maintenance. Copper foam with porosity of 0.935 and pore density of 7 pores per inch (PPI) augmented the melting rate by 2.1 times and maximum temperature drop of 30 °C was achieved. Zhang et al. [139] explored the charging behavior of paraffin saturated with copper foam under varying centrifugal accelerations. By the centrifugal acceleration, the heat transfer coefficient and the PCM convection have been improved. The melting rate of PCM increased asymmetrically in the direction of centrifugal acceleration, resulting in phase interface with a curve pattern. Hu and Gong [140] investigated the thermal regulation of RT 62HC PCM infiltrated with 3D printed porous aluminium material. The research used a variety of porosities and heat fluxes with various thermocouples (TC) positions. The results revealed that for 80% porosity of porous material, the highest increase in PCM melting time and temperature reduction of 78% relative to the case without porous material was achieved (charging/discharging figure shown in Fig. 6).

Ferfera et al. [141] used numerical simulations to analyze the microcellular foam impregnated in paraffin PCM. The effect of porosity on PCM based heat sink thermal conductivity and thermal management was examined. The thermal conductivity of copper and nickel foams with porosities of 96.5%, saturated in PCM, was 28 and 8 times higher than pure PCM, respectively. Porosity reduction improved phase transition rate and increased thermal diffusivity and effusivity. Dinesh and Bhattacharya [142] numerically explored the effect of metal foam pore

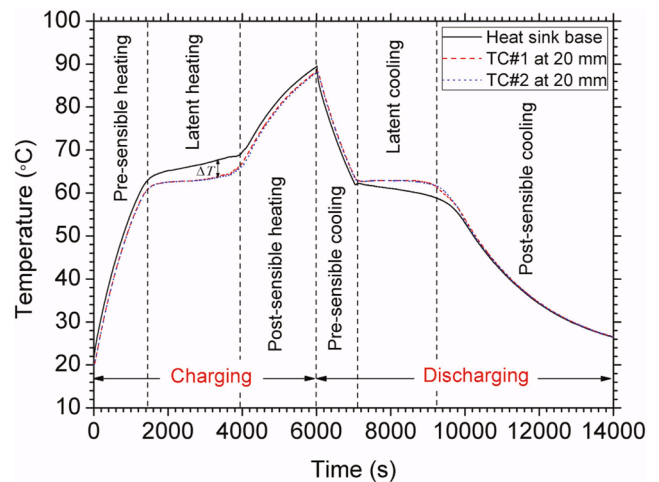


Fig. 6. Charging/discharging profile of PCM heat sink with 80% porosity metal foam for power input of 8 W at various thermocouples positions [140].

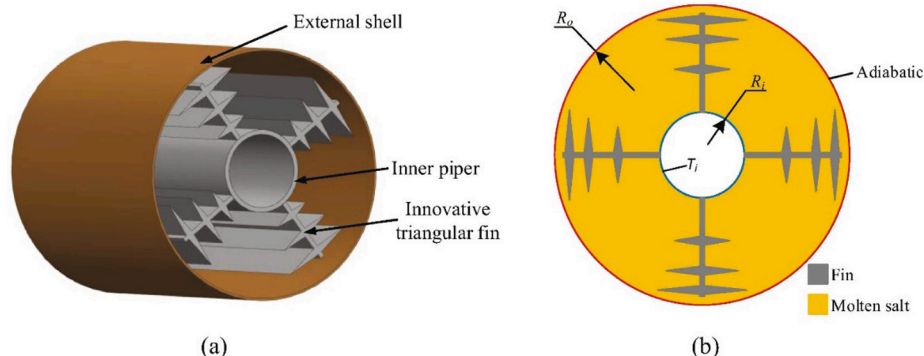


Fig. 5. Latent Heat Storage unit with novel triangular fins [136].

size on PCM thermal behavior. According to the findings, the pore size had a significant impact on the melting rate of PCM with the same porosity, with a smaller pore radius resulting in quicker melting and energy storage. Although the melting rate for large pore sizes was initially high, it gradually decreased. Lower porosity resulted in a higher rate of energy storage for pore sizes of equal scale.

Lolasiello et al. [78] explored the melting process of PCM saturated with aluminium metal foam under various heating loads, porosity, number of PPI and orientations. Results revealed that the lower porosity decreased the melting time, while PPI had no effect and orientation had marginal effect on melting time. Cumulative energy analysis showed that higher energy stored by PPIs under vertical direction and porosity had little impact on the energy storage as low porosity meant less PCM. Hasan et al. [143] performed thermal investigation on paraffin metal foam composite, prepared by prepared vacuum impregnation method. Nickel and aluminium metal foams were used and maximum impregnation ratio of 98% was achieved. The results showed that paraffin/nickel and paraffin/aluminum composite thermal conductivity was 6 times and 18 times higher than PCM respectively. An empirical model for thermal conductivity prediction has also been developed by the author.

Shape-stabilized PCMs

Shape-stabilized PCM is another process in which molten PCM is held in place by capillary action in porous material. Shape-stabilized materials include mineral materials like diatomite, vermiculite and perlite, as well as polymeric materials such as polyethylene, polyurethane, or acyclic resins. Mineral porous materials are mostly used to hold organic PCMs, while polymeric porous materials are used to contain both organic (mainly paraffins) and inorganic PCMs. The methods adopted to prepare the shape-stabilized PCMs include physical adsorption, electrospinning, centrifugal spinning, and intercalation. The selection of suitable porous material is influenced by porosity, thermal conductivity, availability, chemical consistency with the PCM, and preparation techniques [144,145].

Atinafu et al. [146] developed a composite PCM of n-dodecane PCM that used biochar-CNTs as a supporting material. According to the findings, biochar-CNTs and CNT assisted PCM had PCM loading capacities of 70.2% and 83.2%, as well as latent heat of 127.4 J/g and 152.3 J/g, respectively. Besides composite PCM showed better chemical uniformity, form stability, and thermal stability up to 600 °C with considerable weight loss. Also, Atinafu and colleagues [147] investigated the shape-stabilized PCM of eicosane and biochar harvested from wheat straw and softwood. According to the results, the composite PCM of eicosane-wheat straw biochar had the highest TES capacity of 75 J/g and 37.1% encapsulation capacity, while the composite PCM of eicosane-softwood biochar had the lowest TES capacity of 52 J/g and 26.4% encapsulation capacity.

Li et al. [148] used silicon carbide powder, silicon carbide whisker, and silicon carbide nanofiber supports to prepare composite stearic acid PCMs by vacuum impregnation and ethanol techniques. According to findings, the thermal conductivities of composite PCMs were 61.80% to 169.29% more as compared to pure PCM. Besides, the composite PCM demonstrated improved charging/discharging behavior and TES capacity. Chriaa et al. [149] prepared the shape-stabilized PCM by using hexadecane and various styrene-*b*-(ethylene-co-butylene)-*b*-styrene (SEBS) copolymer concentrations with polyethylene coating. The composite PCM was reported to be melted at 25 °C with latent heat of 162.3 J/g. With good form stability, PCM loading approached 80% and thermal deterioration of the composite PCM commenced above 200 °C.

Shen et al. [150] produced shape-stabilized PCM by impregnating cellulose sponge with an emulsified hydrated salt/paraffin composite. According to the findings, shape stabilized PCM increased thermal stability, provided high storage density, decreased super cooling, and had negligible leakage. The phase shift enthalpy of shape stabilized PCM was found to be 227.3 J/g. Xie et al. [151] prepared shape-stabilized

composite PCM by physically mixing the eutectic mixture ($\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$ and $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) with diatomite and coating it with polyurethane acrylate. Due to polymer coating on the composite PCM, leakage was repressed, and salt hydrate supercooling was precluded. High latent heat of fusion of 102.6 J/g, as well as increased thermal reliability and stability was attained. Shape-stabilized composite PCM also minimized high peak temperatures and temperature variations.

Performance enhancement using capsulation of PCM

Addition of NPs increases the viscosity and causes agglomeration in the base fluid due to a stabilization problem [69,86]. One approach to increase PCM performance include encapsulation of PCM [152]. PCM encapsulation is a recent advancement in the field of TES that addresses problems in PCMs such as phase change leakage, low thermal conductivity, low thermal stability, corrosion, super cooling, and environmental deterioration. There are several kinds of encapsulation based on capsule size, such as macroencapsulation, microencapsulation, and nanoencapsulation, although the microencapsulation is the most researched size.

The shell material selection is influenced by several factors, including the operating temperature range, thermal conductivity, toxicity, the encapsulation method, permeability, and compliance with the base PCM. Organic shell materials include melamine-formaldehyde, urea-formaldehyde, polystyrene, polyurethane, and polymethyl methacrylate, among many others. Inorganic shell materials include silica, calcium carbonate, and titania. Some of the methods to prepare encapsulated PCMs are polymerization, coacervation, sol-gel, ionic gelation, spray drying, electrostatic, and solvent evaporation. Each these PCM encapsulation methods has its own combination of benefits and drawbacks, the encapsulation approach should be chosen based on the type of PCM, cost, desired encapsulation efficiency, thermal stability, and reagent accessibility [152,153].

Nikpourian et al. [154] using interfacial polymerization process prepared nano encapsulated paraffin wax with polyurethane. The aim of the study was to examine nano capsules' chemical composition, thermal properties, and morphology. The nano capsules had an energy storage efficiency and capacity of 80.2% and 97.5%, respectively, as well as melting/freezing phase transition enthalpies of 153.9/142.3 J/g. After 100 heating/cooling cycles, the prepared core shell structure demonstrated high thermal stability. Zhang et al. [155] performed thermal analysis of nano encapsulated PCM of octadecane core with pentafluorostyrene shell. According to results nano encapsulated PCM presented encapsulation efficiency of 76%, melting enthalpy of 171.8 J/g, and the best thermal performance at shell/core mass ratio of 1:2. Pandey et al. [156] prepared encapsulated PCM of n-eicosane core and shell of amphiphilic copolymer of poly methyl methacrylate and poly 2-hydroxyethyl methacrylate by double emulsion technique (SEM image shown in Fig. 7). According to the findings, the encapsulated PCM had a 95% loading efficiency, 160 J/g latent heat, uniform thermal behavior after 500 heating/cooling cycles, and no leakage issue.

Abdeali et al. [157] prepared nano encapsulated PCM by using a mini emulsion polymerization process, with paraffin wax as the core and polyurethane as the shell. The small particle size distribution was observed when the sonication time was increased from 15 to 20 min and the reaction temperature was raised from 60 to 80 °C. It was observed that the narrow particle size distribution of core/shell resulted in uniform melting and a lower temperature rise. Tian et al. [158] used coacervation method to prepare MEPCM paraffin with gelatin-sodium alginate as shell material. According to the findings, encapsulation ratio of 82.12% and encapsulation efficiency of 72.5% was attained with melting latent heat and temperature of 126.8 J/g and 60.4 °C and freezing latent heat and temperature of 51.3 °C and 133.8 J/g, respectively. As compared to pure PCM, MEPCM gave high thermal stability, no leakage, and less decomposition. Yu et al. [159] prepared MEPCM

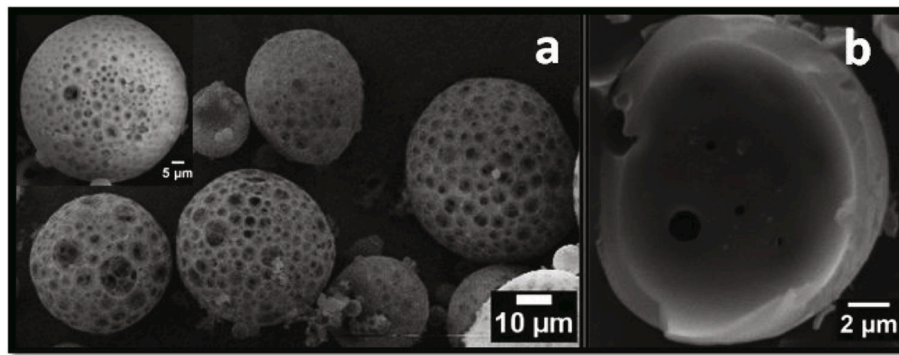


Fig. 7. (a) SEM image of microencapsulated PCM (b) cross-sectional SEM image of microencapsulated PCM [156].

containing paraffin core and double-layer shell with first layer tetraethyl orthosilicate (TEOS) hydrolyzed Silica and second layer of modified spherical silica. Results demonstrated the MEPCM improved thermal conductivity by 133.33% and sustained sealing at 65 °C. The PCM loss was postponed by 47 °C and latent thermal energy approached 166.61 J/g.

Suresh and Pethurajan [160] investigated the thermal characterization of MEPCMs fused with GNPs in a finned heat sink. The PCM shell was made of polymer, with NPs concentrations of 0.5, 1, and 3 wt%. Results revealed that the thermal response rate of MEPCM with GNPs increased by 29.6 % during heating at the set point temperature of 60 °C. At 10 W, the time delay to reach 80 °C was 22.5 % for microencapsulated PCM and 18 % for PCM with 3 wt% GNPs.

Rakkappan et al. [161] studied PCM encapsulated in urea-formaldehyde shells modified with GNP and CuO NPs at concentrations of 2, 6, and 10 wt%. Results revealed that as opposed to the unencapsulated PCM scenario, the encapsulated PCM with 6 wt% GNPs and 10 wt% CuO NPs resulted in charging, discharging time reductions of 25.22 %, 31.01 % and 31.91 %, 37.02 %, respectively.

Bergamo et al. [162] prepared nano encapsulated PCM by coating TiO₂ shell on soybean oil derivative, which was then tested for its potential use in TES system. Because of the TiO₂ shell, the thermal resistance of encapsulated PCM was reduced by 70% compared to unencapsulated PCM. While the cooling time of water increased by 18%, the thermal energy retained in TES system was 10.6% higher than blank TES.

Wang et al. [163] explored the thermal performance of various PCM encapsulating shapes for free and forced convection. The studied shapes were spherical, cylindrical, tetrahedral, pyramidal and biconcave disc red blood cell. The tetra-n-butylammonium bromide based PCM with phase transition range of 0–12 °C was employed as PCM. Results showed that red blood cell shaped encapsulated PCM outperformed other shapes due to rapid melting/freezing processes which were further improved by free stream flow of heat transfer fluid.

Mohaghegh et al. [164] investigated the thermal properties of n-octadecane PCM encapsulated in a pear-shaped geometry. To establish the restriction in geometry, a thin insulated cylindrical rod was positioned in the centre of pear-shaped cavity. According to findings, the PCM encapsulated in pear-shaped geometry took 178 min to melt in constrained scenario and 97 min in unconstrained case. Furthermore, the pear-shaped design took less time to melt (97 min) than a standard cylindrical cavity (108 min).

Ishak et al. [165] synthesized MEPCM using the sol gel technique, utilizing stearic acid PCM as the core and SiO₂ as the shell material. Tetraethyl orthosilicate was employed as precursor to create the SiO₂ layer on the stearic acid. Thermal investigation revealed that MEPCM possessed melting/solidification temperatures and latent heats of 70.37 °C, 64.27 °C and 182.53 J/g, 160.12 J/g, respectively. It was reported that the maximum encapsulation efficiency was 86.68%.

Chen et al. [166] prepared composite eutectic salt PCM by infusing

the sodium thiosulfate pentahydrate and sodium acetate trihydrate eutectic salt with macro-porous carbonized melamine sponge. The composite eutectic salt PCM was then encapsulated in silicon carbide modified polyurethane shell. The macro encapsulated PCM was subjected to comprehensive thermal and mechanical evaluation. The results indicated that macro encapsulated PCM displayed improved mechanical strength and thermal conductivity, as well as outstanding thermal stability and thermal reliability.

Shi et al. [167] utilized acetylated cellulose nanofibrils as the shell and paraffin PCM as the core to prepare MEPCM. PCM and acetylated cellulose nanofibrils content was altered in various proportions. The leakage proof MEPCM demonstrated high energy storage capacity of 173 J/g, high PCM loading of 80%, outstanding thermal stability after 100 heating/cooling cycles, decreased super cooling, and effective shape stability.

Xing et al. [168] prepared MEPCM by employing eutectic of capric and palmitic fatty acids as the core and polyvinyl chloride as the shell. For encapsulating, the author utilized the solvent evaporation technique. Using a core shell ratio of 1:2, the MEPCM displayed melting/solidification temperatures of 17.1 °C 18.8 °C, latent heat values of 92.1 J/g, 91.3 J/g respectively, and the encapsulation ratio of 57%. MEPCM demonstrated outstanding thermal reliability after 500 thermal cycles, and improved thermal stability.

Cai et al. [169] used dodecanol dodecanoate PCM as core material to prepare MEPCM via interfacial polymerization of Toluene 2, 4-diisocyanate, diethylene triamine. Solvent-free technique was used in the polymerization of MEPCM. According to the results, the MEPCM had spherical shape, latent heat range of 103.4–140.3 J/g, strong thermal resistance with commencing evaporation temperature of 234 °C, and enhanced thermal conductivity.

Sari et al. [170] studied the heptadecane MEPCM with calcium carbonate shell for leakage prevention and thermal conductivity enhancement. Results showed that the prepared MEPCM had outstanding thermal durability, enhanced thermal conductivity, long term stability, high latent heat capacity. The MEPCM was observed as promising candidate for TES applications. A detailed summary of various advancements in PCMs is presented in Table 2.

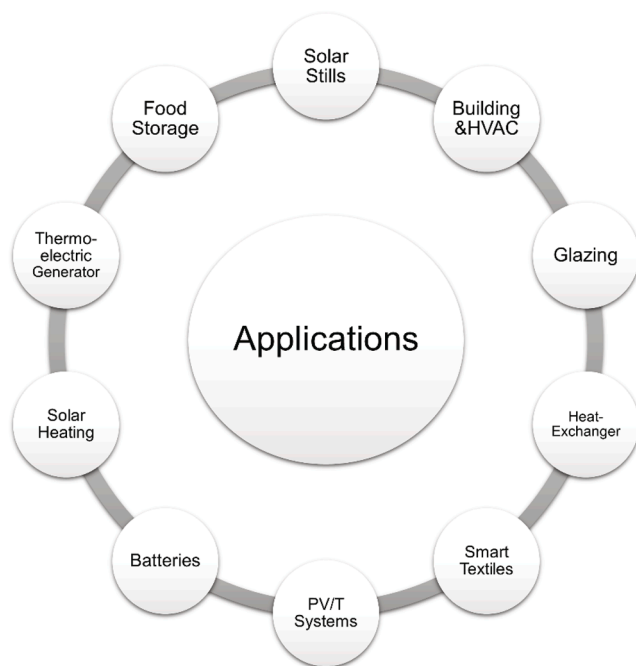
Applications of advanced PCMs in thermal management field

PCMs come with a variety of melting temperatures and enthalpies to meet the needs of different applications. TES systems are utilized in heat absorption and dissipating applications using PCMs for efficient utilization of energy [182]. PCMs are incorporated in solar PV systems, heat exchangers, building, glazing, textiles, packaging, automotive, waste heat recovery, and thermoelectric generators, (depicted in Fig. 8) for optimum operation and to keep system in an optimum temperature range [183,51,184,48,185,186,3]. The potential applications of advanced PCMs for TES and thermal management are discussed below in detail.

Table 2

Summary of various methods of Advancements in PCMs invested by different researchers.

Reference	Study type	PCM	Additive for PCM enhancement	Additive quantity	Key Findings
[171]	Numerical	Paraffin	CuO NPs	0.05 %	Solidification rate enhanced by about 38%
[172]	Experimental	Capric acid	MWCNTs NPs	0.01, 0.02, and 0.025%	Heat transfer rate augmented by 30.97% with 0.02% MWCNTs in PCM.
[173]	Experimental	Paraffin wax, RT35	Al ₂ O ₃ , CuO, and MWCNTs NPs	0.5, 1.0, 2.0, and 4.0 %.	NEPCM with 1.0 wt% Al ₂ O ₃ NPs gave 17.4% lesser temperature than pristine paraffin at 8 W.
[174]	Numerical	Erythritol	Copper NPs	2.5%	15% surge in surface temperature and 108% reduction in melting time was noticed.
[175]	Experimental	Paraffin	Nanographene NPs	1, 2 and 3%	At 3% nanographene, thermal conductivity of PCM presented a maximum of 146%
[176]	Experimental	Paraffin	SWCNT NPs	0.5, 1 and 3%	The inclusion of NPs in PCM enhanced thermal conductivity of paraffin by 12%, while melting and solidification point reduced by 4.5 °C and about 9 °C respectively.
[177]	Experimental	Paraffin	CNTs NPs	5, 10 and 15%	Maximum thermal conductivity enhanced by 323.2% with 15% CNTs and temperature reduced from 3.1 to 6.6°.
[178]	Numerical	Paraffin	Aluminium metal foam	50, 60, 70% porosity, various pore sizes	Effect of convection was higher at high porosity and high pore size.
[179]	Numerical	Paraffin	Copper foam fins	97% porosity and the pore density of 25PPI	Phase transition time was reduced by 43.16% with the inclination angle from 0° to 180°.
[180]	Experimental and Numerical	Paraffin wax	Al ₃ O ₃ and CuO NPs and fins	5 wt%	Melting time reduced by 63% with the introduction of fin in the heat storage cavity.
[91]	Numerical	Octadecane	GNPs	0.2, 0.4, 0.5, 1 and 2%	Adding 0.5 wt% reduced melting time by 9.7% and enhance heat storage rate by 12.6%
[181]	Experimental	RT-35HC	TiO ₂	0.5, 1, 1.5 and 2 wt%	Thermal conductivity enhanced by 59.5 % at 2 wt% NPs in PCM.

**Fig. 8.** Various applications of advanced PCMs.

PCMs usage in solar stills/desalination systems

Energy and water scarcity are two of the world's most pressing challenges, all of which influence a country's overall economic situation. Around a third of the world's countries face water scarcity, and about 97% of the earth's water is seawater, which is unfit for consumption or crop cultivation. 2% of available freshwater is frozen, and 1% is underground, canals, rivers, and reservoirs [187]. However,

microbes, trace elements, industrial waste have polluted this water [188–190]. Solar distillation appears to be a feasible choice, and solar stills are a research priority due to the geographical position of many countries with significant solar energy capacity [191]. The usable heat from the sun can be used in the solar still in the same way as rain does. After evaporation, saline water in a solar still can be drained and used as freshwater. By varying different parameters, a lot of research has been done to advance the effectiveness of traditional solar stills. Because of its high energy storage capacity, PCM has been considered in improving solar still efficiency [192]. Recently, plenty of experimental and theoretical research has been done in the literature which is discussed below.

For fresh water generation, Kumar et al. [193] compared the performance of single slope solar still using PCM and NEPCM to the standard passive solar still. A solar still holding 7.5 L of water was filled with 5 kg of PCM, and 0.5 percent mass silica NPs were introduced to the paraffin PCM. The incorporation of PCM and nanoPCM increased freshwater production by 51.22% and 67.07%, respectively, compared to simple solar still according to the results. In the Tehran climatic conditions, Jahanpanah et al. [194] examined the inclusion of low-temperature PCM of 3 kg and 6 kg amounts in the desalination solar still shown in the Fig. 9. The insertion of 6 kg PCM in the solar still increased daily productivity by 30.3% and improved efficiency by 8.29%, suggesting that increasing PCM quantity improved solar still performance.

The solar still with external condenser enclosing paraffin wax PCM and evacuated tube collector was tested experimentally by Faegh and Shafii [195]. During the daytime, PCM extracted waste heat from condensed vapor from saline water, and in the evening, it released heat, assisting the desalination operation. The results showed that using solar still with an external condenser enclosing PCM improved distillate output by 86%, with a gross yield of 6.55 kg/m²/day, a 50% efficiency, and a cost of 0.015 USD/L/m². Harahsheh et al. [196] explored the effectiveness of a solar still system by employing inorganic salt PCM and solar collector. Different parameters such as hot and cold-water circulation rate and basin water levels for distillation productivity were

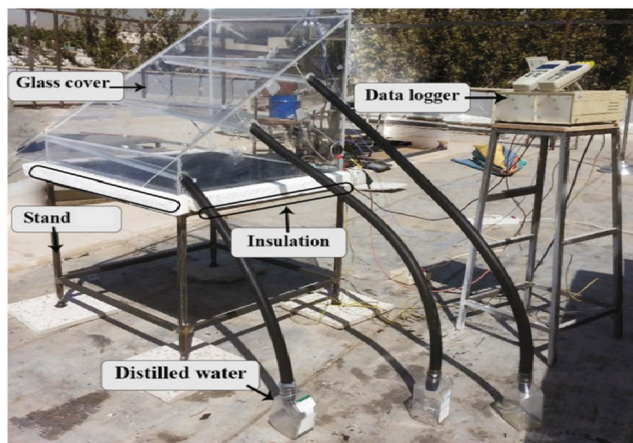


Fig. 9. Pictorial view of Experimental Setup [194].

evaluated. The findings indicated that the higher ambient temperature and high hot water flow rate were closely connected with improved solar still efficiency, the optimum cooling water flow rate of being 10 m/s, while an increase in water level in the basin decreased the productivity. The maximum daily yield of 4.3 L/m²/day was obtained, and 40% yield obtained after sunset. Arabi et al. [197] theoretically investigated a solar still with a solar collector and various PCMs enclosed in tubes (as presented in Fig. 10) to improve the solar still efficiency at various parameters. The findings showed that increasing solar irradiation, installing an additional collector, increasing coolant flow rate, and increasing PCM mass improved the solar still system's efficiency substantially. Solar still productivity for simple solar still, with external collector, and with external collector and PCM tubes was 2.8, 5.1, and 6.31 L/m² respectively, after 24 h of operation.

Kateshia and Lakhera [198] examined the efficiency improvement of single slope solar still with PCM and aluminium pin fins. The output, energy efficiency and exergy efficiency of the solar system still with PCM/pin fins augmented by 30%, 47% and 63% relative to a conventional scenario; and costs per liter was 0.017 USD/L/m².

Vigneswaran et al. [199] inspected the thermal efficiency of a conventional solar still combined with single PCM and a double PCMs to optimize the distillate yield. The latent heat of PCM was used to provide contingency yield during periods of low irradiation time. Finding showed that the solar still with two PCMs enhanced the yield discharge time by 25% as compared to the single PCM case. Multiple PCM solar stills had a 46.29% efficiency and yield of 4.40 L/m²/day, which was higher than a single PCM and conventional solar still case.

Kabeel et al. [200] examined the enhancement of distillate output of

tubular solar still by using copper tubes packed with PCM, during the day and night time. According to results, the output and daily efficiency of traditional solar stills were approximately 4.3 L/m²/day and 33%, respectively, while solar stills with copper tubes enclosing PCM generated output of about 9 L/m²/day and daily efficiency of around 72%. Mousa et al. [201] investigated the candle wax (tricosane) filled in copper tubes of solar desalination system. The experiment used 3 kg of tap water and variable mass of PCM. Presence of PCM resulted in charging and discharging zones, in which temperature was highly influenced by PCM, with the first zone resulting from PCM melting and the second arising from PCM solidification. According to the findings, PCM had a negative effect on water output during the day but a positive impact at nighttime.

Elashmawy and colleagues [202] designed novel paraffin filled aluminium tubes for performance enhancement of tubular solar still. To boost radiation absorption, a solar tracking device was mounted on the parabolic solar concentrator, and copper rods were inserted into paraffin wax to improve thermal behavior. Solar still with PCM tubes improved output and efficiency by 40.51% and 38.25%, respectively as shown in the Fig. 11. PCM tubes solar still resulted in output, efficiency, and cost of 5.55 L/m²/day, 44%, and 0.00782 USD/liter, respectively. As opposed to current solar desalination methods, a 21.8% cost reduction was achieved.

Sarhaddi et al. [203] numerically studied the exergy and energy efficiency of the PCM based weir type cascade solar still. The influence of solar radiation and ambient temperature for sunny and cloudy days were studied. Results showed that the total energy and exergy efficiency of solar still without PCM on a sunny day and with PCM on a cloudy day were 76.69%, 6.5% and 74.35%, 8.59%, respectively. For cloudy days, PCM-based solar still was recommended, while for sunny days, solar still without PCM was preferred. The largest irreversibility rate of solar still with and without PCM was for absorber plate; according to the author, the absorber plate should be the subject of research for design optimization. The summary of previous studies on solar desalination systems utilizing PCMs is summarized in Table 3.

PCMs as thermal energy storage in building air conditioning/HVAC applications

Building air conditioning is a high energy consumption practice worldwide; with heavy environment impacts i.e. global warming and climate change [213,214]. Since temperature and demand fluctuate on a regular and seasonal basis, conserving energy in buildings is very challenging [215]. Nearly 40% of global energy demand is accounted for by the building sector [216]. Heating and cooling loads account for a substantial portion of overall energy consumption in buildings. To meet peak electricity demand and lower costs, utility companies usually give

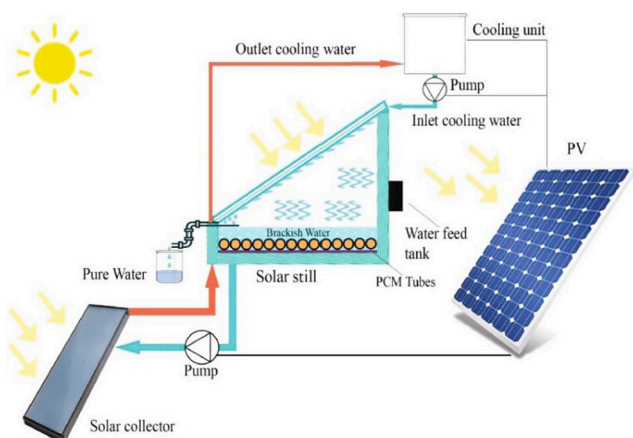


Fig. 10. Schematic Diagram of theoretical study [197].

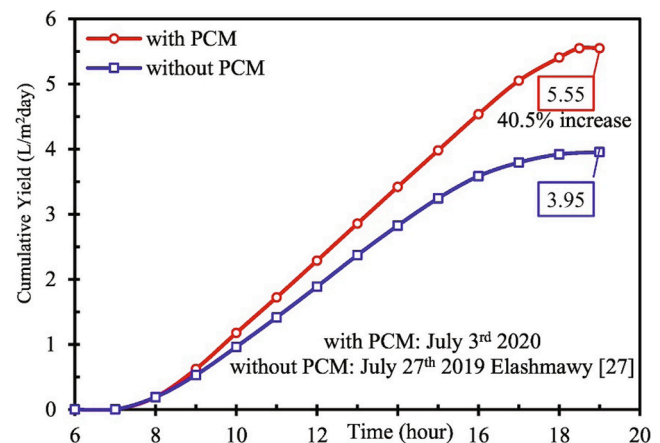


Fig. 11. Cumulative Yield of solar still with PCM and without PCM [202].

Table 3

Summary of previous studies on solar desalination systems utilizing PCMs.

Reference	Type of study	Location	Type of solar still	Maximum Solar Radiation (W/m ²)	Storage Material used	Productivity (L/m ² /day)	Achieved Efficiency (%)	Cost per liter (USD/L)
[204]	Experimental	Egypt	Tubular still with CPC	1050	Paraffin wax and graphite NPs	9.81	45.3	
[205]	Experimental	Egypt	Pyramid solar still with hollow cylindrical fins	1065	Paraffin wax	8.1	64.3	0.019
[206]	Experimental	Bangladesh	Double slope solar still with external condenser and rectangular fins	1100	Paraffin wax	3.07	39.74	0.0117
[207]	Experimental	Egypt	Single slope solar still coupled with two parabolic trough collectors	1100	Paraffin wax	11.14	34	0.0154
[208]	Experimental	India	Single basin solar still	1000	Paraffin wax enriched carbon nanotubes (CNT)	12.7	46.45	
[209]	Numerical	India	Single slope solar still with evacuated tubes	600	Paraffin wax, Gallium, Paraffin RT44HC	10.03	35.78	
[210]	Experiment	Egypt	V-corrugated absorber single-basin solar still	913	Paraffin wax	8.2	37.1	0.08369
[211]	Experimental	China	Pyramid solar still	800	Shape-stabilized paraffin wax with 5 wt% expanded graphite	3.41	43.3	0.0298
[212]	Experimental	Egypt	Single slope solar still with pin fin heat sink	932	paraffin wax	8.4		

customers discounts based on time-of-use pricing. During off-peak hours, the time of use market system provides lower power costs, while during peak hours, energy is more costly [217].

Various researches have been done research using TES materials, to increase building thermal comfort, decrease peak load, and minimize electricity demand [218]. Thermal energy storage materials can reduce the peak energy demand by storing extra energy and providing excess energy at peak hours. Because of the high heat storage capability and feasible melting and freezing temperatures, PCM is an efficient TES medium in this regard [219]. Studies have shown that enclosing PCM in homes [41], such as HVAC modules [220,221], solar water heaters [222,223], and other building systems like envelopes [218,224,225] improves energy storage and utilization significantly. PCM undergoes phase transition due to daytime temperature variations in PCM-integrated building envelopes, reducing the drastic impact of ambient temperature variations of indoor temperatures, lowering energy demand for space conditioning, and improving occupant thermal comfort [226].

Various innovations such as inclusion of NPs [75,227–229] extended surfaces [230], metal foams[231–233], expanded graphite in PCM [234–238], PCM encapsulation [16,239–242], and eutectic PCMs techniques [243–247] have been used in research to improve PCM efficiency and thermal characteristics for building thermal comfort.

Sarri et al. [248] numerically investigated the impact of macro-encapsulated PCM in aluminum rigid slabs using shaded or unshaded lightweight buildings in various Algerian cities. Simulations were carried out using EnergyPlus and GenOpt tool to determine to optimum melting temperature of PCM enclosed in building envelope. Results concluded that the use PCM improved the overall thermal comfort and optimized the annual energy consumption during summer and winter season. The use of shading devices along with HVAC system was favorable during summer season with optimal melting temperature of 28 °C of PCM, while for winter season shaded device was not recommended and ideal melting temperature of PCM was between 19 °C and 21 °C.

Kishore et al. [249] explored the thermal load regulation by embedding PCM in building envelope (walls shown in Fig. 12). Various parameters of PCM were studied including location in the wall, melting

temperature and layer thickness etc. The results showed that the inclusion of PCM in the wall reduced heat gain during peak hours by 70%. Maximum thermal load regulation occurred when PCM was adjacent to interior of building and transition temperature was 24 °C to 25 °C with the transition temperature being the most important determining factor.

Li et al. [250] explored the thermal characteristics of hybrid PCM wallboards and compared with the gypsum wallboard for the climate of Shanghai. Three types of PCMs were microencapsulated in the wallboards in different configurations. Results showed that the PCM inclusion in wallboard reduced power consumption by 30% while maintaining optimal thermal comfort. Boussaba et al. [251] performed thermal analysis on non-cocoa vegetable fat bio PCM latent heat TES material for building thermal management. The bio PCM was inserted in 56 wt% in the board and recycled graphite introduced in board to improve thermal conductivity. Results concluded that the novel bio PCM could be potential candidate in buildings application for hot and cold climates due to suitable performance. Li et al. [252] prepared composite sodium sulfate decahydrate (SSD) PCM for building floor heating application. Expanded graphite (EG) was inserted in composite PCM in various proportions by vacuum impregnation to enhance the thermal conductivity stability. Results revealed that wit 7% expanded graphite in composite PCM, heating and cooling time reduced by 36.2% and 44.5% compared to pure sodium sulfate decahydrate as shown in Fig. 13 and thermal conductivity enhanced by 3.6 times compared to pure sodium sulfate decahydrate.

Wi et al. [253] prepared shape-stabilized PCM by inserting expanded perlite and expanded vermiculite into n-octadecane by vacuum impregnation method and inserted this composite PCM into red clay in various concentrations to perform thermal analysis. The peak temperature dropped by 1.6 °C during phase change of PCM, and the time lag in phase transition was around 1.33 h. Hekimoglu et al. [254] prepared novel shape stable composite PCM using silica fume and capric acid/stearic acid eutectic mixture to investigate the thermal storing and releasing properties in building applications. The composite PCM was introduced in ordinary cement mortar in weight fraction of 10%, 15% and 20%. Results revealed that latent heat-based cement mortar had temperature difference of 2.48 °C and 1.71 °C for heating and cooling stage respectively compared to ordinary cement mortar and presented

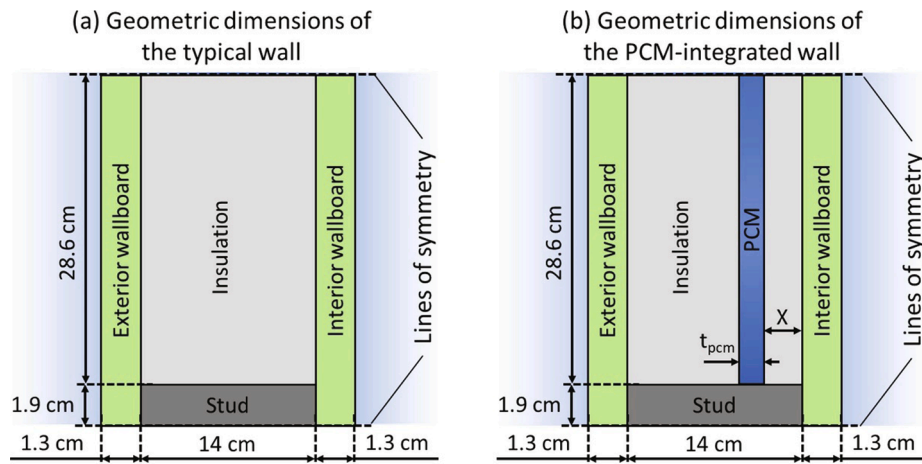


Fig. 12. Two dimensional geometry of considered building walls (a) Common building wall (b) PCM integrated wall [249].

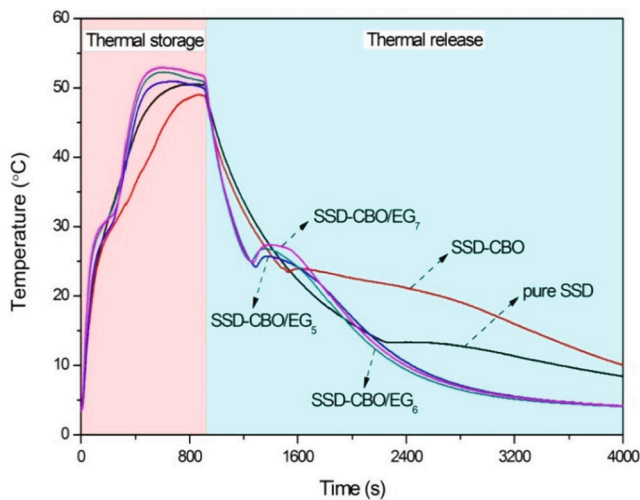


Fig. 13. Charging and discharging curve of various configuration of PCM [252].

suitable properties for indoor thermal regulation.

Yu et al. [255] prepared two novel shape-stabilized composite PCMs by physical blending of organic alcohols 1-tetradecanol and 1-hexadecanol into diatomite. PCM based wallboards were constructed and attached to outside surface of test room to compare its thermal

performance against polystyrene plastic wallboard. Results concluded that polystyrene wallboard was efficient during daytime energy savings and composite PCM based wallboard was effective for nighttime energy savings. Kong et al. [256] coupled passive composite PCM wallboard and active solar thermal energy storage system to form a hybrid heat supply system for winter. Shape-stabilized composite PCM of paraffin and expanded perlite was prepared by vacuum absorption method. Result concluded that hybrid system shortened daily energy usage by 44.16% compared to without PCM board and potentially improved indoor thermal comfort.

Alzoubi et al. [257] prepared nano composite by encapsulating cellulose nano fiber within n-octadecane PCM nanocapsules (FESEM photograph is shown in Fig. 14). The composite nano PCM was incorporated in the cement mortar to analyze compressive strength and thermal conductivity. Results indicated that addition of 0.5–1.5 wt% nanocomposites enhanced compressive strength and reduced thermal conductivity of cement mortar.

Rakkappan et al. [258] inspected the thermal behavior of composite PCM 1-decanol/expanded graphite to assess its applicability in air conditioning of cold storage system. Results showed that composite PCM had an incorporation efficiency of 84.99%, good thermal reliability, and thermal conductivity improvement of 16.33 times compared to pristine PCM. The charging and discharging time of composite PCM was 81.85% and 83.27% respectively, lower than pure PCM. Bondareva et al. [259] investigated the paraffin filled brick under unsteady thermal loads using numerical simulations. It was discovered that increasing the amount of PCM improved heat transfer, and the increase in considered block size

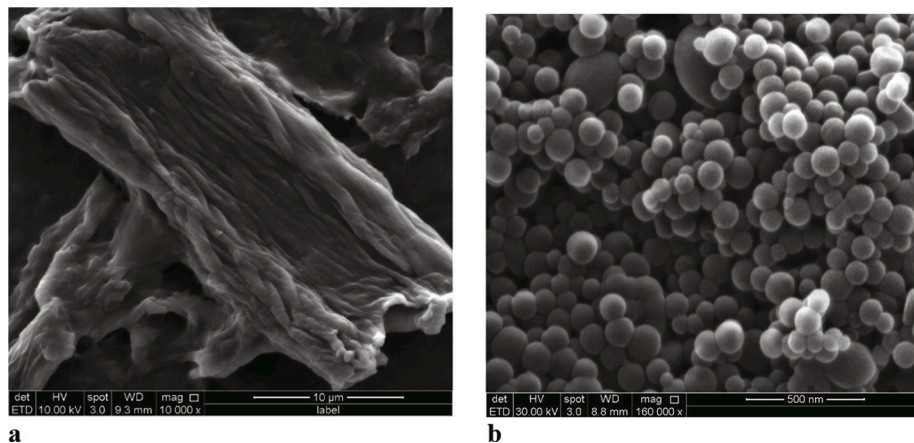


Fig. 14. FESEM images of (a) cellulose nano fibers (b) PCM nano capsules [257].

reduced the indoor temperature by 20% of the maximum temperature difference.

Yasiri and Szabób [260] examined the position of macro-encapsulated PCM inserted in composite roof in Iraqi climate conditions. According to findings, the best PCM position was close to outside of the roof. The maximum room temperature reduction of 9 °C, time delay of 140–180 min, heat flux reduction of 47.9–64.6% was noticed and PCM of 44 °C melting temperature was considered efficient for hot climates. Juarez et al. [261] analyzed the impact of reflecting grey and white coatings with a PCM layer in concrete roof in Mexico City during a summer week. Results showed that for grey roof with PCM, the maximum reduction in temperature of interior and cooling load was 6.4 °C and 22.2% respectively while white coating along with PCM reduced interior temperature by 15.3 °C and cooling load by 62.7%. Hu et al. [262] explored the ventilation window with PCM for preheating and precooling, as well as building energy savings. Experimental and numerical results suggested that energy savings of 62.3% and 9.4% were obtained during summer and winter climates respectively, compared to conventional control system.

Rashed et al. [263] loaded 20 mm PCM layer to building roof and wall to investigate the decrease of heat transfer. Three PCM employed in the study were RT-31, RT-35 and RT-42 to study the effect of phase transition temperature. Results showed that utilization of RT-31 was better choice as it reduced the heat gain by 15.25%, lowered annual heat transfer by 25.71% and achieved maximum carbon saving of 198.65 kg/m² per year. Tafakkori and Fattahi [264] numerically examined the double glazed window with different configurations along with paraffin PCM. It was found that suggested window configuration reduced the energy consumption by 20%, while PCM inclusion improved the thermal performance thereby reducing Nusselt number by a factor of 4 which was calculated by following Eq. (1).

$$Nu = \frac{hL}{k} \quad (1)$$

where h is the heat transfer coefficient, L is the characteristics length and k is the thermal conductivity. Lin et al. [265] optimized the PV panel, Trombe wall, and PCM based building life cycle and indoor thermal comfort in four steps. Results showed that effective solutions lessened the life cycle costs upto 45.51% and boosted thermal comfort up to 43.81%. The recommended heating and cooling temperature setpoints were 18 °C and 24 °C, respectively. Hamdeh et al. [266] utilized seven PCMs of phase transition range of 3–20 °C in building wall for heat loss reduction and energy saving purpose in the months of December, January and February. With PCM of 3 °C melting point, maximum heat transfer rate reduction and maximum power savings of 11.25% and 11.73% respectively were achieved in month of December.

Tyagi et al. [267] reviewed TES PCM-based building heating and cooling systems for different building applications. The author discussed about the various types of PCM, advantages and constraints in the use of PCM in building applications, advancement and cooling/heating applications, significance of the TES in emission limitation, economic analyses and future scope. Jiang et al. [268] studied the energy consumption of various window designs with PCM using EnergyPlus tool in cold climate of Northeast China. According to findings, the triple pane glass and PCM frame showed best results and energy savings of 20% were obtained.

Wen et al. [269] studied the thermal behavior stearic acid/carbonized maze straw composite PCM for its application in TES in buildings. As compared to pure PCM, the thermal conductivity of composite PCM was increased by 87.5%, and the melting and freezing times were decreased by 50% and 73%, respectively. The author concluded that form-stable composite PCM had strong viability for solar energy storage and energy efficient building applications. The summary of recent studies of utilizing advanced PCMs in buildings is summarized in Table 4.

PCMs applications in glazing

In buildings, conventional heating, cooling and glazing units are the reason of excessive energy use and higher costs. The unwanted heat loss and heat gain cause excessive energy consumption for cold and hot climates respectively. The reduction in energy demand for indoor thermal comfort is directly linked to implementation of the reasonable building thermal management systems. Recently the inclusion of PCM in building components like roof, walls, windows, floors and glazing units is being investigated with intent to determine the optimal PCM and building component characteristics. According to studies, the filling of PCMs in building envelopes has shown potential to improve air conditioning, reduce the HVAC costs and greenhouse gas emissions [287].

Sevault et al. [288] explored the active PCM-based building technologies that have surfaced in the last decade. Author explored the various PCM integrated systems in buildings, such as solar thermal heating, space heating and cooling, and household water heating application. The different types of PCMs utilized in the research were also addressed. Furthermore, a step-by-step design method was used to demonstrate the implementation of an active PCM-based latent heating system. The challenges and prospects were also addressed.

Bolteya et al. [289] investigated the thermal efficiency of RT 28HC PCM filled double glazing unit in dry arid region of Egypt. Results showed that for double glazing unit filled with PCM, the internal surface temperature was 7.6 °C less and internal layer's peak temperature time lag was 5 h more than the conventional air filled double glazing unit. The optimal thickness of PCM layer was noticed as 30 mm, further increase in PCM thickness showed poor latent heat utilization.

Wieprkiewicz and Heim [290] studied several optical and thermal characteristics of triple glazed PCM window. Author observed the optical properties of glazing window for solid and liquid phases of PCM. For PCM in solid state, additional 90 W of energy was required for lightening purposes; while limited solar energy transmission through solid PCM lowered the cooling energy demand. The conventional window and window with liquid PCM gave clear sky view and visual comfort while PCM in solid state negatively affected these conditions. Similarly glare effect was lessened due to low transmittance of light for solid PCM case.

Arranz et al. [291] designed and explored the performance of modular active glazing and PCM heat recovery units based window. Results showed that heating consumption in winter lowered by 48% and solar radiation usage reduced the monthly air conditioning consumption by 25% during cold months.

Mao and Yang [277] evaluated the interior thermal comfort and energy utilization of a double-glazed roof filled with PCM. The optimal PCM transition temperature was 26–30 °C under Wuhan's cold and hot climate. It was revealed that increasing the thickness of the PCM and glazing layer improved the roof's thermal insulation performance. Furthermore, smaller value of roof angle lowered the quantity of solar energy collected by the roof and higher latent heat energy of PCM absorbed more solar radiation, thereby increasing the roof's thermal performance.

Li et al. [292] explored the experimental and numerical studies for optical and thermal performance improvement of glazing units containing PCMs in buildings. Author discussed the thermophysical properties of PCMs used in glazing units, different glazing arrangements, and optical and thermal performance modeling of glazing units. Furthermore, challenges and future works in glazing units incorporating PCM were highlighted.

Ahmed et al. [293] performed numerical and experimental thermal analysis of macro-encapsulated PCM for building heating application. Author used magnesium chloride hexahydrate PCM as TES and encapsulated it into aluminium cylinder. Experimentation results showed that the energy and exergy stored in the TES during charging were 39.6 kJ and 6 kJ respectively. Furthermore, the time of heat source accessibility enhanced by 46% compared to heater without TES system.

Table 4

Summary of previous studies of advanced PCMs in buildings.

Reference	Year of Study	Location	PCM Material	Optimal PCM melting point (C)	Type of cooling/ Heating	PCM incorporation approach	Study Type	Observations
[270]	2020	Morocco	n-Nonadecane	32	Passive	PCM impregnation into brick wall	Numerical	High latent heat and melting temperature of 32 C were found to be optimal for better thermal performance of brick wall.
[271]	2021	India	S25, RT 26, S27, FS 28, HS 29, S30, E32		Passive	Roof integrated with PCM	Numerical	FS 28 and HS 29 PCMs using measure of Key response Index gave more than 50% reduction in heat gain.
[272]	2020	America	Encapsulated PCM	22 and 24	Passive	PCM inserted in building wall	Numerical	The insertion of PCM resulted in maximum heat gain of 47.2% and heat loss of 8.3%.
[273]	2020	New Zealand	Macroencapsulated PCM–RT25HC		Active	PCM energy storage unit Inside building	Experiment	Accumulated energy saving of 40% was observed in summer and 30% in winter for one month time.
[274]	2020	China	Shape-stabilized paraffin with polyethylene		Passive	Outer layer of Roof	Numerical	Optimum PCM layer found to be 30 mm and for hot summer, cold winter region optimal PCM melting temperature observed to be 36–38 C.
[275]	2020	Baltic States	Paraffin RT22HC		Active	PCM storage units incorporated under the ceiling slab	Numerical	Results indicated that PCM thermal energy storage system reduced indoor air temperature by an average of 9.5 C.
[276]	2020	China	Composites PCMs of Na ₂ HPO ₄ ·12H ₂ O and CaCl ₂ ·6H ₂ O with expanded graphite		Active	Double PCMlayer radiant floor systems	Experimental	In winter and summer climates, energy savings of up to 58.6% and 64% were observed, respectively.
[277]	2020	China	Paraffin wax	26–30 °C	Passive	PCM layer inserted in glazed roof	Numerical	The increase in PCM layer from 6 mm to 12 mm improved the thermal performance by 23%.
[264]	2021	Iran	Paraffin wax		Passive	PCM inserted in various double glazed windows	Numerical	Building energy consumption reduced by 20%.
[52]	2021	Egypt	RT10 HC, SP 24E		Active	PCM coupled with air conditioning unit	Numerical	Maximum coefficient of performance achieved in summer and winter was 88% and 22% respectively.
[278]	2020	China	Microencapsulated Na ₂ SO ₄ ·10H ₂ O		Passive	PCM layer coupled with wallboard	Experiment	Maximum indoor temperature reduced by 3.1 C.
[279]	2020	Turkey	Various paraffin wax PCMs		Passive	PCM layer inserted in exterior wall	Numerical	Annual heating energy requirements reduced by 6% with the addition of RT21HC.
[280]	2021	Iraq	Paraffin wax		Passive	PCM macroencapsulated in steel panels into roof	Experimental	Maximum room temperature

(continued on next page)

Table 4 (continued)

Reference	Year of Study	Location	PCM Material	Optimal PCM melting point (C)	Type of cooling/ Heating	PCM incorporation approach	Study Type	Observations
[281]	2020	China	Paraffin wax		Passive	Porous brick filled with PCM	Numerical	reduction of 9 C was observed. Heat flux during air-conditioning operation reduced by 73.4%.
[282]	2020	Mexico	Paraffin wax MG29, N-Eicosane and Salt Hydrates		Passive	PCM on interior of roof	Numerical	Thermal load reduction of 57% was observed for paraffin PCM based roof.
[283]	2020	China	Inorganic ternary mixture (CaCl ₂ ·6H ₂ O–NH ₄ Cl–SrCl ₂ ·6H ₂ O) and expanded graphite		Active	PCM integrated in ventilation system	Numerical	Air flow rate of 11.47 kg/h, PCM layer thickness of 12 mm resulted in very small temperature fluctuations of 22.5–27 °C.
[284]	2020	China	Micronal DS 5038 X- Composite PCM		Passive	Concrete panel prepared using artificial aggregates (concrete waste powder) and PCM	Experimental	The artificial aggregate and PCM based panel reduced peak temperature by 1C.
[285]	2021	China	Composite PCM of sodium sulfate decahydrate/expanded vermiculite	25.5 °C	Passive	Composite PCM inserted in multilayer structure	Experimental	Prolonged holding time of indoor temperature was noticed.
[286]	2021	Mediterranean	MEPCM Micronal DS 5001X, Composite PCM of paraffin and calcium carbonate		Passive	PCMs in 5 wt% were inserted in rigid polyurethane foam.	Numerical and experimental	Thermal amplitude reduction of 1C to 3.5C was obtained, a delay of 2–3 h in peak temperature was observed.

Kozelj et al. [58] analyzed the thermal storage tank with encapsulated PCM for building heating purposes. Paraffin was used as PCM and represented 15% of total volume of water thermal storage tank. Results showed that use of encapsulated PCM in water tank, the heat storage was enhanced by 70% compared to conventional case of water based thermal storage system. Further, author did numerical simulation using TRANSYS and validated the experimental results.

PCMs incorporation into building as envelope for thermal management and energy savings is influenced by factors like PCM layer thickness, PCM melting temperature, PCM position, PCM latent heat, ambient temperature, seasonal variations, and active or passive cooling types. PCM inclusion methods into buildings includes direct insertion, immersion, shape-stabilized PCMs, and encapsulation. The modeling and simulation tools used in the studies of thermal and energy optimization of PCM loaded building components includes ANSYS Fluent, COMSOL Multiphysics, EnergyPlus software, TRANSYS, Matlab software, and DIANA-finite element analysis [294].

Jia et al. [295] investigated the use of PCM in prefabricated building for improving the indoor thermal comfort in five different climates using EnergyPlus tool. Results revealed that application of PCM inside the building wall or roof, in east or west side of building enhanced the energy saving rate and interior thermal comfort. Out of five PCMs, cetene gave presented best energy saving rate of 86.4%. The optimal PCM layer thickness was different for different climatic conditions.

Zhu et al. [57] investigated the trombe wall packed with PCM for indoor thermal comfort in Wuhan climate conditions. The author utilized the TRANSYS heat transfer model and GenOpt optimization tool to determine the best value of critical parameters and energy consumption rate. Trombe wall optimization findings indicated that the ideal air gap thickness was 0.05 m, optimal wall thickness was 0.68 m, optimal vent

area was 0.6 m², and optimal melting temperatures of lower and higher temperature PCM layers were 16.5 °C and 27.75 °C respectively. The annual building load was lowered by 13.52% in optimized PCM packed trombe wall as compared to conventional trombe wall.

Qu et al. [59] conducted a multi-factor analysis with PCMs incorporated in building envelopes for attaining thermal comfort and energy savings using EnergyPlus software. The most influential parameter for energy usage and interior thermal comfort was found to be PCM envelope type, followed by PCM layer layout, PCM type, and PCM layer thickness. According to author, the appropriate selection of PCMs based on local climatic conditions could result in energy savings ranging from 4.8% to 34.8%.

Guo and Zhang [60] examined the performance of PCM-packed building envelope for seasonal fluctuations. The PCM placed in the centre of the envelope was evaluated using enthalpy porous model. The results revealed that the application of PCM layer reduced the energy consumption rate, but exhibited varied performances for seasonal fluctuations with negative performance in the cold season. The melting point and volume fraction of PCM showed significant influence on the optimal PCM selection for different climate conditions.

Yang et al. [61] prepared flexible shape-stabilized PCMs by using cross linked polymer swelling strategy for thermal management of building and electronic equipment. According to the findings, the prepared composite PCM demonstrated reliable mechanical properties, and its thermal storage and passive radiative cooling capacity could lower electronic device and building temperatures by 5.5 °C and 9.3 °C respectively.

Dobri et al. [296] used a semi-analytical model to study the transient heating of MEPCM filled in gypsum plaster walls. The author carried out simulations for a constant heat flux and compared the findings to

existing model results. The abruptness of the PCM phase transition was influenced by particle sizes of MEPCM and interfacial thermal resistance, but increase in volume fraction of PCM seemed to postpone the phase transition. Energy savings of 15–20% were achieved during the summer months with MEPCM loading as low as 5%.

Zhu et al. [297] developed composite coatings incorporating MEPCM for thermal comfort and energy savings in buildings. The interface polymerization technique was used to synthesize MEPCM with n-octadecane as the core and polyurea as the shell. The SEM analysis indicated that MEPCM kept its original morphology and distributed evenly in coating with an MEPCM loading of 30%. MPCM and heat-storage coatings exhibited latent heat capacities of 151.89 J/g and 45.5 J/g, respectively. At constant heat load, the chamber with MPCM coatings was able to sustain a steady temperature of 26–28 °C for about 77 min, compared to 13 min for the chamber without PCM coating.

Kabdrakhmanova et al. [298] investigated the influence of climate related variables on the use of PCMs in buildings for humid subtropical climates. Author used panel data regression analysis to explore the influence of climate related factors on the energy demand of PCM-integrated buildings, and to develop equations for estimating energy demand. The results revealed that the PCMs with melting temperatures of 24 °C and 27 °C performed best, achieving yearly energy savings of up to 12,635 kWh and a 19.9% reduction in energy usage. The suggested equations were able to predict the energy consumption for buildings loaded with PCM under humid subtropical climatic conditions.

Staszczuk and Kuczynski [299] explored the thermal performance of PCM integrated wall and roof surfaces and compared it to traditional reinforced concrete. The results indicated that filling PCM with a transition temperature of 21.7 °C was unable to lower the room temperature below 23 °C. According to author, it was not possible to predict the ideal melting temperature range of PCM in a building without air conditioning during the summer. Furthermore, during severe heat wave period, by increasing the thermal mass using reinforced concrete lowered the peak indoor temperature by 2.5 °C.

Choi et al. [300] investigated the thermal performance of PCM integrated walls by introducing air bubbles into the walls. Visualizations were developed by shadowgraph and particle image velocimetry to examine the patterns of phase change and evaluate the interior flow pattern. The injection of bubbles produced a density difference between the PCM and the bubbles, causing bubbles to rise in the bulk PCM. According to the findings, the usage of the bubble injection approach resulted in 11% higher latent heat storage and 28% less heat penetration through the PCM wall. Bubble injection increased the flow rate of the liquid PCM by 19 times compared to a PCM wall without bubble injection.

Piselli et al. [301] investigated the PCM integrated envelope-based passive cooling system in conjunction with natural ventilation mitigating cooling loads multiple Italian climate conditions. To determine the optimal PCM transition temperature for various regions, simulations and optimization analyses were carried out using the EnergyPlus and GenOpt tools. In moderate climatic conditions, the incorporation of PCM in the building envelope reduced the cooling load by more than 65%. Furthermore, natural ventilation control and night time ventilation improved the effectiveness of the PCM TES charging discharging cycle. Li et al. [302] investigated the thermal performance of glass window combined with silica aerogels and PCM for cold climatic conditions of China. Physical model (Fig. 15 shown below) describing heat transfer process of PCMs along with transparent wall was established and proved. Results established that the thermal conductivity and thickness of silica aerogel had notable influence on thermal performance of glazing unit. Silica aerogel thickness of 20–30 mm deemed appropriate for PCM latent heat utilization.

Liu et al. [303] explored the thermal and optical behavior of non-ventilated multilayer glazing facade filled with PCM for severe cold climatic conditions. Author validated the numerical model with the experimental data successfully. Results indicated that PCM thickness

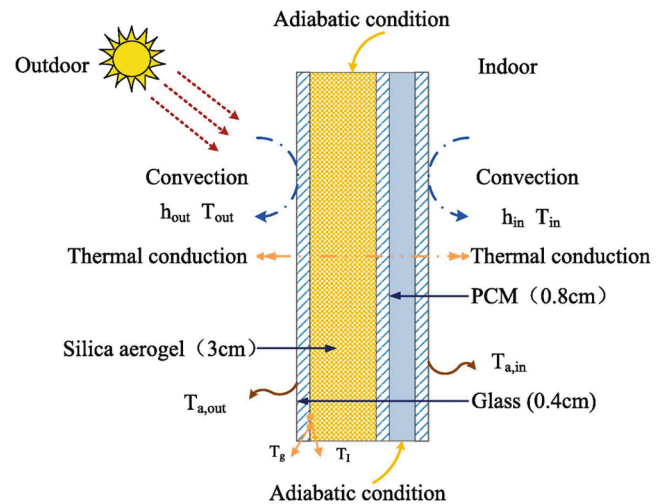


Fig. 15. Physical model of glass window incorporating PCM/silica aerogel [302].

play a crucial role in thermal and optical performance of multi-layer glazing facade and increase in PCM thickness reduced heat loss from glazing facade, increased time lag and interior surface temperature as depicted in Fig. 16. The thickness of the PCM layer was recommended not to exceed 20 mm as solar transmission was significantly reduced.

Yang et al. [304] studied the photothermic characteristics of ZnO and CuO augmented paraffin in order to thermally control and save energy in glazed structures. Findings indicated that the inclusion of NPs in PCM resulted in the scattering effect of light and reduced transmittance, though the transmittance improved by increasing the temperature and decreasing the concentration of NPs. CuO/paraffin composite resulted in better performance in terms of solar absorption. Optimum photothermal performance was realized with the NPs volume fraction from 5×10^{-4} to 1.5×10^{-3} .

Zhang et al. [305] numerically studied the thermal performance of glazing window unit by incorporating TiO₂, ZnO, Al₂O₃ NPs enhanced PCM in different seasonal conditions. The addition of NPs improved PCM melting/solidification, extended time of liquid state of PCM, enhanced interior glass surface temperature and raised the peak temperature difference. Al₂O₃ addition in PCM enhanced glass inner surface temperature, however, with the introduction of ZnO and TiO₂ in PCM, the inside surface temperature of the glazing window reduced in summer and increased in other seasons, thereby enhancing interior thermal

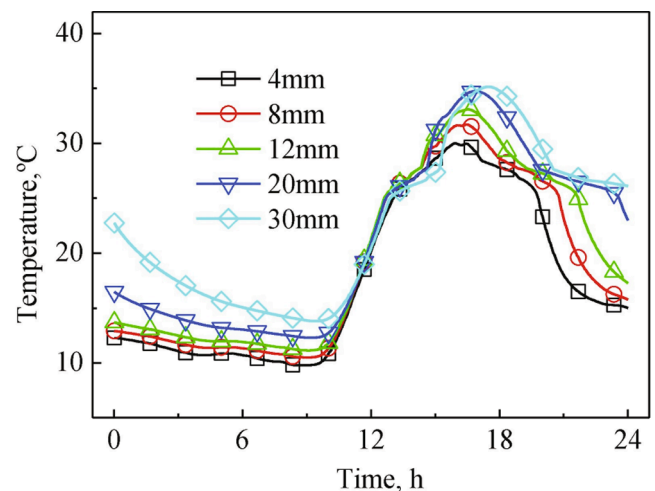


Fig. 16. Temperature profile of indoor surface of double glazing facade loaded with PCM [303].

comfort. The NPs remained in solid phase in winter, still the indoor temperature increased. Zhang et al. [306] examined the thermal behavior of the glazing unit loaded with silica aerogel and PCM for severe cold weather of China. The PCM integration in the glass window deteriorated the thermal performance of window in winter; however, the addition of the silica aerogel with PCM of appropriate melting point enhanced thermal comfort. With appropriate optical glass characteristics, the radiation beyond $2.5 \mu\text{m}$ substantially enhanced the energy efficiency of the glass window. Energy saving rate of 56.67% was achieved with optimal three-glazing window, for optical glass absorption coefficient and refractive index of 3.4 and 10 respectively, silica aerogel of 12 mm thickness and PCM melting temperature of 283 K. Li et al. [307] studied the consumption of heat load by comparing the sunspace with and without PCM at various inside temperatures of typical rural residence in the cold climate of China. According to findings, annual energy savings of 22.73% was accomplished at interior design temperature of 18°C . Additionally, PCM louvers integration with sunspace further saved heating energy by 5.27% and 13.88% at indoor temperature of 18°C and 20°C respectively throughout heating period. But due to poor exploitation of latent heat, PCM louvers exhibited enhanced heating power consumption at 14°C and 16°C indoor design temperatures. For the internal design temperature of 18°C or more, PCM louvers were proposed for better thermal comfort in sunspace.

Latent heat TES in heat exchanger

The latent heat of PCMs is a property defined by the phase change of the PCM from solid to liquid and vice versa during which heat energy can be stored and released for useful purposes. Since PCMs can store more latent heat than sensible heat, there has been a lot of research in this field for a variety of applications and purposes [182,184]. Flow and heat transfer properties of PCMs have been researched by many authors for heat exchangers [308] applications.

Hosseini et al. [309] studied a dimensionless model to predict the effective thermal conductivity of nanofluids. The model defined nanofluid thermal conductivity as a function of thermal conductivity of NPs and base fluid, NPs size, volume fractions, interfacial shell characteristics and nanofluid temperature. As a result of the modeling, a non-linear correlation was established for thermal conductivity, and the present model and prior experimental results were found to be in good agreement. Sarafraz et al. [310] dispersed graphene nanoplatelets (GNPs) in water/ethylene glycol base fluid and studied its thermal and flow behaviour as coolant in compact heat exchanger. It was shown that 0.3 wt% GNPs at 60°C , augmented the thermal conductivity of base fluid by 32.1%. Furthermore, the thermal performance of system enhanced by 21% while the pressure drop increased by 12.1%.

Sarafraz et al. [310] dispersed GNPs in water/ethylene glycol base fluid and studied its thermal and flow behavior as coolant in compact heat exchanger. It was shown that 0.3 wt% GNPs at 60°C , augmented the thermal conductivity of base fluid by 32.1%. Furthermore, the thermal performance of system enhanced by 21% while the pressure drop increased by 12.1%.

Tian et al. [311] explored the nanofluid flow and thermal attributes through spiral double pipe heat exchanger by using the k-epsilon RNG model. Different operating parameters like nanofluid inlet temperature and velocity, NPs type and volume fraction were observed for thermal performance evaluation. With the increase of Reynolds number from 16,000 and increment of volume fraction of NPs, the variation in pressure drop was observed for different types of NPs. Results showed that $\text{Al}_2\text{O}_3/\text{water}$ and $\text{SiO}_2/\text{water}$ nanofluids gave best performance for Reynolds number values of 10,551–17,220 and 17,220–31,910, respectively which was calculated by following Eq. (2).

$$Re = \frac{\rho \nu L}{\mu} \quad (2)$$

where ρ is the density of fluid, ν is the velocity, L is the characteristics length and μ is the dynamic viscosity of fluid.

Sardari et al. [312] theoretically examined the composite RT54HC PCM with metal foam mounted on the backside of radiator to store excess heat and release it during peak hours to turn off boiler or radiator. Results demonstrated the composite system provided constant temperature of 54°C for 11-hour period during discharging and charging occurred in less than 15 min, time saving in charging and discharging for composite PCM case was 98.8% and 48.01% respectively. The suitable porosity of metal foam was observed to be 97%.

Zhang et al. [313] investigated the discharging performance of lauric acid PCM in shell and tube latent heat storage system using novel tree shaped fins. Results indicated that the solidification and melting time was lessened by 66.2% and 4.4% respectively compared to radial fin system. Maher et al. [314] prepared composite PCM by using paraffin wax and silicon carbide and silver nanomaterials for TES applications. According to findings, 15 wt% of silicon carbide and silver NPs into PCM resulted in thermal conductivity enhancement of 58.2 % and 31.2% respectively. At lower concentrations of NPs, a decrease in latent heat of fusion, melting point, and specific heat power was detected, which became more evident at higher concentrations.

Zhang et al. [315] investigated the use of a cascaded PCM storage device with multiple PCMs and uneven fins to reduce charging time and increase TES rate as shown in Fig. 17. The vertical system consisted of a shell and a tube, with PCM (RT45, RT50) in the shell and water in the tube. The findings revealed that using multiple PCMs improved TES and reduced charging time by 9% and 29%, respectively, while using fins enhanced thermal energy storage and reduced charging time by 7% and 16%.

Asip Khan and Mahabat Khan [316] numerically and experimentally analyzed different fin orientations in shell and tube heat exchanger with stearic acid PCM filled shell and water as heat transfer fluid (HTF) in tube. Results concluded that the y shaped fin enhanced the melting rate by 50.7% and increase in 7.6% temperature in HTF improved the melting rate by 80%. Mehta et al. [317] studied vertical shell and finned tube system using spiral fins for assessment of charging and discharging characteristics. PCM was stearic acid and working fluid was water for

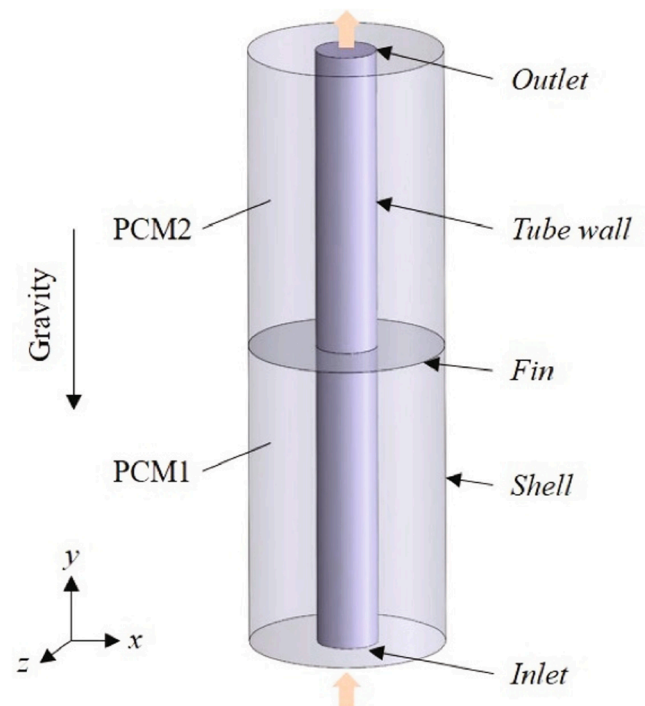


Fig. 17. Physical model of composite latent heat TES system [315].

experimental and numerical study. Results revealed that assistance of spiral fins enhanced the charging and discharging rate by 41.48% and 22.16% respectively and average cycle time for energy storing and retrieve reduced by 31.82%. Nie et al. [318] numerically studied concentric latent heat storage system using lauric acid PCM, to investigate the various fins parameters i.e. fin number, fin length, outer shell material and four types of fins arrangements. Results suggested that the increase in fins number and length gave reduction in charging-discharging rate by 67.6% and 50.6% respectively for angled fins as depicted in Fig. 18.

To research the melting activity of RT58 PCM, Yu et al. [319] incorporated gradient fins into the horizontal shell and tube system. The optimal fin thickness and central angle were found to be 1.846 mm and 14.96° respectively, and the charging rate of the LHS unit was reduced by 30.5%. Zhang et al. [320] investigated vertical and horizontal shell and tube latent heat system having annular, helical, longitudinal fins with RT35 PCM as storage material and water as HTF. Results suggested the maximum time saving in complete melting of PCM was 31.0% and 10.0% for the vertical and horizontal helical finned LHS systems, respectively.

Hajizadeh et al. [321] studied the RT35 PCM solidification in a tank with and without fins, by dispersing CuO NPs in 5% volume fraction. Results showed that addition of fins and NPs in latent heat storage system reduced by discharging time by 70.06% compared to simple case. Tang et al. [322] investigated numerically horizontal shell and tube latent TES unit with non-uniform fins, Paraffin RT50 as TES material, and water as HTF. Study showed that melting time lowered by 83.9% with fin angle of 25° and fin length of 40 mm and thermal storage density enhanced by 466%.

Shalaby et al. [323] experimentally investigated the flat plate solar heater with shell and finned tube paraffin wax based latent heat system. Water was pumped from the solar heater to the latent heat system to heat PCM, and the accumulated heat in PCM was to be used to heat domestic water via another loop. According to the findings, the solar heater combined with the latent heat system had a daily efficiency of 65% and supplied water at a temperature of $50\text{--}60.4^\circ\text{C}$. Chananipoor et al. [324] used nano-encapsulated PCM and water to prepare nano enhanced slurry for thermal evaluation of double pipe heat exchanger. The nano-encapsulated PCM composed of n-dodecanol core, polymethacrylate shell and graphene-oxide nanosheet as additional protective layer. Results showed that the optimum nanoencapsulated PCM concentration, inlet temperature, and Reynolds number were 14%, 83°C , and 5046, respectively, with an efficiency factor of 4.5. The summary of recent studies on usage of TES materials in heat exchangers is summarized in Table 5.

Advanced PCMs applications in smart textiles

Textiles play a significant role in the heat flow between the body and

the surrounding environment. Textile study on personal thermal control, on the other hand, has received insufficient consideration for a long time. Lately, a substantial research has been done on smart textiles to regulate for human thermal comfort [74,340]. Research is underway to design and optimize textiles for adjusting heat dissipation according to hot or cold climate for human thermal comfort and to curtail the ever-increasing energy consumption. Smart textile research is emerging as a promising solution to provide passive and responsive adaptive thermal comfort according to external stimuli [341].

Four important factors for thermal comfort are thermal radiation, temperature, air humidity and wind speed. HVAC systems are used to maintain these parameters within optimal range but consume a lot of energy and cause environment pollution and greenhouse gas emissions. Therefore, to lessen the building energy consumption for thermal comfort has become an important topic for development of smart textiles [342].

Heat exchange between the human body and surroundings take place by conduction, convection and radiation, the primary heat transfer route for human body is radiation, so the basic design principle of smart textiles is to govern the radiation heat transfer by transmitting as much body radiation as possible in the summer and blocking as much as possible in the winter. In this part, the use of PCM in smart textiles for passive and responsive thermal management is being discussed thoroughly [343].

Wang et al. [344] performed experimental study on twelve young males for 90 min to investigate the hybrid personal cooling system integrating PCM and electric fans. Results revealed that electric fans maintained the temperature at 34°C for more than 15 min, while PCM removal after melting reduced the physical load by 37.3% enhancing the evaporative cooling. It was suggested to turn-off fan during onset of hot exposure and switch on when consumer felt warm. Baykara and Taylan [345] prepared nanofibers using polyvinyl alcohol as core and Nigella seed oil as shell by coaxial electrospinning as depicted in Fig. 19. Results showed that the nanofiber morphology was influenced by parameters like flow rate, voltage difference and solution concentration. With 18–20% polyvinyl alcohol concentrations, homogenous and lesser bead nanofibers were obtained while with polyvinyl alcohol concentration of more than 22% resulted in microdroplets formation rather than nanofiber.

Fonseca et al. [346] studied numerically the PCM based firefighting suit for transitory period of fire exposure, postfire exposure and resting phases. Five different PCMs were chosen to carry out the analysis. The results concluded that the minimum mass of PCM sodium acetate trihydrate was suitable for medium and low intensity and long duration thermal exposures while magnesium nitrate hexahydrate deemed as best candidate for high intensity heat exposure firefighting suits due to lowest cooling time and lower mass requirements. Calleja et al. [347] experimented to check the heat dissipation characteristics of cellulosic fiber with the coatings of polyurethane, microcapsules of graphene and

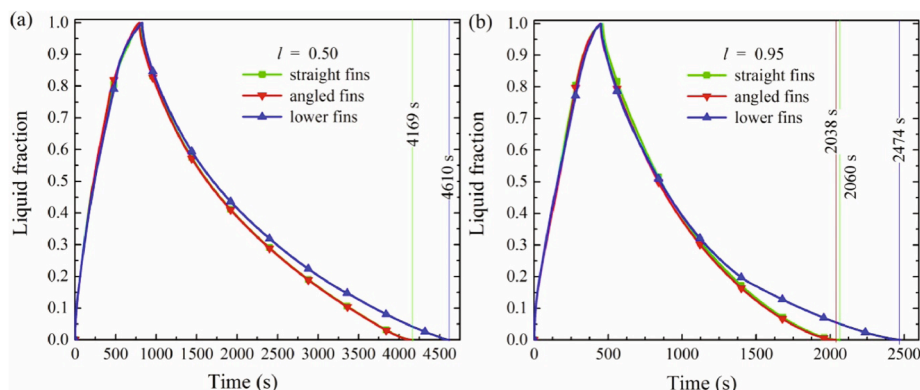


Fig. 18. PCM liquid fraction profiles for various fins arrangements lengths [318].

Table 5

Summary of previous studies on latent heat storage in heat exchangers.

Reference	PCM material	PCM enhancement	Latent heat system Configuration/Structure Type, Material	Nature of Study	Findings
[325]	RT82	Longitudinal fins and strips fins and Al ₂ O ₃ NPs	Triplex-tube heat exchanger	Numerical	A combination 2% strip fins and 2% NPs led to improvement in melting time by 83.75%.
[326]	RT82	Spiral and longitudinal fins	Finned cylindrical annulus	Numerical	Spiral fins saved melting and solidification time by 57.6% and 74.13% respectively.
[327]	Paraffin	Spiral fins	Vertical shell and spiral fin tubes, copper	Experimental	Increase in flow rate from 0.5–2 L/m resulted in reduction of charging time by 47.79% at inlet temperature of 80 °C.
[328]	Paraffin	Graphene nano-platelets, longitudinal, circular, and wire-wound fin types	Vertical shell and tube latent TES unit	Numerical	Wire-wound fins gave 94.68% improvement in charging compared to tube without fins while 1% concentration of GNP was found to be optimal.
[329]	Water	SWCNTs, Y-shaped fins	Triplex-tube storage unit, copper	Numerical	Solidification time reduction of 48.53% and 44.03% was obtained for optimized fin and 4% NPs respectively compared to pure PCM case.
[330]	RT82	Copper metal foam, frustum tube and conical shell	Vertical shell and tube latent TES unit, Stainless steel	Numerical	Maximum enhancement in melting process observed to be 32.5% than baseline case.
[108]	Potassium hydroxide, potassium nitrate, and sodium nitrate	Regular and irregular fin distribution, multi PCM system	Vertical shell and tube latent TES unit	Numerical	Multi PCM system lessened the charging and discharging time by 30% and 9%, respectively.
[331]	Paraffin (n-octadecane)	CuO NPs, y shaped metallic fins	Double pipe system	Numerical	The charging mechanism was accelerated by using long and thinner fins incorporated in inner and outer cylinders.
[332]	RT82	Al ₂ O ₃ NPs and Y-shaped fins	Triplex-tube heat exchanger, copper	Numerical	The usage of NPs and fins led to the reduction in solidification and melting time by 12%, 6.4%, and 5%, 37.8% respectively.
[233]	RT28	CuO NPs and aluminum foam	Wavy shape of duct	Numerical	Introduction of porous foam resulted in discharging time drop by about 19.5%.
[333]	RT-50	Copper fins	Horizontal shell and tube	Numerical	Melting time improved by 50% and 40% by optimizing height and angle respectively of fins.
[334]	Paraffin wax	Different inclination angles of LHS system	Double-pipe system, inner pipe copper, outer pipe galvanized steel	Experimental and numerical	The latent heat system with 45° inclination and PCM layer thickness of 14.5 mm gave maximum melting time reduction.
[335]	Sodium acetate trihydrate	Novel spiral tubes	Spiral tubes heat exchanger, stainless-steel material	Experimental and numerical	The higher flow rate of working fluid enhanced the charging and discharging of PCM.
[336]	Paraffin wax	Longitudinal fins	Horizontal shell and tube system, copper	Numerical	Optimization of fin length and thickness reduced the melting time of PCM by 54.1% compared to base line case.
[337]	RT-25	Rectangular fins	Rectangular shell and tube, copper	Numerical	Thermal conductivity in melting and solidification enhanced by 1.38 and 4.75 times respectively, and heat transfer efficiency of 72.4% was attained.
[338]	Paraffin wax	Straight, punched and slit-fins.	Shell and tube heat exchanger, copper	Numerical	For slit-and punched fin cases, the reduction in the total melting time was almost 14.0% and 10.1% respectively compared to straight fins
[339]	Lauric acid	Spiderweb-like fins, plate fins	Shell and tube heat exchanger, aluminum 6061	Numerical	Solidifying time reduced by 47.9% and heat release rate surged by 1.44 times.

PCM. Results concluded that the PCM heat storage capacity along with high thermal conductivity of graphene displayed a promising behavior for heat dissipation and thermoregulation. Feng et al. [348] experimentally explored the microfibers prepared by coaxial electrospun with polyethylene glycol (PEG) as core material and polyurethane as shell. It was revealed that the polyurethane/PEG porous membrane resulted in high mechanical strength of 125 times mass of load, thermal actuated shape memory effect and exceptional thermal regulation in terms of cooling and drying function with 7.6 °C less temperature than conventional fabric.

Dong et al. [349] prepared microtubule composite fiber by merging kapok and PET fiber, and encapsulated PCM (fatty acid ester of myristic acid/tetradecyl alcohol) into the composite fiber by vacuum impregnation method. Results revealed that kapok/PET with PCM retained 76.8 J/g and emitted 77.4 J/g of PCM latent heat in the temperature

range of 24–42.3 °C and 29.7–10.5 °C respectively. In heating situation, the kapok/PET fiber reached at the temperature of 49.6 °C and kapok/PET with PCM reached 43.4 °C in 420 s as represented in Fig. 20.

Fang et al. [350] investigated a polyethylene glycol (PEG) based composite PCM comprising hydroxylated boron nitride, cellulose nanofiber, and chitosan. Composite PCM showed improved shape and thermal stability and 22.56 times higher thermal conductivity than pure PCM and research concluded its viability in using temperature regulating textiles. Hemmatian et al. [351] using coaxial electrospinning prepared core shell nanofiber structure for thermoregulation of body temperature. Eutectic mixture (lauric acid/palmitic acid) and polyvinyl alcohol was employed as core and nylon 6 as shell. According to results, the melting temperature of the nanofibers' cores was found to be 32.17 °C, which is near actual body temperature, and the overall thermal analysis revealed that PCM-based nanofibers improved thermal

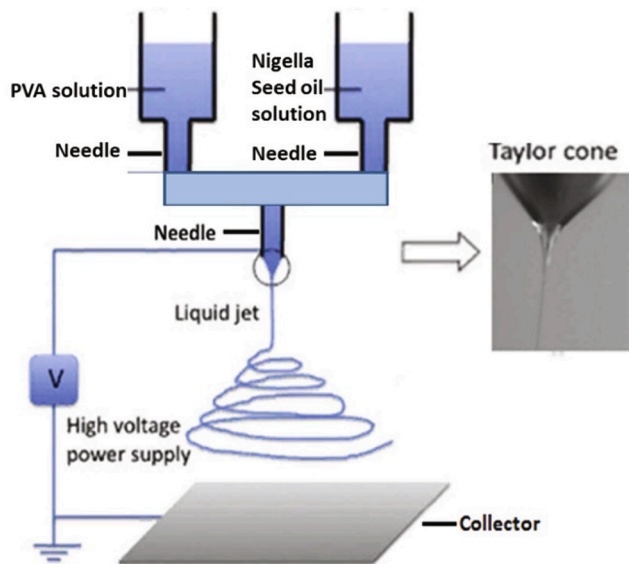


Fig. 19. Schematic of the coaxial electrospinning setup [345].

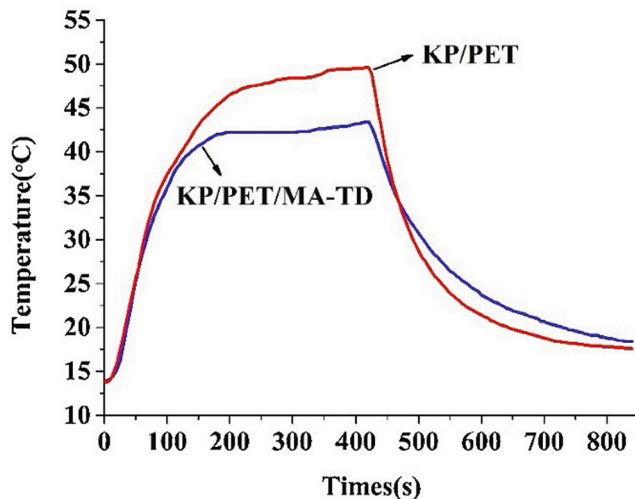


Fig. 20. Thermal regulation curves of kapok/PET and kapok-PET/PCM [349].

management characteristics, especially in the cooling period.

Haghighat et al. [352] prepared the core shell fibrous layer by coaxial electrospinning with hexadecane and octadecane PCMs as core and polyvidone as shell (SEM image shown in Fig. 21). Thermoregulating properties of PCM based fibrous layer were studied and author concluded that PCM with phase transition temperature near ambient temperature was optimum choice to attain high thermal performance.

Su et al. [353] inspected the thermal properties of various paraffin coated fabrics using bidirectional thermal regulation. The author concluded that the optimal PCM for improved thermoregulation was with the higher thermal capacity and low heat rejection attributes. PCM coating was well suited for thermal protective clothing at a transitional temperature of 35 °C and a 45% PCM material. Hou et al. [354] investigated numerically and experimentally the PCM filled and water cooled jacket for thermoregulation in hot environment. Results revealed that the PCM based vest gave effective cooling for 2 h, maintaining the thermal comfort for indoor working conditions. The summary of latest studies on smart textiles using advanced PCMs is presented in Table 6.

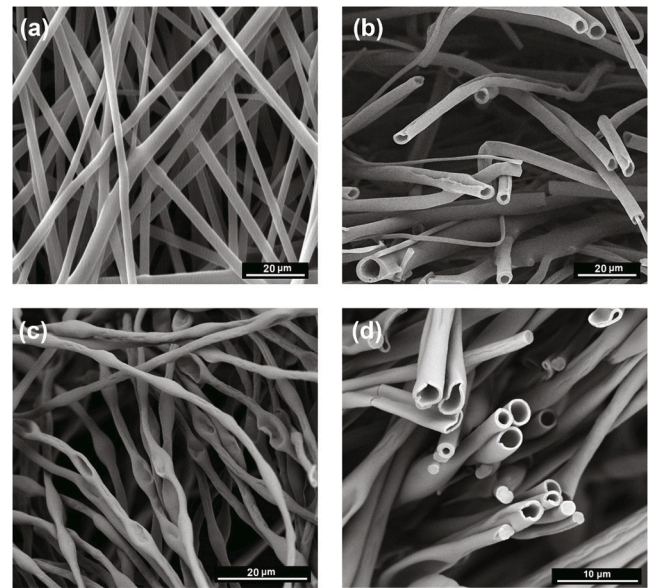


Fig. 21. Cross sectional and longitudinal FESEM images of fibers containing (a, b) octadecane and (c, d) hexadecane [352].

Applications of advanced PCMs in PV/thermal system

Rapid degradation of energy resources and increase in environmental pollution has prompted the world to go for renewable and environmental friendly technologies [374]. Solar energy harvesting for electricity generation by photovoltaic panels is gaining worldwide attention [375]. But the increase in temperature photovoltaic panels due to continuous irradiance has negative effect on the efficiency of PV panel, that's why research is being done for thermal and electrical optimization and performance improvement of PV panels [376,377].

Akshayveer et al. [378] performed numerical study for PV thermal management with encapsulated RT27 PCM enclosed in rectangular and non-rectangular containers. While the temperature of conventional PV module reached 56.78 °C, the introduction of PCM in rectangular enclosures reduced the temperature 17%, and electrical conversion efficiency of PV approached 12%. Al-Waeli et al. [379] explored the thermal and electrical performance of NEPCM and nanofluid based PV thermal unit. System consisted paraffin wax enriched with 0.1 wt% silicon carbide (SiC) NPs and water/SiC based nanofluid was circulated at flow rate of 0.17 kg/s to transfer heat from NEPCM and to augment thermal effectiveness. The schematic diagram is represented in Fig. 22. The findings showed that by using nanofluid and NEPCM decreased PV cell temperature by 30 °C, boosted electrical efficiency upto 13.7%, and increased thermal efficiency to 72%.

Similarly, Al-Waeli et al. [380] performed thermal and economic analysis of PV thermal system integrated with NEPCM storage unit and nanofluid HTF. A tank filled with SiC-PCM was combined at rear side of PV panel and cooled by water-SiC nanofluid circulated at 0.175 kg/s flow rate. The power output, electrical, and thermal efficiencies of the designed PV module were 12.7 W, 13.7%, and 72%, respectively. Electricity cost was calculated as 0.125 USD/kWh with payback periods of 5–6 years. Hassan et al. [381] experimentally studied the PV system PCM as TES material and water-graphene nanofluid as HTF with different flow rates and different NPs concentrations and RT-35HC PCM as heat storage medium as experimental setup is shown in Fig. 23. As compared to a traditional PV panel, the NEPCM/PV system caused 23.9 °C temperature drop and a 23.9% augmentation in electrical efficiency, and the overall efficiency of the PV module increased by 12%.

Islam et al. [382] explored the thermal and electrical performance of NEPCM based PV thermal system by energy and exergy analysis. PV thermal system consisted of RT-44 PCM and multi wall carbon

Table 6
Summary of previous studies on smart textiles using advanced PCMs.

Reference	PCM	PCM enhancement	PCM/fabric Preparation method	Phase change/ melting enthalpy (KJ/kg or J/g)	Melting temperature (°C)	Results
[355]	PEG	Lignin	Microwave assisted blending	100.91	65.6	Prepared PCM showed exceptional thermal stability and reliability.
[356]	Methyl palmitate and isopropyl palmitate	Polyacrylonitrile and poly (methacrylic acid-co-ethyl acrylate) shell	Coaxial electrospinning			Transitory heat discharge efficacy of methyl palmitate nanoweb in 24 and 22 °C temperature range was 51 % more as compared to nanoweb without PCM,
[357]	Octadecane	Acrylic based microcapsule	Suspension polymerization	175	26.4	Microencapsulated PCM gave melting enthalpy around 175 J/g, and was thermally stable up to 170 °C.
[358]	n-Eicosane	Polysiloxane elastomer	Mechanical co-emulsification/ padding method	34.3	37.3	In dry mass of nonwovens, PCM amount was 37 wt%. and energy storage capacity exceeded 98.9%.
[359]	Paraffin	Polyacrylonitrile (PAN) nanofibers	Electrospinning	58.74	26.77–29.55	Composite nanofiber revealed enthalpy of 58.74 J/g and 57.41 J/g during heating and cooling respectively.
[360]	Octadecane	Pectin—fatty acids	Impregnation	100.38	30.26	Treated cotton fabric with PCM composite gave high stability and enthalpy and good thermoregulating effect.
[361]	PEG and Poly (vinyl alcohol)	Glutaraldehyde/ Ethanol	Electrospinning	72.3	24.6	Maximum enhancement of 59.0% and 89.5% in heating and freezing processes were obtained, for phase change fiber with 50% PEG.
[362]	Poly (hexadecyl acrylate) (PA16)	Cellulosic fibers	Radical polymerization			High phase-change enthalpy (73 J/g), exceptional tensile strength (10.5 MPa), and super hydrophobicity were obtained by cellulose base PCM fibers.
[363]	Quinary fatty acid eutectics	SiO ₂ nanofibers	Microemulsion electrospinning	123.80	17.83	Highest melting/freezing enthalpy maintained by hollow SiO ₂ nanofibrous mats was 123.80/ 120.50 J/g.
[364]	Eutectic fatty acids	Iron oxide NPs, polyester (PET) fibers	Single step method	40.3–59.7	36.3–54.1	The composite fiber revealed improved stability, enhanced thermal conductivity, leakage proof, optimal melting/freezing temperatures, and latent heats.
[365]	Paraffin	Polypropylene hollow fiber membrane and expanded graphite	Double encapsulating	110.24	53.72	Composite fiber showed the highest thermal conductivity (0.35 W/m K) and maximum paraffin encapsulation potential of 85.31 wt%.
[366]	PEG based PCMs	Poly acrylonitrile	Coaxial electrospinning	29–60	25–29	Enhancement in specific heat capacity and thermal effusivity was observed for PCM based nanofibers.
[367]	PEG	Polyamide 6	Coaxial electrospinning	121.86	211.1	The increase in concentration of PEG, higher peak temperature latent heat enthalpy was achieved.
[368]	Capric acid, palmitic acid and stearic acid	Polyacrylonitrile/silver NPs	Electrospinning	138.1–156.7	24–30	Due to addition of Ag- NPs, melting/freezing times of form-stable phase change composite fibers were shortened by about 44% and 52% respectively.
[369]	PEG	Carbon nanotubes, polyvidone	Centrifugal spinning	72.45	48.2	Composite phase change fiber presented improved stability, thermal conductivity, heat storage capacity and phase transition.
[370]	PEG and Polyvinyl alcohol	AgNO ₃	Electrospinning	67.5	45.9	Composite fiber with 15 wt% of AgNO ₃ , resulted in heat storage/release time reduction by 23.5 and 26% compared to fiber without AgNO ₃ .
[371]	Paraffin wax	Polyacrylonitrile and hexagonal cesium tungsten bronze	Coaxial electrospinning	60.31	35.57	Smart textile presented good stability, encapsulation efficiency of 54.3% and latent heat capacity of 60.31 J/g.
[372]	Organic and inorganic PCMs	Polyamide6, fumed silica	Melt spinning technique	9.44	45	Phase change fiber with enthalpy of 9.44 J/g maintained cooling and heating for 591 s and 786 s respectively and endured washing at various temperatures for one hour.
[373]	Paraffin wax	Polyacrylonitrile and phosphors as sheath	Coaxial electrospinning	64.08	42.14	Smart textiles possessed the encapsulation efficiency of 52.1%, enthalpy of 64.08 J/g.

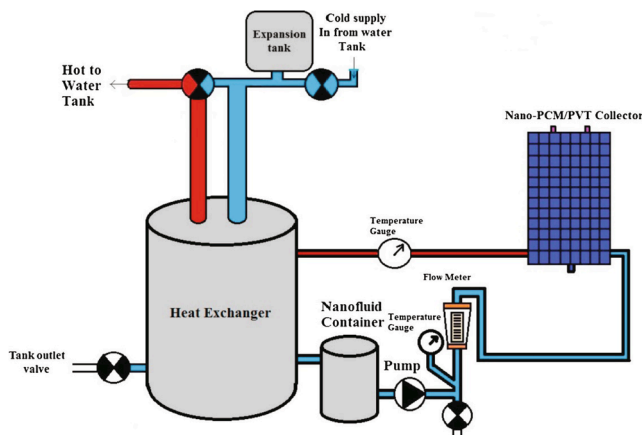


Fig. 22. Schematic diagram of nano-PCM based PV system [379].



Fig. 23. Front view of experimental setup [381].

nanotubes NPs (MWCNTs) and water as heat transfer fluid (HTF) was circulated for cooling of nano-PCM. Results showed that MWCNTs-PCM based system reduced the PV cell temperature by 4 °C, augmented water exit temperature by 46 °C and provided overall energy and exergy efficiency of 85.3% and 12%, respectively. Essa et al. [383] investigated the influence of porous metallic foam of stainless steel filled with paraffin wax and active cooling on the performance of PV panels. The water as HTF was circulated at flow rate of 0.2, 0.3 and 0.4 L/m. Results showed that PV panel with porous metal foam and PCM gave maximum electrical efficiency of 23% at 0.4 L/m and maximum thermal efficiency and overall efficiency of 74% and 95% at 0.3 L/m.

Naderi et al. [384] numerically investigated energy generation capacity PV integrated with PCM (paraffin) and thermoelectric generator (TEG) module. The introduction of PCM-PV reduced the temperature from 74.43 °C to 53.72 °C and electricity output increased by 100%. Output power at nighttime was low and increase in wind velocity increased the efficiency of solar cell up to 16.42%. Mahdi et al. [385] investigated numerically the PV panel with metal foam layer at rear of PCM layer at different PCM layer thickness and PV tilt angles. Results showed that inclusion of metal foam increased the PCM melting time and PV thermal management time by 32 and 55% respectively. PCM thickness in the range of 20–30 mm enhanced the melting time by 54%.

Sarafraz et al. [386] experimentally investigated thermal and electrical performance of PV thermal system with MWCNT/Paraffin NEPCM filled cooling jacket attached at backside of PV panel for cooling purposes. Nanofluid of MWCNT/water-ethylene glycol act as heat transfer

fluid for removing heat from NEPCM. Results showed that the designed system caused enhancement in electrical power and thermal power by 20% and 130% respectively. The 0.2 wt% MWCNTs based nanofluid gave highest thermal and electrical performance of 292.1 W/m².

Feng et al. [387] studied the thermal and electrical attributes of the PV thermal unit using OP35E nano emulsion and water as HTF. The nanoemulsion was formed in 10 and 20 wt% of PCM in water. Maximum average temperature of the PV module after 3000 s with 10 wt% nanoemulsion at a volume flow rate of 10 L/h was 45.87 °C compared to conventional PV temperature of almost 88.4 °C as shown in Fig. 24. It was found that PCM based nanoemulsion coolant required 50% less pumping power compared to water for PV cooling.

Sakhaei et al. [388] performed thermal analysis of the flat plate solar collector coupled with PCM as TES medium and corrugated heat collecting tubes. Use of corrugated tubes reduced the heat loss by 39.8% compared simple tubes, discharging of PCM in evening provided hot water and PCM inclusion improved the efficiency of flat plate solar collector from 41.5% to 48.9%. Maximum outlet temperature of HTF obtained was 40.8 °C at 5150 Reynolds number.

Pawar and Sobhansarbandi [389] studied heat transfer performance of heat pipe assisted evacuated tube solar collector with various PCMs. Findings revealed that optimization of heat pipe accelerated the phase change process of PCM by 48 min. Trtriacontane paraffin displayed total TES of 295 kJ/kg at water mass flow rate of 45 L/h, erythritol PCM showed highest TES capacity of 413.15 kJ/kg while xylitol gave fin temperature of 10 °C more than other PCM filled tubes. Avargani et al. [390] investigated numerically the solar water heater coupled with paraffin PCM unit, depicted in Fig. 1. Single solar water heater collector generated 300 L of hot water during 7 h of operation, while 1200 L of water was obtained by employing two consecutive collectors for 8 h operation. Bhagwat et al. [391] evaluated the latent heat TES heat pipe system with fins in conjunction with the parabolic trough collector under Northeast India climate. Parabolic trough collector improved the solar insolation density and enhanced the charging rate by 81%. The average charging and discharging efficiency of LHS with fins was 44.7% and 32.7% more than system without fins.

Rahmanian et al. [392] fabricated building integrated concentrated PV panel attached with PCM encapsulated aluminium single/multiple packs immersed in nanofluid filled vessel. For single pack PCM arrangement, PV temperature was retained below 78 °C, electrical and thermal efficiency enhanced by 12% and 23% compared to conventional case. In active cooling mode 97% of PCM remained solidified for 60 min and silicon temperature was controlled at 43 °C. Yadav and

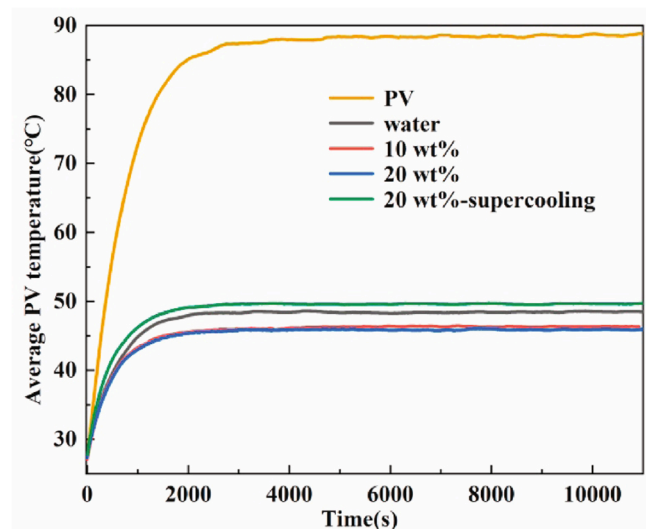


Fig. 24. Average temperature reduction of PV module with various weightages of nano emulsion compared to conventional PV module [387].

Chandramohan [393] developed two dimensional numerical model of paraffin TES based solar dryer system with and without fins. Average outlet temperature of was 5.47 K and 9.1 K higher than inlet temperature for cases of without and with fin respectively at 1 m/s air velocity. Maximum heat gain by air for fin case was 55.3% more than case of without fin. Latest studies to improve performance of PV panels using advanced PCMs are summarized in Table 7.

Thermal regulation of batteries using advanced PCMs

Mohammad Parhizi [406] carried out analytical model and optimized the geometry of lithium ion batteries using phase change thermal management. Yihuan Huang [407] used flexible PCM block to control the temperature of lithium ion batteries pack and results concluded that battery temperature was dropped to 18 C Zou et al.[408] improved the performance of lithium ion batteries by using multiwall carbon nanotubes (MWCNTs), graphene and MWCNTs/graphene based PCM. The composite PCM were prepared and characterized for optimal concentration. According to findings, the value of “k” enhanced by 31.8%, 55.4% and 124% at mass ratio of ¼ of MWCNTs/graphene based PCM compared to MWCNTs/PCM, graphene/PCM and pure PCM respectively, as shown in Fig. 25. Highest increment and decrement of temperature was also observed by MWCNTs/graphene based PCM which was shortened by 63.3% and 50% respectively compared to pure PCM.

Heydat et al.[409] investigated the thermal performance of lithium ion battery using PCM. The NPs, fins and metal foams were also used to enhance the performance of phase change material. The numerical work was done and was validated with literature work. The experimental results concluded that porous metal form having PCM indicted better performance compared to nanoPCM and fin PCM. The 4–6 K drop in battery temperature was observed compared to pristine PCM. Sun et al. [410] used fin structure composite PCM for battery thermal control. The longitudinal and cylindrical ring shape fins were used, and numerical simulations were also conducted to validate the experimental data. It was found that fin shape structure increased the thermal performance and heat transfer area. The optimum number of rings and fins were 1 and

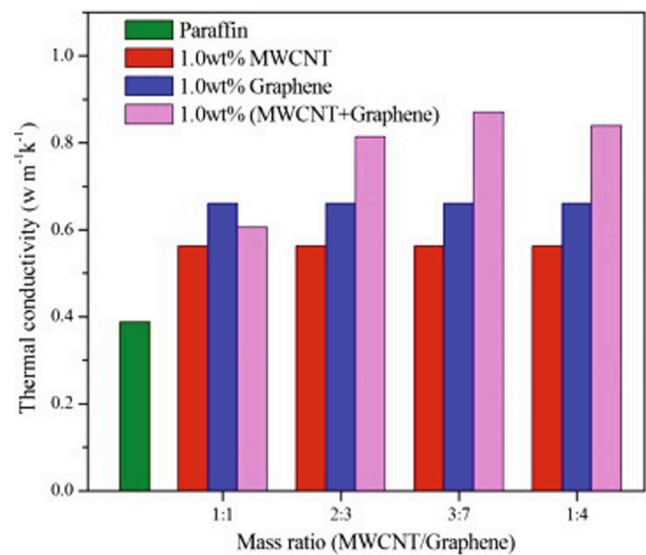


Fig. 25. Thermal conductivity of composite PCM [408].

8 and PCM fin system controlled the battery temperature even at high power rate of 20 W.

Safdari et al.[411] investigated three various configurations for battery thermal optimization using PCM. The PCM was surrounded by circular, rectangular and hexagonal vessel cross sections. The results indicated that hexagonal and circular PCM vessel exhibited same behavior. Similarly, the high charging or discharging was seen when in rectangular shape as compared to other shape vessels. Kiani et al. [412] proposed hybrid thermal control system for lithium ion batteries using numerical and experimental analysis (shown in Fig. 26). The passive cooling was suggested using with copper foam and paraffin integrated with active cooling of alumina nanofluids. At different heating loads, the efficiency was determined and various Reynolds number. The nanofluids enhanced the operating rate of battery as compared to water-

Table 7

Summary of previous studies to improve performance of PV panels using advanced PCMs.

Reference	Type of study	Type of PCM	Method for increasing PCM performance	Type of Cooling	Temperature reduction (°C)	Maximum Thermal efficiency (%)	Maximum Electrical efficiency (%)
[394]	Experimental	PT-58	MWCNTs, GNPs, and magnesium oxide NPs	Passive	9.94		12.10
[395]	Experimental	Paraffin	Polystyrene-carbon nanotube foam	Passive and active		66.8–82.6%	14%
[396]	Experimental	RT-35	Plate with finned tubes	Passive and active		17.22	14.14
[397]	Experimental	Composite PCM of OM35 and biochar	Aluminium metal powder 5 wt%	Active and Passive		60.3–71.2	13.47
[398]	Experimental	TH-SL35 Fatty acid	Copper fins	Active and passive		73	13.1
[399]	Experimental	Paraffin wax	Copper foam matrix and MWCNTs NPs	Passive	9.5		10.5
[400]	Numerical and experimental	Paraffin wax	Expanded graphite	Passive	6.7		14.75
[401]	Experimental	Salt Hydrate	Aluminum shaving	Passive	20	60	15.9
[402]	Experimental	Paraffin wax	Aluminium external fins	Passive	22.3		6.43
[403]	Experimental	OM47	Aluminum matrix of honeycomb structure	Passive	20	7.1–48.5	15.2
[404]	Numerical	RT25HC	Aluminium sheets	Passive			18
[405]	Experimental	Palm wax	Grooved, tubed, and finned containers	Passive	6.1		9.82



Fig. 26. Proposed Experimental Setup for battery thermal management [412].

based cooling system. By using PCM with foam, the melting temperature was high resulting in decreasing the rate of temperature rise.

Behi et al. [413] suggested heat pipe-based cooling system for discharging of lithium titanate battery cell. The temperature was evaluated during discharging rate using various cooling methods. The results depicted that maximum cell temperature was observed to be 56.8 °C in natural convection. Moreover, for forced cooling and heat pipe arrangement, the maximum temperature was observed to be 37.8 °C which was 33.4% less than the natural convection case.

Wang et al. [414] proposed compact cooling system by heat pipe and PCM for thermal control of cylindrical lithium ion batteries. The PCM utilized was paraffin and heat transfer fluid of heat pipe was acetone. It was found that hybrid system controlled the highest and maximum temperature difference in the battery by 47.7 °C and 2.5 °C respectively during discharging process. The heat pipe was observed to be more efficient in terms of performance in battery thermal management. Wu et al. [415] carried out battery thermal management using heat pipe based phase change material used in electric and hybrid vehicles. The heat pipe assisted phase change material was most suitable for cooling system. The results concluded that the temperature distribution of battery was greatly influenced by heat pipe under high discharge rate. The highest temperature was to be controlled below 50 °C under highest discharge rate. The results were in good agreement in optimization of battery thermal management.

Xiao et al. [416] prepared solid–solid PCM namely polymer PCM for efficient battery thermal management. Such PCM exhibited highest and stable heat dissipation. The maximum temperature that was observed 45.3 °C and temperature difference of below 3.1 °C during charging–discharging process. The insertion of such PCM was much feasible in battery thermal management. Smith et al. [417] proposed heat pipe based thermal control system for battery having eight cells (presented in Fig. 27). The heat pipe extracted heat from the battery cell and then

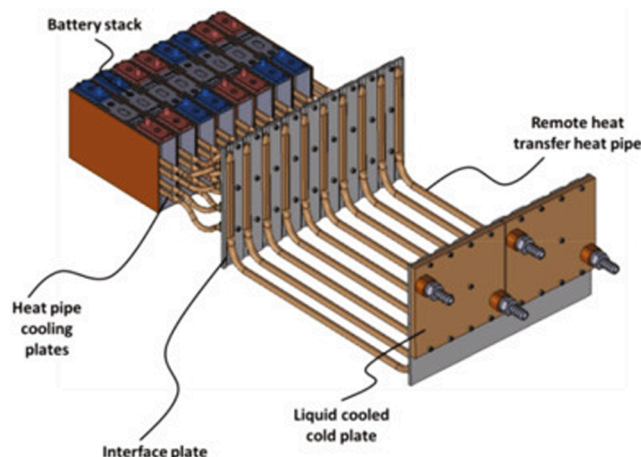


Fig. 27. Proposed heat pipe system for battery thermal management [417].

delivered heat to liquid cooled plates. The modeling was done for the complete system and single-phase analysis for cold plate system. Experiments were carried out, and the proposed system dissipated 50 W of heat from each cell while maintaining the temperature below 55 °C with a coolant inlet temperature of 25 °C.

Huang et al. [418] determined the thermal performance of PCM based heat pipe for efficient thermal control of lithium ion batteries. Three various configurations namely pure PCM, heat pipe along with liquid assisted PCM and heat pipe with air assisted PCM. The heat pipe-based system is better technology for high performance achievement. The experiments were performed at various discharge rates and results indicated that current system controlled the highest temperature upto 50 °C. Moreover, the highest decrement of temperature 3 °C was observed by using heat pipe based liquid PCM as compared to other configurations.

Advanced PCMs in solar heating applications

Solar energy use is being extensively investigated for heating and cooling purposes, as well as power generation. Since solar energy is intermittent, the researcher opted to use PCMs to store thermal energy during peak hours and supply the accumulated latent heat during off-sun hours [419]. With a minor temperature difference between heat store and release, PCMs' latent heat storage results in higher storage density than sensible heat storage. Furthermore, the PCMs contribute to smooth performance and increased reliability. Solar water and air heaters, solar cookers, and winter building heating are among the various solar heating applications [3,41,420,421,42].

Xiao et al. [422] investigated a solar water heater unit with a collector and two composite PCMs of sodium acetate trihydrate/expanded graphite (SAT/EG) and SAT/EG/graphene oxide (GO) produced by physical hybrid approach. Results showed that the solar heat absorbing efficiency and solar to thermal translation efficiency of the novel device with SAT/GO/EG composite PCM were 63.7% and 54.5%, respectively. The temperature of 10 L water enhanced from 25 °C to 38.2 °C in 25 min during a heat exchange operation as depicted in Fig. 28.

For solar energy water heating applications, Liu et al. [423] prepared a shape stable composite PCM using paraffin OP70/EG blended with Ethylene-Propylene-Diene Monomer. To prevent heat loss on the surface, the SiO₂ aerogel of low thermal conductivity and high transmittance was used in a solar thermal system. Results revealed that the composite PCM had enthalpy of 126.8 J/g and thermal conductivity of 1.106 W/m.k. The solar thermal system with composite PCM was able to heat more than 200 L of water. Chopra et al. [424] investigated heat pipe based glass tube solar collector with the integration of stearic acid PCM

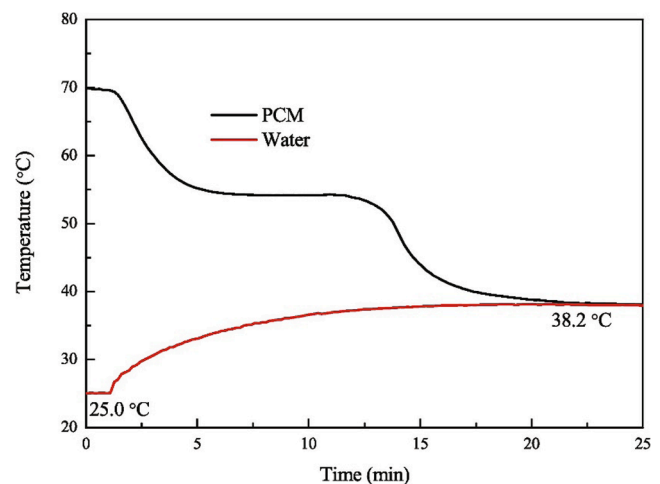


Fig. 28. Temperature evolution of SAT/GO/EG PCM and water in the heat exchange process [422].

with different flow rates of HTF water. Results showed that solar collector without PCM resulted in HTF maximum outlet temperatures of 83°C and maximum temperature gradient of 38.8°C , at flow rate of 8 L/h . Maximum energy efficiency of 78.36% and 54.10% and exergy efficiency of 23.15% and 20.06% were observed for solar collector without and with PCM respectively, at 24 L/h flow rate of HTF. This study showed that solar collector without PCM performed better compared to with PCM in terms of thermal performance and cost of production. Ding et al. [425] explored at the use of paraffin wax in a solar water heating unit (depicted in Fig. 29) in various climate zones. The solar water heating system included a solar collector, a PCM storage tank, and an electric auxiliary heater for maintaining a fixed temperature in the event of low solar radiation. The maximum power savings of 31% was observed in Lhasa, and the maximum heat loss savings of 9.9% were observed in Guangzhou. In all cities, lower auxiliary heater power consumption was observed for PCM of 52°C .

The thermal behavior of an air type evacuated solar thermal collector with a PCM rod inserted in the vacuum tube was designed and investigated by Luo et al. [426]. The engineered solar collector with PCM absorbed 0.247 MJ of heat, of which 0.245 MJ was supplied to air at a conversion efficiency of 99.2% , and PCM insertion extended the collector's operating period by 8 h . During the day, the thermal efficiency of increased from 13% to 397% , with the highest corresponding thermal efficiency of PCM reaching 484% . Kumara and Mysamy [427] studied the thermal behavior of evacuated tube solar heater by using different concentration of CeO_2NPs inserted in paraffin. According to findings, the maximum energy and exergy efficiency of solar water heater with 1% NPs in paraffin was 79.2% and 5.1% respectively and 1% mass of NPs in PCM was declared as optimum in terms of thermal energy storage.

The thermal efficiency of the parabolic dish solar cooker was investigated by Mawire et al. [428] who used sunflower oil and erythritol PCM in the cooking pots. For heat retaining during off-sunshine hours, cooking pots were put in an insulated wonderbag. Results showed that the sunflower cooking pot had a faster cooking time and a higher storage temperature than the PCM pot, while during off-sunshine period, the PCM cooking pot temperature reduced by $0.1\text{--}9.7^{\circ}\text{C}$ and the sunflower oil pot temperature dropped by $8.3\text{--}34^{\circ}\text{C}$. Sunflower oil outperformed water as a cooking fluid due to high storage temperature. Bhavne and Kale [429] studied solar cooker equipped with a concentrated parabolic dish and solar salt PCM (eutectic combination of sodium nitrate and potassium nitrate). Vertical fins were integrated in the PCM enclosure to improve thermal efficiency. It was discovered that the designed solar cooker stored heat up to the melting point of PCM 220°C in 110 min . During indoor cooking, temperature of $170\text{--}180^{\circ}\text{C}$ for the oil was easily attained. 0.25 kg potato and 0.6 kg rice took 17 min and 20 min respectively, to fry. The solar energy collection and storage efficiency found to be $11\text{--}14\%$.

Saxena et al. [430] explored the thermal performance of the solar cooker by incorporating copper tubes packed TES materials. Grainy carbon powder (GCP), paraffin wax, and combination of GCP and

paraffin wax were chosen as TES materials for the experiment. When compared to a traditional solar box cooker, the modified solar box cooker increased thermal efficiency by 53.81% , cooking capacity by 68.81 W , and heat transfer coefficient by $56.78\text{ W/m}^2\text{C}$. Coccia et al. [431] investigated a compact solar cooker equipped with a double-walled stainless steel TES containing 2.5 kg erythritol PCM. The aim of the experiment was to test the cooker's efficiency in the absence of solar radiation during the heating and cooling cycles. As compared to the system without TES, the designed solar cooker with PCM-based TES extended the average cooling time by 351% .

Usage of advanced PCMs in thermoelectric generator

To limit emissions and the use of fossil fuels, researchers are exploring and improving clean energy technologies. One of the emerging technologies to transfer heat energy into electricity is the TEG. Small size, light weight, no moving parts, no noise, no pollution, and a long service life are some of the benefits. TEG is made up of thermoelectric units wedged between a heat source and a cooling source. TEG modules produce seebeck voltage because of temperature differences. The energy produced can be stored in a battery and used to power small electronic devices [432]. TEG can utilize waste heat from various sources e.g. automotive [433], waste heat in industries [434], solar energy [435], to generate electricity. But the output of TEG is very low which can be improved by maintaining high temperature gradient across its hot and cold sides. The use of PCM as energy storage material on cold side has advantages as it can provide thermal management as well as can store extra energy which can be supplied for heating applications [436]. Huan et al. [437] investigated the intermittent efficacy of automotive Bi_2Te_3 TEG by using pentaerythritol PCM on the hot side of TEG as presented in Fig. 30. According to results, the introduction of PCM in TEG resulted in voltage and power production improvement of 0.7% and 1.16% , respectively.

Borhani et al. [438] examined TEG with paraffin RT35 on the cold side as a heat sink and RT69 on the hot side to store thermal energy and minimize voltage fluctuations. To improve thermal efficiency, porous copper foam with different porosities and different PPI was embedded in PCM enclosures. Results showed that the use of metal foam on the cold side of TEG with high PPI and low porosity increased electricity generation by 25.47% . TEG with eutectic PCM of silver-indium-tin as thermal interface materials, and heat exchanger, was investigated experimentally by Liu [439] for thermal control and improved power efficiency. Results showed that the TEG combined with PCM increased the power output from 1.22 W to 4.02 W during the PCM phase transition and reduced the time to achieve stable cooling from 700 s to 200 s . According to fitting analysis, TEG with PCM could obtain a continuous power of 3.51 W for 395 s , maximum electrical energy of 1386.45 J under ideal working conditions.

Rezania et al. [440] used five crucial parameters to investigate the TEG integrated with PCM and heat sink for thermal control and enhancement of electrical efficiency. The study revealed that the TEG length to height ratio had the greatest impact on conversion performance, while the length to width ratio had the least impact. The average estimated conversion efficiency of the optimal TEG/PCM system was 4.28% after 6000 s of heating and cooling cycles. Luo et al. [441] studied solar TEG with composite PCM of paraffin/EG and micro channel heat

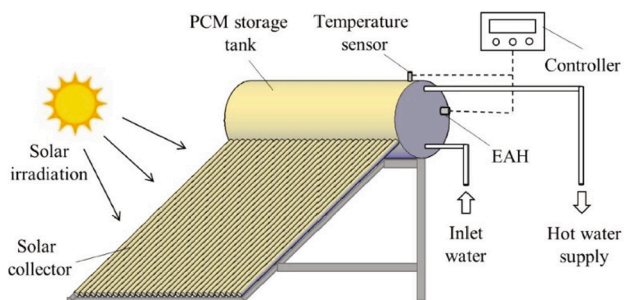


Fig. 29. Schematic diagram of solar water heater unit with PCM storage tank [425].

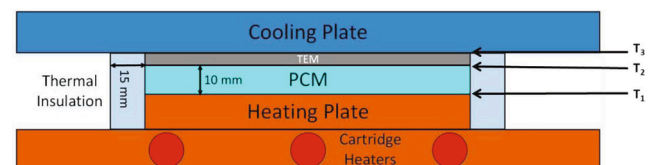


Fig. 30. Main test section along with TEG having PCM [437].

sink on hot side and fins on cold side, for thermal and electrical optimization. Solar TEG was studied using PCMs with various melting points. Results showed that, in the case of 80 °C composite PCMs, the maximum output voltage was 2.697 V, and even in nighttime, the TEG produced electricity due to latent heat TES of composite PCM.

Tuoi et al. [442] explored numerically and experimentally, the solar assisted TEG with PCM for energy harvesting under atmospheric temperature fluctuations. Under ambient temperature variations of 5–35 °C, solar TEG with PCM produced maximum output power up to 0.6 mW, corresponding to a power density of 37.5 W/cm² (using the temperature-controlled oven). The proposed device was seen to be a viable applicant for miniature systems such as wireless Internet of Things sensing systems. Montero et al. [443] investigated the solar TEG in conjunction with paraffin PCM/copper heat sink on the TEG's cold side (as shown in Fig. 31) in desert climate conditions. The effect of PCM volume fraction, heat sink mass, and container geometry on thermal management and electricity production was investigated. Results revealed that temperature difference of 120 °C was noticed between the hot and cold sides of the TEG with 6 kg of PCM and 2.5 kg of heat sink. While at nighttime latent heat storage of PCM generated 0.6% of electricity compared to daytime.

Santiago Madruga [444] numerically investigated the micro energy harvester TEGs in conjunction with a heat storage unit loaded with PCMs in aluminium metal foams on TEG cold side. TEG with PCM/metal foam increased power production by 3.4, 8.5, and 22.5 times for energy harvesting cases of the fuselage, aft pylon, and solar, respectively (Fig. 32).

Applications of advanced PCMs in storage of food

Owing to the risk of spoilage and deformation, food storage and shipping is a challenging task. Conventional packaging's poor thermal insulation and buffering capability have little protection against temperature variations in various environmental conditions and at various times of the day, putting food quality at risk [446]. Today, a wide range of trucks with refrigerators and freezers are manufactured, for food transportation, including the use of PCMs. PCMs are used as TES materials to increase the thermal buffering capacity and maintain optimum temperature of food packages. Because of the high storage density and ability to adjust temperature in response to temperature variations,

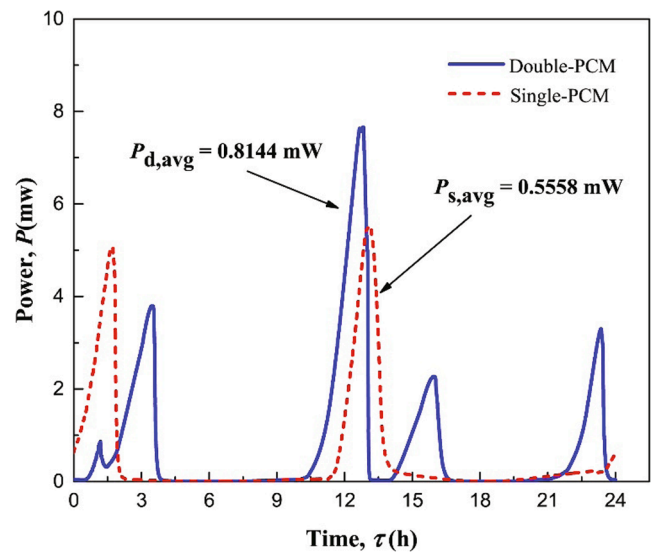


Fig. 32. Comparison of output power of double-PCM and single-PCM based TEG device [445].

PCMs are a preferred method for food packaging [447].

Arjenaki et al. [448] prepared a novel packaging system that used tetradecane PCM to extend shelf life and maintain meat package quality. Temperatures of 12.5 °C and 30 °C were applied to the meat package. According to findings, PCM was able to keep the meat temperature uniform for 133–173 min while exposed to 12.5 °C. While for 30 °C exposure, PCM showed no impact on delaying the increase in meat temperature. Rakshamuthu et al. [449] studied the use of PCM in compact solar dryer for food drying and preservation. In experiment, gooseberries were used as food and zinc nitrate hexahydrate was used as PCM. According to findings, the addition of PCM to solar dryer increased moisture and temperature removal rate and attained full drying in two days, compared to three days for solar drying without PCM and four days for open drying.

To make polyethylene-based nanocomposite, Tas and Unal [450] integrated PCM/halloysite nanotubes into polyethylene films. According to the findings, the nanocomposite film demonstrated adequate mechanical properties and thermal buffering capability in cold chain temperatures. As compared to polyethylene films, the defrost time of frozen and chilled samples was delayed by 18 and 20 min, respectively. Vennapusa et al. [451] used a direct impregnation process to prepare caprylic acid-based composite PCMs using six different porous materials. The composite PCMs were characterized using thermal analysis in terms of enthalpy, charging, discharging, and thermal buffering. In comparison to other composites, the PCM/Silica gel composite had the lowest charging and discharging times compared to other composites but had a lower enthalpy value. When wrapped inside flexible packaging, PCM/expanded perlite composite with an enthalpy value of 81.1 kJ/kg demonstrated thermal buffering of 349 min to raise its temperature from 8.6 °C to 25 °C with 360 g chocolate.

Chen et al. [452] investigated a polyvinyl alcohol PVA and silica aerogel-based chocolate packaging film. Different concentrations of silica aerogel were introduced to PVA. It was discovered that adding 3% silica aerogel to the PVA provided the best film with good thermal energy storage capability, better insulation, better thermal stability, and good gas barrier properties. Besides, the developed film was effective in preventing the softening of chocolate. Kim et al. [453] prepared polypropylene composite packaging by combining octadecane and zeolite for studying cherry tomatoes. The variations in gas concentrations, durability of cherry tomatoes and gas permeabilities were studied using composite polypropylene packaging at storage temperatures ranging from 10 °C to 23 °C. According to the findings, the composite packaging

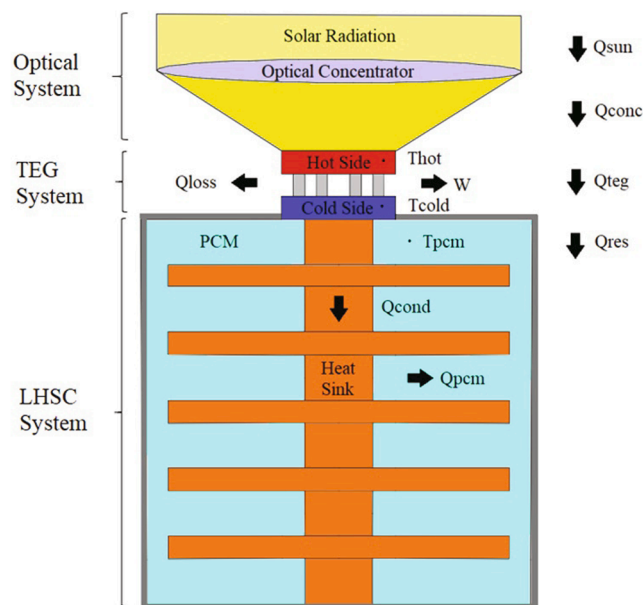


Fig. 31. Schematic diagram of Solar TEG coupled with latent heat storage system [443].

reduced O_2 and CO_2 concentrations in the enclosure while maintaining the consistency of cherry tomatoes in terms of overall soluble solid content, stiffness, and color. Furthermore, by regulating accumulated gas inside the packaging, the composite film based on temperature dependent permeability could be used to prolong the shelf life of agricultural goods. Sandoval et al. [454] developed a fiber coating of polystyrene/PCM by electro hydrodynamic processing and applied it on polystyrene tray by mild heat treatment. Results showed that the polystyrene encapsulated RT5 PCM, with storage capacity of about 34 wt% of PCM and encapsulation efficiency of 78%. With storage temperature as deciding factor, the reduction in melting and solidification enthalpy of 56–58% and 30–40% were observed at 25 °C and 4 °C respectively.

Under normal room conditions, Abdolmaleki et al. [455] tested a household freezer equipped with a eutectic mixture of different grades of PEG with frozen food as depicted in Fig. 33. The study's key goal was to find the right amount of PCM and melting temperature within the freezer to save energy and maintain temperature uniformity for food storage. According to results, the temperature variations were reduced by 40.59% for PCM of 2 kg and -20 °C melting temperature. According to the findings, temperature variations were reduced by 40.59% for PCM of 2 kg and of -20 °C melting temperature. The overall energy savings of 8.37% was observed with PCM of 1.5 kg and -20 °C melting temperature.

Reddy et al. [456] using blending ultra-sonication process prepared eutectic PCM of OM21 and dodecanol with a composition of 60 wt% and 40 wt%, respectively, for thermal buffering applications. Results showed that the charging and discharging times for 180 g of eutectic PCM in a thermal control packaging box were 65 and 48 min, respectively, while the buffering time for 240 g chocolate in a thermal control packaging box exposed to 27 °C was 390 min.

The current status of advanced energy storage materials

Advanced energy storage materials with superior properties are promising for potential energy storage applications because of recently evolving heat transfer fluids. With the assistance of some statistics from the previous decade, this section addresses the current research interests of thermal energy storage materials around the world (2010–2020).

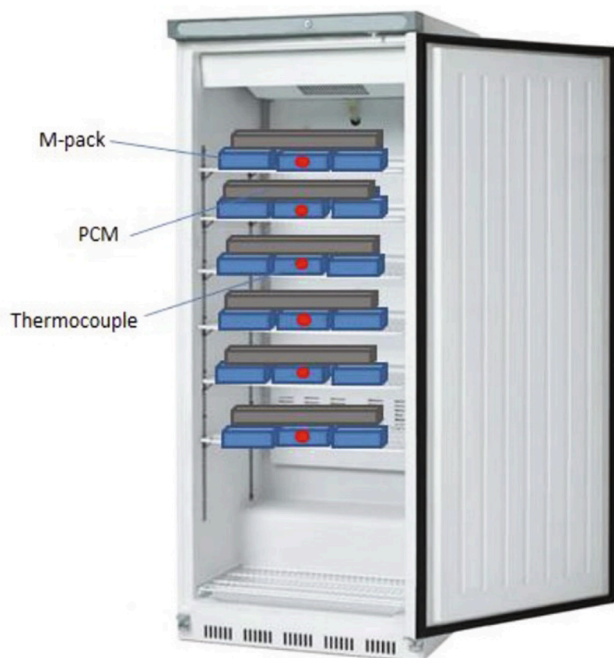


Fig. 33. Diagram viewing the thermocouples M-packs, PCMs positions in refrigerator [455].

According to Web of Science statistics, a search for the key word “advanced energy storage materials” yielded 12,388 results. “CABEZA LF”, “LIU”, and “SARI A”, the most energetic research group, compiled the studies on heat transfer behavior of energy storage materials.

Fig. 34 represents the top ten studied done during the last decade (2010–2020). The countries list from “Web of Science” indicated that China is the most top country where this study is more dominating. It is cleared that number of publications is high in this country. Fig. 34 shows the graphical illustration of the statistics of leading investigators, countries, journals, and organizations published the work on advanced energy storage materials.

Major challenges and future recommendations

The use of phase change materials (PCM) in energy storage applications has received a lot of attention. Following are some challenges recommendations for careful consideration based on this study. These suggestions can be summed up as follows:

- Even though a great deal of research is being conducted on the thermophysical and chemical properties of PCMs for their use in different applications, there is still no international standard for PCM testing and analysis. Furthermore, the thermophysical properties of PCMs are the subject of much debate in the literature.
- The materials with the best thermal properties, such as paraffin wax or fatty acids, are highly flammable.
- Organic PCMs have major drawback of low thermal conductivity which negatively effects its utilization in applications.
- The cost of nanomaterials, instability, and agglomeration are all significant obstacles to the widespread use of nanomaterials based PCM.
- In recent studies, nanoparticles have demonstrated to have promising as a PCM additive. However, there is a shortage of studies into hybrid nanomaterials' implementations. As a result, further research into the thermal properties and performance analysis of hybrid NPs based PCM is needed.
- NEPCM preparation is time-consuming and expensive processes. Although studies have shown that NEPCMs have higher thermal conductivity than single PCMs, it has also been found that the phase enthalpy of PCMs decreases when nanoparticles are added. Their long-term stability is also difficult to maintain compared to initial stabilization.
- Encapsulation of PCMs is costly and time consuming, give poor thermal characteristics. Further research needs to be done to optimize the PCM encapsulation process.
- Although most PCMs and NEPCMs have defined melting points, but thermophysical properties of liquids and solids vary amongst studies. A standard test procedure and database should be established to yield correct results and records.
- For several years, researchers have tested and analyzed the thermal conductivity of PCMs using a variety of techniques. The usage of various values of PCM thermophysical properties revealed a major problem with research criteria.
- To maintain stable performance of the storage systems, future work must avoid phase separation and supercooling of PCMs.
- It is suggested that the use of PCMs in various sectors should be assessed in terms of their effect on the atmosphere and human health.
- The fundamental feature of an energy storage system is phase transition enthalpy. So, detailed examination of the existence of interactions between PCM and nanostructures, as well as their potential impact on the performance of these systems is needed.
- Techniques for preparing composite PCMs are complex and sensitive, and they have a direct effect on performance improvement. As a result, it is strongly advised to report the preparing procedures and methods in as much depth as possible.

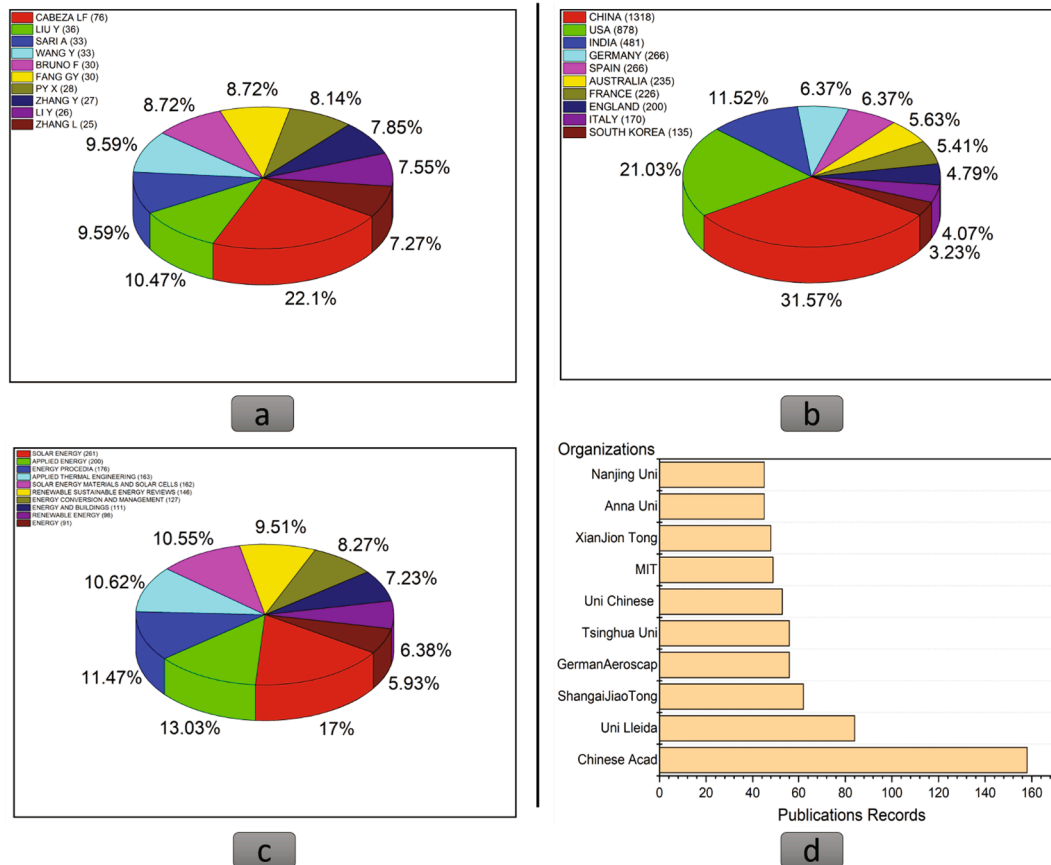


Fig. 34. Web of Science Stats of the top ten (a) authors, (b) countries (c) journals (d) organizations published the work of the entire domain (searched with the keyword “energy storage materials” and refined with a word of “advanced”).

- PCM micro-encapsulation can provide solution to the corrosion problem; a standard leakage test method must be developed, since currently most enclosed PCMs are not tested. Since micro-encapsulation can improve heat transfer and avoid phase separation during the melting process, the stability test is critical for the final assessment of the impacts of encapsulation process.

Conclusion

Researchers are eager to come up with a variety of solutions to address the global energy crisis. High energy consumption levels, along with the depletion of traditional energy resources, have compelled the world to investigate green energy technologies. PCMs are classified as a category of special materials with excellent potential in latent heat storage technologies, as they can retain and release a significant amount of energy when experiencing phase transition process. The major concluded points are:

- PCMs have a high phase transition enthalpy but have a poor thermal conductivity. Various methods have been used to solve the problem of poor thermal conductivity, and studies have shown that dispersing different high thermal conductivity nanoparticles to PCMs accelerate the phase transition rate and improve the energy storage/release capacity of PCMs.
- In most studies, adding nanoparticles lowers the melting temperature and raises the freezing temperature.
- Composite PCMs are developed by impregnating PCMs with porous materials such as metallic foams, carbon foams, and expanded

graphite. Prepared composite PCMs are important for improving the PCM's thermophysical properties and preventing leakage issue.

- Despite composite PCMs' high thermal efficiency, there is leakage issue of PCM, particularly at high temperatures. As a result, it is suggested that PCMs be encapsulated before being used. However, further research is needed to improve current encapsulation techniques and make them commercially available.
- Recent studies have focused on improving performance of PCMs for both residential and industrial applications. The selection of PCM for a specific application is dependent on latent heat, thermal conductivity, sensible heat, melting point, and chemical, kinetic properties.
- Due to better properties of advanced PCMs such as high “k” and high HT, these are being utilized in broad range of thermal applications that desalination, building envelopes, solar heaters, smart textiles, and food packaging applications, for improving thermal and electrical efficiency in PV thermal systems, thermoelectric generators, and lithium-ion batteries as well as glazing in for building applications.
- According to recent studies on the use of PCM for heat energy storage, energy consumption was reduced in all conditions and the utilization of advanced PCMs to retain thermal energy significantly improve the performance. More type of advanced PCMs should be explored by researchers to bring new areas of applications for their usage for good thermal performance in that field.
- However, still leakage and little better performance of performance of advanced PCMs compared to conventional PCMs is still the bottleneck of PCMs which should be considered by future researchers, which will bring down in thermal management field.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Solarin SA. An environmental impact assessment of fossil fuel subsidies in emerging and developing economies. *Environ Impact Assess Rev* 2020;85: 106443. <https://doi.org/10.1016/j.eiar.2020.106443>.
- [2] Sinsel SR, Riemke RL, Hoffmann VH. Challenges and solution technologies for the integration of variable renewable energy sources—a review. *Renew Energy* 2020; 145:2271–85. <https://doi.org/10.1016/j.renene.2019.06.147>.
- [3] Javadi FS, Metselaar HSC, Ganesan P. Performance improvement of solar thermal systems integrated with phase change materials (PCM), a review. *Sol Energy* 2020;206:330–52. <https://doi.org/10.1016/j.solener.2020.05.106>.
- [4] Tao YB, He Y-L. A review of phase change material and performance enhancement method for latent heat storage system. *Renew Sustain Energy Rev* 2018;93:245–59. <https://doi.org/10.1016/j.rser.2018.05.028>.
- [5] Jouhara H, Żabnieńska-Góra A, Khordehghah N, Ahmad D, Lipinski T. Latent thermal energy storage technologies and applications: A review. *Int J Thermofluids* 2020;5–6:100039. <https://doi.org/10.1016/j.ijft.2020.100039>.
- [6] Veerakumar C, Sreekumar A. Phase change material based cold thermal energy storage: Materials, techniques and applications – A review. *Int J Refrig* 2016;67: 271–89. <https://doi.org/10.1016/j.ijrefrig.2015.12.005>.
- [7] Velmurugan K, Kumarasamy S, Wongwuttanasatian T, Seiththanabutra V. Review of PCM types and suggestions for an applicable cascaded PCM for passive PV module cooling under tropical climate conditions. *J Clean Prod* 2021;293: 126065. <https://doi.org/10.1016/j.jclepro.2021.126065>.
- [8] Lin Y, Jia Y, Alva G, Fang G. Review on thermal conductivity enhancement, thermal properties and applications of phase change materials in thermal energy storage. *Renew Sustain Energy Rev* 2018;82:2730–42. <https://doi.org/10.1016/j.rser.2017.10.002>.
- [9] Wang T, Almarashi A, Al-Turki YA, Abu-Hamdeh NH, Hajizadeh MR, Chu Y-M. Approaches for expedition of discharging of PCM involving nanoparticles and radial fins. *J Mol Liq* 2021;329:115052. <https://doi.org/10.1016/j.molliq.2020.115052>.
- [10] Li F, Sheikholeslami M, Dara RN, Jafaryar M, Shafee A, Nguyen-Thoi T, et al. Numerical study for nanofluid behavior inside a storage finned enclosure involving melting process. *J Mol Liq* 2020;297:111939. <https://doi.org/10.1016/j.molliq.2019.111939>.
- [11] Hajizadeh MR, Alsabery AI, Sheremet MA, Kumar R, Li Z, Bach Q-V. Nanoparticle impact on discharging of PCM through a thermal storage involving numerical modeling for heat transfer and irreversibility. *Powder Technol* 2020;376:424–37. <https://doi.org/10.1016/j.powtec.2020.08.031>.
- [12] Shafee A, Sheikholeslami M, Jafaryar M, Babazadeh H. Utilizing copper oxide nanoparticles for expedition of solidification within a storage system. *J Mol Liq* 2020;302:112371. <https://doi.org/10.1016/j.molliq.2019.112371>.
- [13] Jin X, Xiao Q, Xu T, Huang G, Wu H, Wang D, et al. Thermal conductivity enhancement of a sodium acetate trihydrate–potassium chloride–urea/expanded graphite composite phase–change material for latent heat thermal energy storage. *Energy Build* 2021;231:110615. <https://doi.org/10.1016/j.enbuild.2020.110615>.
- [14] Wang T, Wang K, Ye F, Ren Y, Xu C. Characterization and thermal properties of a shape-stable Na₂CO₃-K₂CO₃/coal fly ash/expanded graphite composite phase change materials for high-temperature thermal energy storage. *J Energy Storage* 2021;33:102123. <https://doi.org/10.1016/j.est.2020.102123>.
- [15] Sakai H, Sheng N, Kurniawan A, Akiyama T, Nomura T. Fabrication of heat storage pellets composed of microencapsulated phase change material for high-temperature applications. *Appl Energy* 2020;265:114673. <https://doi.org/10.1016/j.apenergy.2020.114673>.
- [16] Rathore PKS, Shukla SK. Potential of macroencapsulated PCM for thermal energy storage in buildings: A comprehensive review. *Constr Build Mater* 2019;225: 723–44. <https://doi.org/10.1016/j.conbuildmat.2019.07.221>.
- [17] Nong H, Hajizadeh MR, Babazadeh H. Numerical modeling of paraffin melting expedition with considering nanoparticles through wavy duct. *J Mol Liq* 2020; 305:112807. <https://doi.org/10.1016/j.molliq.2020.112807>.
- [18] Christopher S, Parham K, Mosaffa AH, Farid MM, Ma Z, Thakur AK, et al. A critical review on phase change material energy storage systems with cascaded configurations. *J Clean Prod* 2021;283:124653. <https://doi.org/10.1016/j.jclepro.2020.124653>.
- [19] Lakshmi Narasimhan N. Assessment of latent heat thermal storage systems operating with multiple phase change materials. *J Energy Storage* 2019;23: 442–55. <https://doi.org/10.1016/j.est.2019.04.008>.
- [20] Singh P, Sharma RK, Ansu AK, Goyal R, Sari A, Tyagi VV. A comprehensive review on development of eutectic organic phase change materials and their composites for low and medium range thermal energy storage applications. *Sol Energy Mater Sol Cells* 2021;223:110955. <https://doi.org/10.1016/j.solmat.2020.110955>.
- [21] Hashem Zadeh SM, Mehryan SAM, Ghalambaz M, Ghodrati M, Young J, Chamkha A. Hybrid thermal performance enhancement of a circular latent heat storage system by utilizing partially filled copper foam and Cu/GO nano-additives. *Energy* 2020;213:118761. <https://doi.org/10.1016/j.energy.2020.118761>.
- [22] Zhang X, Su G, Lin J, Liu A, Wang C, Zhuang Y. Three-dimensional numerical investigation on melting performance of phase change material composited with copper foam in local thermal non-equilibrium containing an internal heater. *Int J Heat Mass Transf* 2021;170:121021. <https://doi.org/10.1016/j.ijheatmasstransfer.2021.121021>.
- [23] Gao Y, Liu F, Wang J, Ye W, Meng X. Influence of the copper foam shape on thermal performance of phase-change material. *J Energy Storage* 2021;36: 102416. <https://doi.org/10.1016/j.est.2021.102416>.
- [24] Nie C, Liu J, Deng S. Effect of geometric parameter and nanoparticles on PCM melting in a vertical shell-tube system. *Appl Therm Eng* 2021;184:116290. <https://doi.org/10.1016/j.applthermaleng.2020.116290>.
- [25] Zeng S, Chen S, Wu G. Effect of inclination angle on melting process of phase change materials in a square cavity under mechanical vibration. *J Energy Storage* 2021;36:102392. <https://doi.org/10.1016/j.est.2021.102392>.
- [26] Maldonado JM, de Gracia A, Cabeza LF. Systematic review on the use of heat pipes in latent heat thermal energy storage tanks. *J Energy Storage* 2020;32: 101733. <https://doi.org/10.1016/j.est.2020.101733>.
- [27] Xiong Q, Tlili I, Dara RN, Shafee A, Nguyen-Thoi T, Rebey A, et al. Energy storage simulation involving NEPCM solidification in appearance of fins. *Phys A Stat Mech Its Appl* 2020;544:123566. <https://doi.org/10.1016/j.physa.2019.123566>.
- [28] Huu-Quan D, Sheremet M, Kamel MS, Izadi M. Investigation of thermal-hydro dynamical behavior on nano-encapsulated PCM suspension: Effect of fin position, fractioning and aspect ratio. *Chem Eng Process - Process Intensif* 2020;157: 108122. <https://doi.org/10.1016/j.ccep.2020.108122>.
- [29] Safaei MR, Goshayeshi HR, Chaer I. Solar still efficiency enhancement by using graphene oxide/paraffin Nano-PCM. *Energies* 2019;12. <https://doi.org/10.3390/en12102002>.
- [30] Sarafriz MM, Tian Z, Tlili I, Kazi S, Goodarzi M. Thermal evaluation of a heat pipe working with n-pentane-acetone and n-pentane-methanol binary mixtures. *J Therm Anal Calorim* 2020;139:2435–45. <https://doi.org/10.1007/s10973-019-08414-2>.
- [31] Ahmadi MH, Mohseni-Gharyehsafa B, Ghazvini M, Goodarzi M, Jilte RD, Kumar R. Comparing various machine learning approaches in modeling the dynamic viscosity of CuO/water nanofluid. *J Therm Anal Calorim* 2020;139: 2585–99. <https://doi.org/10.1007/s10973-019-08762-z>.
- [32] Bahiraei M, Jamshidmofid M, Goodarzi M. Efficacy of a hybrid nanofluid in a new microchannel heat sink equipped with both secondary channels and ribs. *J Mol Liq* 2019;273:88–98. <https://doi.org/10.1016/j.molliq.2018.10.003>.
- [33] Bahmani MH, Sheikhzadeh G, Zarringhalam M, Akbari OA, Alrashed AAAA, Shabani GAS, et al. Investigation of turbulent heat transfer and nanofluid flow in a double pipe heat exchanger. *Adv Powder Technol* 2018;29:273–82. <https://doi.org/10.1016/j.appt.2017.11.013>.
- [34] Giwa SO, Sharifpur M, Goodarzi M, Alsulami H, Meyer JP. Influence of base fluid, temperature, and concentration on the thermophysical properties of hybrid nanofluids of alumina–ferrofluid: experimental data, modeling through enhanced ANN, ANFIS, and curve fitting. *J Therm Anal Calorim* 2021;143:4149–67. <https://doi.org/10.1007/s10973-020-09372-w>.
- [35] Bagherzadeh SA, Jalali E, Sarafriz MM, Ali Akbari O, Karimipour A, Goodarzi M, et al. Effects of magnetic field on micro cross jet injection of dispersed nanoparticles in a microchannel. *Int J Numer Methods Heat Fluid Flow* 2020;30: 2683–704. <https://doi.org/10.1108/HFF-02-2019-0150>.
- [36] Bagherzadeh SA, D'Orazio A, Karimipour A, Goodarzi M, Bach Q-V. A novel sensitivity analysis model of EANN for F-MWCNTs–Fe₃O₄/EG nanofluid thermal conductivity: Outputs predicted analytically instead of numerically to more accuracy and less costs. *Phys A Stat Mech Appl* 2019;521:406–15. <https://doi.org/10.1016/j.physa.2019.01.048>.
- [37] Akbari OA, Safaei MR, Goodarzi M, Akbar NS, Zarringhalam M, Shabani GAS, et al. A modified two-phase mixture model of nanofluid flow and heat transfer in a 3-D curved microtube. *Adv Powder Technol* 2016;27:2175–85. <https://doi.org/10.1016/j.appt.2016.08.002>.
- [38] Goshayeshi HR, Safaei MR, Goodarzi M, Dahari M. Particle size and type effects on heat transfer enhancement of Ferro-nanofluids in a pulsating heat pipe. *Powder Technol* 2016;301:1218–26. <https://doi.org/10.1016/j.powtec.2016.08.007>.
- [39] Shahsavari A, Goodarzi A, Mohammed HI, Shirmeshan A, Talebizadehsardari P. Thermal performance evaluation of non-uniform fin array in a finned double-pipe latent heat storage system. *Energy* 2020;193:116800. <https://doi.org/10.1016/j.energy.2019.116800>.
- [40] Zhang S, Feng D, Shi L, Wang L, Jin Y, Tian L, et al. A review of phase change heat transfer in shape-stabilized phase change materials (ss-PCMs) based on porous supports for thermal energy storage. *Renew Sustain Energy Rev* 2021;135: 110127. <https://doi.org/10.1016/j.rser.2020.110127>.
- [41] Douvi E, Pagkalos C, Dogkas G, Koukou MK, Stathopoulos VN, Caouris Y, et al. Phase change materials in solar domestic hot water systems: A review. *Int J Thermofluids* 2021;100075. <https://doi.org/10.1016/j.ijft.2021.100075>.
- [42] Aramesh M, Shabani B. On the integration of phase change materials with evacuated tube solar thermal collectors. *Renew Sustain Energy Rev* 2020;132: 110135. <https://doi.org/10.1016/j.rser.2020.110135>.
- [43] Jaguemont J, Omar N, Van den Bossche P, Mierlo J. Phase-change materials (PCM) for automotive applications: A review. *Appl Therm Eng* 2018;132:308–20. <https://doi.org/10.1016/j.applthermaleng.2017.12.097>.
- [44] Jaguemont J, Van Mierlo J. A comprehensive review of future thermal management systems for battery-electrified vehicles. *J Energy Storage* 2020;31: 101551. <https://doi.org/10.1016/j.est.2020.101551>.

- [45] Zichen W, Changqing D. A comprehensive review on thermal management systems for power lithium-ion batteries. *Renew Sustain Energy Rev* 2021;139: 110685. <https://doi.org/10.1016/j.rser.2020.110685>.
- [46] Nejman A, Cieślak M. The impact of the heating/cooling rate on the thermoregulating properties of textile materials modified with PCM microcapsules. *Appl Therm Eng* 2017;127:212–23. <https://doi.org/10.1016/j.applthermaleng.2017.08.037>.
- [47] Jaworski M. Mathematical model of heat transfer in PCM incorporated fabrics subjected to different thermal loads. *Appl Therm Eng* 2019;150:506–11. <https://doi.org/10.1016/j.applthermaleng.2019.01.019>.
- [48] Selvn H, Allouche Y, Manescu RI, Hafner A. Review on cold thermal energy storage applied to refrigeration systems using phase change materials. *Therm Sci Eng Prog* 2021;22:100807. <https://doi.org/10.1016/j.tsep.2020.100807>.
- [49] Ko J, Thu K, Miyazaki T. Investigation of a cascaded CO₂ refrigeration system using phase change materials for energy-saving potentials. *Appl Therm Eng* 2021; 182:116104. <https://doi.org/10.1016/j.applthermaleng.2020.116104>.
- [50] Anagnostopoulos A, Navarro ME, Stefanidou M, Ding Y, Gaidajis G. Red mud-molten salt composites for medium-high temperature thermal energy storage and waste heat recovery applications. *J Hazard Mater* 2021;413:125407. <https://doi.org/10.1016/j.jhazmat.2021.125407>.
- [51] Du K, Calautit J, Eames P, Wu Y. A state-of-the-art review of the application of phase change materials (PCM) in Mobilized-Thermal Energy Storage (M-TES) for recovering low-temperature industrial waste heat (IWH) for distributed heat supply. *Renew Energy* 2021;168:1040–57. <https://doi.org/10.1016/j.renene.2020.12.057>.
- [52] Said MA, Hassan H. Impact of energy storage of new hybrid system of phase change materials combined with air-conditioner on its heating and cooling performance. *J Energy Storage* 2021;36:102400. <https://doi.org/10.1016/j.est.2021.102400>.
- [53] Hajizadeh MR, Abohamzeh E, Tiwari AK, Sheremet MA, Li Z, Bach Q-V. Discharging of PCM for ventilation system incorporating nanoparticles. *J Mol Liq* 2020;315:113696. <https://doi.org/10.1016/j.molliq.2020.113696>.
- [54] Diao Y, Ma C, Chen C, Wang Z, Zhao Y, Liang L. Experimental analysis of a latent heat storage unit with multichannel flat tube and copper foam. *Appl Therm Eng* 2021;186:116503. <https://doi.org/10.1016/j.applthermaleng.2020.116503>.
- [55] Yang L, Jin X, Zhang Y, Du K. Recent development on heat transfer and various applications of phase-change materials. *J Clean Prod* 2021;287:124432. <https://doi.org/10.1016/j.jclepro.2020.124432>.
- [56] Al-Ahmed A, Mazumder MAJ, Salhi B, Sari A, Afzaal M, Al-Sulaiman FA. Effects of carbon-based fillers on thermal properties of fatty acids and their eutectics as phase change materials used for thermal energy storage: A review. *J Energy Storage* 2021;35:102329. <https://doi.org/10.1016/j.est.2021.102329>.
- [57] Zhu N, Deng R, Hu P, Lei F, Xu L, Jiang Z. Coupling optimization study of key influencing factors on PCM trombe wall for year thermal management. *Energy* 2021;236:121470. <https://doi.org/10.1016/j.energy.2021.121470>.
- [58] Kozelj R, Mlakar U, Zavrl E, Stritih U, Stropnik R. An experimental and numerical analysis of an improved thermal storage tank with encapsulated PCM for use in retrofitted buildings for heating. *Energy Build* 2021;248:111196. <https://doi.org/10.1016/j.enbuild.2021.111196>.
- [59] Qu Y, Zhou D, Xue F, Cui L. Multi-factor analysis on thermal comfort and energy saving potential for PCM-integrated buildings in summer. *Energy Build* 2021;241: 110966. <https://doi.org/10.1016/j.enbuild.2021.110966>.
- [60] Guo J, Zhang G. Investigating the performance of the PCM-integrated building envelope on a seasonal basis. *J Taiwan Inst Chem Eng* 2021;124:91–7. <https://doi.org/10.1016/j.jtice.2021.04.066>.
- [61] Yang L-Y, Feng C-P, Bai L, Bao R-Y, Liu Z-Y, Yang M-B, et al. Flexible shape-stabilized phase change materials with passive radiative cooling capability for thermal management. *Chem Eng J* 2021;425:131466. <https://doi.org/10.1016/j.cej.2021.131466>.
- [62] Arıcı M, Bilgin F, Nizetić S, Karabay H. PCM integrated to external building walls: An optimization study on maximum activation of latent heat. *Appl Therm Eng* 2020;165:114560. <https://doi.org/10.1016/j.applthermaleng.2019.114560>.
- [63] Tunçbilek E, Arıcı M, Krajčák M, Nizetić S, Karabay H. Thermal performance based optimization of an office wall containing PCM under intermittent cooling operation. *Appl Therm Eng* 2020;179:115750. <https://doi.org/10.1016/j.applthermaleng.2020.115750>.
- [64] Tunçbilek E, Arıcı M, Bouadila S, Wonorahardjo S. Seasonal and annual performance analysis of PCM-integrated building brick under the climatic conditions of Marmara region. *J Therm Anal Calorim* 2020;141:613–24. <https://doi.org/10.1007/s10973-020-09320-8>.
- [65] Wonorahardjo S, Sutjahja IM, Tunçbilek E, Achsani RA, Arıcı M, Rahmah N. PCM-based passive air conditioner in urban houses for the tropical climates: An experimental analysis on the stratum air circulation. *Build Environ* 2021;192: 107632. <https://doi.org/10.1016/j.buildenv.2021.107632>.
- [66] Li D, Wu Y, Liu C, Zhang G, Arıcı M. Numerical investigation of thermal and optical performance of window units filled with nanoparticle enhanced PCM. *Int J Heat Mass Transf* 2018;125:1321–32. <https://doi.org/10.1016/j.ijheatmasstransfer.2018.04.152>.
- [67] Wang Q, Wei W, Li D, Qi H, Wang F, Arıcı M. Experimental investigation of thermal radiative properties of Al₂O₃-paraffin nanofluid. *Sol Energy* 2019;177: 420–6. <https://doi.org/10.1016/j.solener.2018.11.034>.
- [68] Li Q, Ma L, Li D, Arıcı M, Yıldız Ç, Wang Z, et al. Thermoeconomic analysis of a wall incorporating phase change material in a rural residence located in northeast China. *Sustain Energy Technol Assessments* 2021;44:101091. <https://doi.org/10.1016/j.seta.2021.101091>.
- [69] Tariq SL, Ali HM, Akram MA, Janjua MM, Ahmadlouydarab M. Nanoparticles enhanced phase change materials (NePCMs)-A recent review. *Appl Therm Eng* 2020;176:115305. <https://doi.org/10.1016/j.applthermaleng.2020.115305>.
- [70] Alehosseini E, Jafari SM. Nanoencapsulation of phase change materials (PCMs) and their applications in various fields for energy storage and management. *Adv Colloid Interface Sci* 2020;283:102226. <https://doi.org/10.1016/j.cis.2020.102226>.
- [71] Shen Z-G, Chen S, Liu X, Chen B. A review on thermal management performance enhancement of phase change materials for vehicle lithium-ion batteries. *Renew Sustain Energy Rev* 2021;148:111301. <https://doi.org/10.1016/j.rser.2021.111301>.
- [72] Preet S. A review on the outlook of thermal management of photovoltaic panel using phase change material. *Energy Clim Chang* 2021;2:100033. <https://doi.org/10.1016/j.egycc.2021.100033>.
- [73] Mahian O, Ghafarian S, Sarrafha H, Kasaeian A, Yousefi H, Yan W-M. Phase change materials in solar photovoltaics applied in buildings: An overview. *Sol Energy* 2021;224:569–92. <https://doi.org/10.1016/j.solener.2021.06.010>.
- [74] Farooq AS, Zhang P. Fundamentals, materials and strategies for personal thermal management by next-generation textiles. *Compos Part A Appl Sci Manuf* 2021; 142:106249. <https://doi.org/10.1016/j.compositesa.2020.106249>.
- [75] Rostami S, Afrand M, Shahsavari A, Sheikholeslami M, Kalbasi R, Aghakhani S, et al. A review of melting and freezing processes of PCM/nano-PCM and their application in energy storage. *Energy* 2020;211:118698. <https://doi.org/10.1016/j.energy.2020.118698>.
- [76] Wang XL, Li B, Qu ZG, Zhang JF, Jin ZG. Effects of graphite microstructure evolution on the anisotropic thermal conductivity of expanded graphite/paraffin phase change materials and their thermal energy storage performance. *Int J Heat Mass Transf* 2020;155:119853. <https://doi.org/10.1016/j.ijheatmasstransfer.2020.119853>.
- [77] Ren Q, Guo P, Zhu J. Thermal management of electronic devices using pin-fin based cascade microencapsulated PCM/expanded graphite composite. *Int J Heat Mass Transf* 2020;149:119199.
- [78] Iasiello M, Mameli M, Filippeschi S, Bianco N. Metal foam/PCM melting evolution analysis: Orientation and morphology effects. *Appl Therm Eng* 2021;187:116572. <https://doi.org/10.1016/j.applthermaleng.2021.116572>.
- [79] Opolot M, Zhao C, Liu M, Mancin S, Bruno F, Hooman K. Influence of cascaded graphite foams on thermal performance of high temperature phase change material storage systems. *Appl Therm Eng* 2020;180:115618. <https://doi.org/10.1016/j.applthermaleng.2020.115618>.
- [80] Zhang C, Yu M, Fan Y, Zhang X, Zhao Y, Qiu L. Numerical study on heat transfer enhancement of PCM using three combined methods based on heat pipe. *Energy* 2020;195:116809. <https://doi.org/10.1016/j.energy.2019.116809>.
- [81] Hayat MA, Ali HM, Janjua MM, Pao W, Li C, Alizadeh M. Phase change material/heat pipe and Copper foam-based heat sinks for thermal management of electronic systems. *J Energy Storage* 2020;32:101971. <https://doi.org/10.1016/j.est.2020.101971>.
- [82] Huang X, Zhu C, Lin Y, Fang G. Thermal properties and applications of microencapsulated PCM for thermal energy storage: A review. *Appl Therm Eng* 2019;147:841–55. <https://doi.org/10.1016/j.applthermaleng.2018.11.007>.
- [83] Rodríguez-Cumplido F, Pabón-Gelves E, Chejne-Jana F. Recent developments in the synthesis of microencapsulated and nanoencapsulated phase change materials. *J Energy Storage* 2019;24:100821. <https://doi.org/10.1016/j.est.2019.100821>.
- [84] Arshad A, Jabbar M, Yan Y, Darkwa J. The micro-/nano-PCMs for thermal energy storage systems: A state of art review. *Int J Energy Res* 2019;43:5572–620.
- [85] Qureshi ZA, Ali HM, Khushnood S. Recent advances on thermal conductivity enhancement of phase change materials for energy storage system: a review. *Int J Heat Mass Transf* 2018;127:838–56.
- [86] Eanest Jebasingh B, Valan AA. A detailed review on heat transfer rate, supercooling, thermal stability and reliability of nanoparticle dispersed organic phase change material for low-temperature applications. *Mater Today Energy* 2020;16:100408. <https://doi.org/10.1016/j.mtener.2020.100408>.
- [87] Wong-Pinto L-S, Milian Y, Ushak S. Progress on use of nanoparticles in salt hydrates as phase change materials. *Renew Sustain Energy Rev* 2020;122: 109727. <https://doi.org/10.1016/j.rser.2020.109727>.
- [88] Nizetić S, Jurčević M, Arıcı M, Arasu AV, Xie G. Nano-enhanced phase change materials and fluids in energy applications: A review. *Renew Sustain Energy Rev* 2020;129:109931. <https://doi.org/10.1016/j.rser.2020.109931>.
- [89] Wu X, Gao M, Wang K, Wang Q, Cheng C, Zhu Y, et al. Experimental study of the thermal properties of a homogeneous dispersion system of a paraffin-based composite phase change materials. *J Energy Storage* 2021;36:102398. <https://doi.org/10.1016/j.est.2021.102398>.
- [90] B Kalidasan, Pandey AK, Shahabuddin S, George M, Sharma K, Samykano M, et al. Synthesis and characterization of conducting Polyaniline@cobalt-Paraffin wax nanocomposite as nano-phase change material: Enhanced thermophysical properties. *Renew Energy* 2021;173:1057–69. <https://doi.org/10.1016/j.renene.2021.04.050>.
- [91] Daneshazarian R, Antoun S, Dworkin SB. Performance assessment of nano-enhanced phase change material for thermal storage. *Int J Heat Mass Transf* 2021;173:121256. <https://doi.org/10.1016/j.ijheatmasstransfer.2021.121256>.
- [92] Liu X, Tie J, Wang Z, Xia Y, Wang C-A, Tie S. Improved thermal conductivity and stability of Na₂SO₄-10H₂O PCMs system by incorporation of Al/C hybrid nanoparticles. *J Mater Res Technol* 2021;12:982–8. <https://doi.org/10.1016/j.jmrt.2021.02.096>.
- [93] Reuben Raj C, Suresh S, Vasudevan S, Chandrasekar M, Kumar Singh V, Bhavsar RR. Thermal performance of nano-enriched form-stable PCM implanted in a pin finned wall-less heat sink for thermal management application. *Energy*

- Convers Manag 2020;226:113466. <https://doi.org/10.1016/j.enconman.2020.113466>.
- [94] Benbrika M, Teggari M, Arıcı M, Ismail KAR, Bouabdallah S, Mezaache E-H. Effect of graphene nanoparticles on charging and discharging processes of latent thermal energy storage using horizontal cylinders. *Sustain Energy Technol Assessments* 2021;45:101242. <https://doi.org/10.1016/j.seta.2021.101242>.
- [95] Sun X, Liu L, Mo Y, Li J, Li C. Enhanced thermal energy storage of a paraffin-based phase change material (PCM) using nano carbons. *Appl Therm Eng* 2020;181:115992. <https://doi.org/10.1016/j.applthermaleng.2020.115992>.
- [96] Qu Y, Wang S, Zhou D, Tian Y. Experimental study on thermal conductivity of paraffin-based shape-stabilized phase change material with hybrid carbon nano-additives. *Renew Energy* 2020;146:2637–45. <https://doi.org/10.1016/j.renene.2019.08.098>.
- [97] Shahsavari A, Khanmohammadi S, Karimipour A, Goodarzi M. A novel comprehensive experimental study concerned syntheses and prepare liquid paraffin-Fe3O4 mixture to develop models for both thermal conductivity & viscosity: A new approach of GMDH type of neural network. *Int J Heat Mass Transf* 2019;131:432–41. <https://doi.org/10.1016/j.ijheatmasstransfer.2018.11.069>.
- [98] Arıcı M, Tütüncü E, Yıldız Ç, Li D. Enhancement of PCM melting rate via internal fin and nanoparticles. *Int J Heat Mass Transf* 2020;156:119845. <https://doi.org/10.1016/j.ijheatmasstransfer.2020.119845>.
- [99] Sarani I, Payan S, Nada SA, Payan A. Numerical investigation of an innovative discontinuous distribution of fins for solidification rate enhancement in PCM with and without nanoparticles. *Appl Therm Eng* 2020;176:115017. <https://doi.org/10.1016/j.applthermaleng.2020.115017>.
- [100] Sheikholeslami M, Keshmeli AN, Shafee A. Melting and solidification within an energy storage unit with triangular fin and CuO nano particles. *J Energy Storage* 2020;32:101716. <https://doi.org/10.1016/j.est.2020.101716>.
- [101] Alizadeh M, Nabizadeh A, Fazlollahabbar A, Ganji DD. An optimization study of solidification procedure in a wavy-wall storage unit considering the impacts of NEPCM and curved fin. *Int Commun Heat Mass Transf* 2021;124:105249. <https://doi.org/10.1016/j.icheatmasstransfer.2021.105249>.
- [102] Eslami M, Khosravi F, Fallah Kohan HR. Effects of fin parameters on performance of latent heat thermal energy storage systems: A comprehensive review. *Sustain Energy Technol Assessments* 2021;47:101449. <https://doi.org/10.1016/j.seta.2021.101449>.
- [103] Zayed ME, Zhao J, Li W, Elsheikh AH, Elbanna AM, Jing L, et al. Recent progress in phase change materials storage containers: Geometries, design considerations and heat transfer improvement methods. *J Energy Storage* 2020;30:101341. <https://doi.org/10.1016/j.est.2020.101341>.
- [104] Shademan M, Hossein Nezhad A. Numerical investigation of a vertical finned-tube and shell energy storage system using coupled boundary condition. *J Energy Storage* 2021;41:102831. <https://doi.org/10.1016/j.est.2021.102831>.
- [105] Hosseinzadeh K, Montazer E, Shafii MB, Ganji ARD. Solidification enhancement in triplex thermal energy storage system via triplets fins configuration and hybrid nanoparticles. *J Energy Storage* 2021;34:102177. <https://doi.org/10.1016/j.est.2020.102177>.
- [106] Safari V, Abolghasemi H, Kamkari B. Experimental and numerical investigations of thermal performance enhancement in a latent heat storage heat exchanger using bifurcated and straight fins. *Renew Energy* 2021;174:102–21. <https://doi.org/10.1016/j.renene.2021.04.076>.
- [107] Pu L, Zhang S, Xu L, Li Y. Thermal performance optimization and evaluation of a radial finned shell-and-tube latent heat thermal energy storage unit. *Appl Therm Eng* 2020;166:114753. <https://doi.org/10.1016/j.applthermaleng.2019.114753>.
- [108] Sodhi GS, Muthukumar P. Compound charging and discharging enhancement in multi-PCM system using non-uniform fin distribution. *Renew Energy* 2021;171:299–314. <https://doi.org/10.1016/j.renene.2021.02.084>.
- [109] Kalapala L, Devanuri JK. Effect of orientation on thermal performance of a latent heat storage system equipped with annular fins – An experimental and numerical investigation. *Appl Therm Eng* 2021;183:116244. <https://doi.org/10.1016/j.applthermaleng.2020.116244>.
- [110] Zheng J, Wang J, Chen T, Yu Y. Solidification performance of heat exchanger with tree-shaped fins. *Renew Energy* 2020;150:1098–107. <https://doi.org/10.1016/j.renene.2019.10.091>.
- [111] Zhang T, Lu G, Zhai X. Design and experimental investigation of a novel thermal energy storage unit with phase change material. *Energy Rep* 2021;7:1818–27. <https://doi.org/10.1016/j.egyr.2021.03.029>.
- [112] Ding C, Wang L, Niu Z. Thermal performance evaluation of latent heat storage systems with plate fin-metal foam hybrid structure. *Case Stud Therm Eng* 2021;27:101309. <https://doi.org/10.1016/j.csite.2021.101309>.
- [113] Wołoszyn J, Szopa K, Czerwiński G. Enhanced heat transfer in a PCM shell-and-tube thermal energy storage system. *Appl Therm Eng* 2021;196:117332. <https://doi.org/10.1016/j.applthermaleng.2021.117332>.
- [114] Mehta DS, Vaghela B, Rathod MK, Banerjee J. Heat transfer enhancement using spiral fins in different orientations of Latent Heat Storage Unit. *Int J Therm Sci* 2021;169:107060. <https://doi.org/10.1016/j.ijthermalsci.2021.107060>.
- [115] Yao S, Huang X. Study on solidification performance of PCM by longitudinal triangular fins in a triplex-tube thermal energy storage system. *Energy* 2021;227:120527. <https://doi.org/10.1016/j.energy.2021.120527>.
- [116] Huang X, Yao S. Solidification performance of new trapezoidal longitudinal fins in latent heat thermal energy storage. *Case Stud Therm Eng* 2021;26:101110. <https://doi.org/10.1016/j.csite.2021.101110>.
- [117] Aly KA, El-Lathy AR, Fouad MA. Enhancement of solidification rate of latent heat thermal energy storage using corrugated fins. *J Energy Storage* 2019;24:100785. <https://doi.org/10.1016/j.est.2019.100785>.
- [118] Karami R, Kamkari B. Experimental investigation of the effect of perforated fins on thermal performance enhancement of vertical shell and tube latent heat energy storage systems. *Energy Convers Manag* 2020;210:112679. <https://doi.org/10.1016/j.enconman.2020.112679>.
- [119] Li H, Hu C, He Y, Tang D, Wang K, Huang W. Effect of perforated fins on the heat-transfer performance of vertical shell-and-tube latent heat energy storage unit. *J Energy Storage* 2021;39:102647. <https://doi.org/10.1016/j.est.2021.102647>.
- [120] Rozenfeld A, Kozak Y, Rozenfeld T, Ziskind G. Experimental demonstration, modeling and analysis of a novel latent-heat thermal energy storage unit with a helical fin. *Int J Heat Mass Transf* 2017;110:692–709. <https://doi.org/10.1016/j.ijheatmasstransfer.2017.03.020>.
- [121] Nakhchi ME, Esfahani JA. Improving the melting performance of PCM thermal energy storage with novel stepped fins. *J Energy Storage* 2020;30:101424. <https://doi.org/10.1016/j.est.2020.101424>.
- [122] Desai AN, Shah H, Singh VK. Novel inverted fin configurations for enhancing the thermal performance of PCM based thermal control unit: A numerical study. *Appl Therm Eng* 2021;195:117155. <https://doi.org/10.1016/j.applthermaleng.2021.117155>.
- [123] Nakhchi ME, Hatami M, Rahmati M. A numerical study on the effects of nanoparticles and stair fins on performance improvement of phase change thermal energy storages. *Energy* 2021;215:119112. <https://doi.org/10.1016/j.energy.2020.119112>.
- [124] Skaalum J, Groulx D. Heat transfer comparison between branching and non-branching fins in a latent heat energy storage system. *Int J Therm Sci* 2020;152:106331. <https://doi.org/10.1016/j.ijthermalsci.2020.106331>.
- [125] Abdulateef AM, Jaszczur M, Hassan Q, Anish R, Niyas H, Sopian K, et al. Enhancing the melting of phase change material using a fins-nanoparticle combination in a triplex tube heat exchanger. *J Energy Storage* 2021;35:102227. <https://doi.org/10.1016/j.est.2020.102227>.
- [126] Li F, Muhammad N, Abohamzeh E, Hakeem AKA, Hajizadeh MR, Li Z, et al. Fined unit solidification with use of nanoparticles improved PCM. *J Mol Liq* 2020;314:113659. <https://doi.org/10.1016/j.molliq.2020.113659>.
- [127] Yang X, Xu F, Wang X, Guo J, Li M-J. Solidification in a shell-and-tube thermal energy storage unit filled with longitude fins and metal foam: a numerical study. *Energy Built Environ* 2021. <https://doi.org/10.1016/j.enbenv.2021.08.002>.
- [128] Mahdavi A, Erfani Moghaddam MA, Mahmoudi A. Simultaneous charging and discharging of multi-tube heat storage systems using copper fins and Cu nanoparticles. *Case Stud Therm Eng* 2021;27:101343. <https://doi.org/10.1016/j.csite.2021.101343>.
- [129] Zhao C, Opolot M, Liu M, Bruno F, Mancin S, Hooman K. Numerical study of melting performance enhancement for PCM in an annular enclosure with internal external fins and metal foams. *Int J Heat Mass Transf* 2020;150:119348. <https://doi.org/10.1016/j.ijheatmasstransfer.2020.119348>.
- [130] Chen S-B, Saleem S, Alghamdi MN, Nisar KS, Arsalanloo A, Issakhov A, et al. Combined effect of using porous media and nano-particle on melting performance of PCM filled enclosure with triangular double fins. *Case Stud Therm Eng* 2021;25:100939. <https://doi.org/10.1016/j.csite.2021.100939>.
- [131] Li Z, Shahsavari A, Al-Rashed AAAA, Talebizadehsardari P. Effect of porous medium and nanoparticles presences in a counter-current triple-tube composite porous/nano-PCM system. *Appl Therm Eng* 2020;167:114777. <https://doi.org/10.1016/j.applthermaleng.2019.114777>.
- [132] Ren Q, Xu H, Luo Z. PCM charging process accelerated with combination of optimized triangle fins and nanoparticles. *Int J Therm Sci* 2019;140:466–79. <https://doi.org/10.1016/j.ijthermalsci.2019.03.005>.
- [133] ul Hasnain F, Irfan M, Khan MM, Khan LA, Ahmed HF. Melting performance enhancement of a phase change material using branched fins and nanoparticles for energy storage applications. *J Energy Storage* 2021;38:102513. <https://doi.org/10.1016/j.est.2021.102513>.
- [134] Huang Y, Han Q, Liu X. Experimental investigation on the melting and solidification performance enhancement of a fractal latent heat storage unit. *Int J Heat Mass Transf* 2021;179:121640. <https://doi.org/10.1016/j.ijheatmasstransfer.2021.121640>.
- [135] Yang X, Guo J, Yang B, Cheng H, Wei P, He Y-L. Design of non-uniformly distributed annular fins for a shell-and-tube thermal energy storage unit. *Appl Energy* 2020;279:115772. <https://doi.org/10.1016/j.apenergy.2020.115772>.
- [136] Liu S, Peng H, Hu Z, Ling X, Huang J. Solidification performance of a latent heat storage unit with innovative longitudinal triangular fins. *Int J Heat Mass Transf* 2019;138:667–76. <https://doi.org/10.1016/j.ijheatmasstransfer.2019.04.121>.
- [137] Kothari R, Sahu SK, Kundalwal SI, Sahoo SP. Experimental investigation of the effect of inclination angle on the performance of phase change material based finned heat sink. *J Energy Storage* 2021;37:102462. <https://doi.org/10.1016/j.est.2021.102462>.
- [138] Yao Y, Wu H. Thermal transport process of metal foam/paraffin composite (MFPC) with solid-liquid phase change: An experimental study. *Appl Therm Eng* 2020;179:115668. <https://doi.org/10.1016/j.applthermaleng.2020.115668>.
- [139] Zhang H, Li X, Liu L, Ma T, Wang Q, Liu J. Experimental investigation on paraffin melting in high porosity copper foam under centrifugal accelerations. *Appl Therm Eng* 2020;178:115504. <https://doi.org/10.1016/j.applthermaleng.2020.115504>.
- [140] Hu X, Gong X. Experimental study on the thermal response of PCM-based heat sink using structured porous material fabricated by 3D printing. *Case Stud Therm Eng* 2021;24:100844. <https://doi.org/10.1016/j.csite.2021.100844>.
- [141] Sabrina Ferfera R, Madani B, Serhane R. Investigation of heat transfer improvement at idealized microcellular scale for metal foam incorporated with paraffin. *Int J Therm Sci* 2020;156:106444. <https://doi.org/10.1016/j.ijthermalsci.2020.106444>.

- [142] Dinesh BVS, Bhattacharya A. Comparison of energy absorption characteristics of PCM-metal foam systems with different pore size distributions. *J Energy Storage* 2020;28:101190. <https://doi.org/10.1016/j.est.2019.101190>.
- [143] Hasan MI, Thana HL. Using of phase change materials to enhance the thermal performance of micro channel heat sink. *Eng Sci Technol an Int J* 2018;21: 517–26. <https://doi.org/10.1016/j.jestech.2018.03.017>.
- [144] Zhao Y, Zhang X, Hua W. Review of preparation technologies of organic composite phase change materials in energy storage. *J Mol Liq* 2021;336:115923. <https://doi.org/10.1016/j.molliq.2021.115923>.
- [145] Yu K, Liu Y, Yang Y. Review on form-stable inorganic hydrated salt phase change materials: Preparation, characterization and effect on the thermophysical properties. *Appl Energy* 2021;292:116845. <https://doi.org/10.1016/j.apenergy.2021.116845>.
- [146] Atinafu DG, Wi S, Yun BY, Kim S. Engineering biochar with multiwalled carbon nanotube for efficient phase change material encapsulation and thermal energy storage. *Energy* 2021;216:119294. <https://doi.org/10.1016/j.energy.2020.119294>.
- [147] Atinafu DG, Yeol Yun B, Uk Kim Y, Wi S, Kim S. Introduction of eicosane into biochar derived from softwood and wheat straw: Influence of porous structure and surface chemistry. *Chem Eng J* 2021;415:128887. <https://doi.org/10.1016/j.cej.2021.128887>.
- [148] Li C, Zhao X, Chen J. Diversiform microstructure silicon carbides stabilized stearic acid as composite phase change materials. *Sol Energy* 2020;201:92–101. <https://doi.org/10.1016/j.solener.2020.02.102>.
- [149] Chriaa I, Trigui A, Karkri M, Jedidi I, Abdelmouleh M, Boudaya C. Thermal properties of shape-stabilized phase change materials based on Low Density Polyethylene, Hexadecane and SEBS for thermal energy storage. *Appl Therm Eng* 2020;171:115072. <https://doi.org/10.1016/j.applthermaleng.2020.115072>.
- [150] Shen C, Li X, Yang G, Wang Y, Zhao L, Mao Z, et al. Shape-stabilized hydrated salt/paraffin composite phase change materials for advanced thermal energy storage and management. *Chem Eng J* 2020;385:123958. <https://doi.org/10.1016/j.cej.2019.123958>.
- [151] Xie N, Niu J, Zhong Y, Gao X, Zhang Z, Fang Y. Development of polyurethane acrylate coated salt hydrate/diatomite form-stable phase change material with enhanced thermal stability for building energy storage. *Constr Build Mater* 2020; 259:119714. <https://doi.org/10.1016/j.conbuildmat.2020.119714>.
- [152] Cárdenas-Ramírez C, Jaramillo F, Gómez M. Systematic review of encapsulation and shape-stabilization of phase change materials. *J Energy Storage* 2020;30: 101495. <https://doi.org/10.1016/j.est.2020.101495>.
- [153] Singh Rathore PK, Shukla SK, Gupta NK. Potential of microencapsulated PCM for energy savings in buildings: A critical review. *Sustain Cities Soc* 2020;53:101884. <https://doi.org/10.1016/j.scs.2019.101884>.
- [154] Nikpourian H, Bahramian AR, Abdollahi M. On the thermal performance of a novel PCM nanocapsule: The effect of core/shell. *Renew Energy* 2020;151: 322–31.
- [155] Zhang K, Wang J, Xu L, Xie H, Guo Z. Preparation and thermal characterization of n-octadecane/pentafluorostyrene nanocapsules for phase-change energy storage. *J Energy Storage* 2021;35:102327. <https://doi.org/10.1016/j.est.2021.102327>.
- [156] Pandey K, Ali SF, Gupta SK, Saikia P, Rakshit D, Saha S. Facile technique to encapsulate phase change material in an amphiphilic polymeric matrix for thermal energy storage. *Appl Energy* 2021;292:116917. <https://doi.org/10.1016/j.apenergy.2021.116917>.
- [157] Abdeali G, Abdollahi M, Bahramian AR. Synthesis and characterization of paraffin wax nanocapsules with polyurethane shell (PU/PW); the droplet size distribution: A key factor for thermal performance. *Renew Energy* 2021;163:720–31. <https://doi.org/10.1016/j.renene.2020.09.013>.
- [158] Tian Y, Liu Y, Zhang L, Hua Q, Liu L, Wang B, et al. Preparation and characterization of gelatin-sodium alginate/paraffin phase change microcapsules. *Colloids Surfaces A Physicochem Eng Asp* 2020;586:124216. <https://doi.org/10.1016/j.colsurfa.2019.124216>.
- [159] Yu X, Luan J, Chen W, Tao J. Preparation and characterization of paraffin microencapsulated phase change material with double shell for thermal energy storage. *Thermochim Acta* 2020;689:178652. <https://doi.org/10.1016/j.tca.2020.178652>.
- [160] Praveen B, Suresh S, Pethurajan V. Heat transfer performance of graphene nanoplatelets laden micro-encapsulated PCM with polymer shell for thermal energy storage based heat sink. *Appl Therm Eng* 2019;156:237–49.
- [161] Rakkapann SR, Sivan S, Pethurajan V, Aditya A, Mittal H. Preparation and thermal properties of encapsulated 1-Decanol for low-temperature heat transfer fluid application. *Colloids Surfaces A Physicochem Eng Asp* 2021;614:126167. <https://doi.org/10.1016/j.colsurfa.2021.126167>.
- [162] Bergamo J, Rossi E, Maffi JM, De Angelis L, Errea MI. Preparation and characterization of nanoencapsulated synthetic soybean oil derivative-an abundant and environmentally friendly phase change material-heat transfer analysis and applications. *Sustain Energy Technol Assessments* 2020;41:100794. <https://doi.org/10.1016/j.seta.2020.100794>.
- [163] Wang X, Djakovic U, Bao H, Torres JF. Experimental evaluation of heat transfer performance under natural and forced convection around a phase change material encapsulated in various shapes. *Sustain Energy Technol Assessments* 2021;44:101025. <https://doi.org/10.1016/j.seta.2021.101025>.
- [164] Mohaghegh MR, Alomair Y, Alomair M, Tasnim SH, Mahmud S, Abdullah H. Melting of PCM inside a novel encapsulation design for thermal energy storage system. *Energy Convers Manag* 2021;11:100098. <https://doi.org/10.1016/j.ecmx.2021.100098>.
- [165] Ishak S, Mandal S, Lee H-S, Singh JK. Microencapsulation of stearic acid with SiO₂ shell as phase change material for potential energy storage. *Sci Rep* 2020;10: 15023. <https://doi.org/10.1038/s41598-020-71940-9>.
- [166] Chen W, Liang X, Wang S, Gao X, Zhang Z, Fang Y. Macro-encapsulated 3D phase change material: Na₂S₂O₃·5H₂O-NaOAc·3H₂O/carbonized Melamine sponge composite as core and SiC modified polyurethane thin-layer as shell. *Compos Sci Technol* 2021;214:108981. <https://doi.org/10.1016/j.compscitech.2021.108981>.
- [167] Shi X, Yazdani MR, Ajdary R, Rojas OJ. Leakage-proof microencapsulation of phase change materials by emulsification with acetylated cellulose nanofibrils. *Carbohydr Polym* 2021;254:117279. <https://doi.org/10.1016/j.carbpol.2020.117279>.
- [168] Xing J, Zhou Y, Yang K, Chang J, Yu Y, Cai L, et al. Microencapsulation of fatty acid eutectic with polyvinyl chloride shell used for thermal energy storage. *J Energy Storage* 2021;34:101998. <https://doi.org/10.1016/j.est.2020.101998>.
- [169] Cai C, Ouyang X, Zhou L, Liu G, Wang Y, Zhu G, et al. Co-solvent free interfacial polycondensation and properties of polyurea PCM microcapsules with dodecanoate dodecanoate as core material. *Sol Energy* 2020;199:721–30. <https://doi.org/10.1016/j.solener.2020.02.071>.
- [170] Sari A, Saleh TA, Hekimoğlu G, Tyagi VV, Sharma RK. Microencapsulated heptadecane with calcium carbonate as thermal conductivity-enhanced phase change material for thermal energy storage. *J Mol Liq* 2021;328:115508. <https://doi.org/10.1016/j.molliq.2021.115508>.
- [171] Hajizadeh MR, Selimefendilif F, Muhammad T, Ramzan M, Babazadeh H, Li Z. Solidification of PCM with nano powders inside a heat exchanger. *J Mol Liq* 2020; 306:112892. <https://doi.org/10.1016/j.molliq.2020.112892>.
- [172] Yadav C, Rekha Sahoo R. Experimental analysis for optimum thermal performance and thermophysical parameters of MWCNT based capric acid PCM by using T-history method. *Powder Technol* 2020;364:392–403. <https://doi.org/10.1016/j.powtec.2020.02.008>.
- [173] Wang J, Yu K, Duan R, Xie G, Sundén B. Enhanced thermal management by introducing nanoparticle composite phase change materials for cooling multiple heat sources systems. *Energy* 2021;227:120495. <https://doi.org/10.1016/j.energy.2021.120495>.
- [174] Soni V, Kumar A, Jain VK. Fast waste heat recovery in 100–150 °C using close-contact charging of nano-enhanced PCM composite. *J Mol Liq* 2019;285:347–61. <https://doi.org/10.1016/j.molliq.2019.03.132>.
- [175] Ali MA, Fayaz, Viegas RF, Shyam Kumar MB, Kannapiran RK, Feroskhan M. Enhancement of heat transfer in paraffin wax PCM using nano graphene composite for industrial helmets. *J Energy Storage* 2019;26:100982. <https://doi.org/10.1016/j.est.2019.100982>.
- [176] Habib NA, Ali AJ, Chaichan MT, Kareem M. Carbon nanotubes/paraffin wax nanocomposite for improving the performance of a solar air heating system. *Therm Sci Eng Prog* 2021;23:100877. <https://doi.org/10.1016/j.tsep.2021.100877>.
- [177] Xu C, Xu S, Eticha RD. Experimental investigation of thermal performance for pulsating flow in a microchannel heat sink filled with PCM (paraffin/CNT composite). *Energy Convers Manag* 2021;236:114071. <https://doi.org/10.1016/j.enconman.2021.114071>.
- [178] Parida A, Bhattacharya A, Rath P. Effect of convection on melting characteristics of phase change material-metal foam composite thermal energy storage system. *J Energy Storage* 2020;32:101804. <https://doi.org/10.1016/j.est.2020.101804>.
- [179] Meng X, Liu S, Zou J, Liu F, Wang J. Inclination angles on the thermal behavior of Phase-Change Material (PCM) in a cavity filled with copper foam partly. *Case Stud Therm Eng* 2021;25:100944. <https://doi.org/10.1016/j.csite.2021.100944>.
- [180] Kok B. Examining effects of special heat transfer fins designed for the melting process of PCM and Nano-PCM. *Appl Therm Eng* 2020;170:114989. <https://doi.org/10.1016/j.applthermaleng.2020.114989>.
- [181] Arshad A, Jabbar M, Shi L, Darkwa J, Weston NJ, Yan Y. Development of TiO₂/RT-35HC based nanocomposite phase change materials (NCPs) for thermal management applications. *Sustain Energy Technol Assessments* 2021;43:100865. <https://doi.org/10.1016/j.seta.2020.100865>.
- [182] Jebasingh EB, Arasu VA. A comprehensive review on latent heat and thermal conductivity of nanoparticle dispersed phase change material for low-temperature applications. *Energy Storage Mater* 2020;24:52–74.
- [183] Ali S, Deshmukh SP. An overview: Applications of thermal energy storage using phase change materials. *Mater Today Proc* 2020;26:1231–7. <https://doi.org/10.1016/j.matpr.2020.02.247>.
- [184] Yang L, Huang J, Zhou F. Thermophysical properties and applications of nano-enhanced PCMs: An update review. *Energy Convers Manag* 2020;214:112876. <https://doi.org/10.1016/j.enconman.2020.112876>.
- [185] Ben Romdhane S, Amamou A, Ben Khalifa R, Said NM, Younsi Z, Jemni A. A review on thermal energy storage using phase change materials in passive building applications. *J Build Eng* 2020;32:101563. <https://doi.org/10.1016/j.jobbe.2020.101563>.
- [186] Kalapala L, Devanuri JK. Influence of operational and design parameters on the performance of a PCM based heat exchanger for thermal energy storage – A review. *J Energy Storage* 2018;20:497–519. <https://doi.org/10.1016/j.est.2018.10.024>.
- [187] Katekar VP, Deshmukh SS. A review of the use of phase change materials on performance of solar stills. *J Energy Storage* 2020;30:101398. <https://doi.org/10.1016/j.est.2020.101398>.
- [188] Souza I da C, Morozeski M, Mansano AS, Mendes VAS, Azevedo VC, Matsumoto ST, et al. Atmospheric particulate matter from an industrial area as a source of metal nanoparticle contamination in aquatic ecosystems. *Sci Total Environ* 2021; 753:141976. <https://doi.org/10.1016/j.scitotenv.2020.141976>.

- [189] Kolarova N, Napiórkowski P. Trace elements in aquatic environment. Origin, distribution, assessment and toxicity effect for the aquatic biota. *Ecohydrol Hydrobiol* 2021. <https://doi.org/10.1016/j.ecohyd.2021.02.002>.
- [190] Richir J, Bray S, McAleese T, Watson GJ. Three decades of trace element sediment contamination: The mining of governmental databases and the need to address hidden sources for clean and healthy seas. *Environ Int* 2021;149:106362. <https://doi.org/10.1016/j.envint.2020.106362>.
- [191] Anand B, Shankar R, Murugavel S, Rivera W, Midhun Prasad K, Nagarajan R. A review on solar photovoltaic thermal integrated desalination technologies. *Renew Sustain Energy Rev* 2021;141:110787. <https://doi.org/10.1016/j.rser.2021.110787>.
- [192] Negi P, Dobriyal R, Singh DB, Badhotiya GK. A review on passive and active solar still using phase change materials. *Mater Today Proc* 2021. <https://doi.org/10.1016/j.matpr.2020.12.996>.
- [193] Manoj Kumar P, Sudarvizhi D, Prakash KB, Anupradeepa AM, Boomiha Raj S, Shanmathi S, et al. Investigating a single slope solar still with a nano-phase change material. *Mater Today Proc* 2021. <https://doi.org/10.1016/j.matpr.2020.12.804>.
- [194] Jahanpanah M, Sadatnejad SJ, Kasaiean A, Jahangir MH, Sarrafha H. Experimental investigation of the effects of low-temperature phase change material on single-slope solar still. *Desalination* 2021;499:114799. <https://doi.org/10.1016/j.desal.2020.114799>.
- [195] Faegh M, Shafii MB. Experimental investigation of a solar still equipped with an external heat storage system using phase change materials and heat pipes. *Desalination* 2017;409:128–35.
- [196] Al-harashsheh M, Abu-Arabi M, Mousa H, Alzghoul Z. Solar desalination using solar still enhanced by external solar collector and PCM. *Appl Therm Eng* 2018;128:1030–40. <https://doi.org/10.1016/j.applthermaleng.2017.09.073>.
- [197] Abu-Arabi M, Al-harashsheh M, Ahmad M, Mousa H. Theoretical modeling of a glass-cooled solar still incorporating PCM and coupled to flat plate solar collector. *J Energy Storage* 2020;29:101372. <https://doi.org/10.1016/j.est.2020.101372>.
- [198] Kateshia J, Lakhera VJ. Analysis of solar still integrated with phase change material and pin fins as absorbing material. *J Energy Storage* 2021;35:102292. <https://doi.org/10.1016/j.est.2021.102292>.
- [199] Vigneswaran VS, Kumaresan G, Dinakar BV, Kamal KK, Velraj R. Augmenting the productivity of solar still using multiple PCMs as heat energy storage. *J Energy Storage* 2019;26:101019. <https://doi.org/10.1016/j.est.2019.101019>.
- [200] Kabeel AE, Abdelgaied M, Harby K, Eisa A. Augmentation of diurnal and nocturnal distillate of modified tubular solar still having copper tubes filled with PCM in the basin. *J Energy Storage* 2020;32:101992. <https://doi.org/10.1016/j.est.2020.101992>.
- [201] Mousa H, Naser J, Gujarathi AM, Al-Sawafi S. Experimental study and analysis of solar still desalination using phase change materials. *J Energy Storage* 2019;26:100959. <https://doi.org/10.1016/j.est.2019.100959>.
- [202] Elashmawy M, Alhadri M, Ahmed MMZ. Enhancing tubular solar still performance using novel PCM-tubes. *Desalination* 2021;500:114880. <https://doi.org/10.1016/j.desal.2020.114880>.
- [203] Sarhaddi F, Farshchi Tabrizi F, Aghaei Zoori H, Mousavi SAHS. Comparative study of two weir type cascade solar stills with and without PCM storage using energy and exergy analysis. *Energy Convers Manag* 2017;133:97–109. <https://doi.org/10.1016/j.enconman.2016.11.044>.
- [204] Kabeel AE, Harby K, Abdelgaied M, Eisa A. Augmentation of a developed tubular solar still productivity using hybrid storage medium and CPC: An experimental approach. *J Energy Storage* 2020;28:101203. <https://doi.org/10.1016/j.est.2020.101203>.
- [205] Kabeel AE, El-Maghlany WM, Abdelgaied M, Abdel-Aziz MM. Performance enhancement of pyramid-shaped solar stills using hollow circular fins and phase change materials. *J Energy Storage* 2020;31:101610. <https://doi.org/10.1016/j.est.2020.101610>.
- [206] Tuly SS, Rahman MS, Sarker MRI, Beg RA. Combined influence of fin, phase change material, wick, and external condenser on the thermal performance of a double slope solar still. *J Clean Prod* 2021;287:125458. <https://doi.org/10.1016/j.jclepro.2020.125458>.
- [207] Khairat Dawood MM, Nabil T, Kabeel AE, Shehata AI, Abdalla AM, Elnaghi BE. Experimental study of productivity progress for a solar still integrated with parabolic trough collectors with a phase change material in the receiver evacuated tubes and in the still. *J Energy Storage* 2020;32:102007. <https://doi.org/10.1016/j.est.2020.102007>.
- [208] Chamkha AJ, Rufuss DDW, Kabeel AE, Sathyamurthy R, Abdelgaid M, Manokar AM, et al. Augmenting the potable water produced from single slope solar still using CNT-doped paraffin wax as energy storage: an experimental approach. *J Braz Soc Mech Sci Eng* 2020;42:625. <https://doi.org/10.1007/s40430-020-02703-w>.
- [209] Mazraeh AE, Babayan M, Yari M, Sefidan AM, Saha SC. Theoretical study on the performance of a solar still system integrated with PCM-PV module for sustainable water and power generation. *Desalination* 2018;443:184–97. <https://doi.org/10.1016/j.desal.2018.05.024>.
- [210] Shalaby SM, El-Bialy E, El-Sebaei AA. An experimental investigation of a v-corrugated absorber single-basin solar still using PCM. *Desalination* 2016;398:247–55. <https://doi.org/10.1016/j.desal.2016.07.042>.
- [211] Cheng W-L, Huo Y-K, Nian Y-L. Performance of solar still using shape-stabilized PCM: Experimental and theoretical investigation. *Desalination* 2019;455:89–99. <https://doi.org/10.1016/j.desal.2019.01.007>.
- [212] Yousef MS, Hassan H, Kodama S, Sekiguchi H. An experimental study on the performance of single slope solar still integrated with a PCM-based pin-finned heat sink. *Energy Procedia* 2019;156:100–4. <https://doi.org/10.1016/j.egypro.2018.11.102>.
- [213] Wan H, Cao T, Hwang Y, Andersen SO, Chin S. A comprehensive review of life cycle climate performance (LCCP) for air conditioning systems. *Int J Refrig* 2021. <https://doi.org/10.1016/j.ijrefrig.2021.06.026>.
- [214] Xu L, Dai L, Yin L, Sun X, Xu W, Yang R, et al. Research on the climate response of variable thermo-physical property building envelopes: A literature review. *Energy Build* 2020;226:110398. <https://doi.org/10.1016/j.enbuild.2020.110398>.
- [215] She X, Cong L, Nie B, Leng G, Peng H, Chen Y, et al. Energy-efficient and -economic technologies for air conditioning with vapor compression refrigeration: A comprehensive review. *Appl Energy* 2018;232:157–86. <https://doi.org/10.1016/j.apenergy.2018.09.067>.
- [216] Berardi U. A cross-country comparison of the building energy consumptions and their trends. *Resour Conserv Recycl* 2017;123:230–41. <https://doi.org/10.1016/j.resconrec.2016.03.014>.
- [217] Zhang Z, Zhang P, Zhao Y, Chen X, Zong Z, Wu K, et al. Survey-based air-conditioning demand response for critical peak reduction considering residential consumption behaviors. *Energy Rep* 2020;6:3303–15. <https://doi.org/10.1016/j.egypr.2020.11.242>.
- [218] Mohseni E, Tang W. Parametric analysis and optimisation of energy efficiency of a lightweight building integrated with different configurations and types of PCM. *Renew Energy* 2021;168:865–77. <https://doi.org/10.1016/j.renene.2020.12.112>.
- [219] Borri E, Zsembinszki G, Cabeza LF. Recent developments of thermal energy storage applications in the built environment: A bibliometric analysis and systematic review. *Appl Therm Eng* 2021;189:116666. <https://doi.org/10.1016/j.applthermaleng.2021.116666>.
- [220] Promopattum P, Yao S-C, Hult T, Agee D. Experimental and numerical investigation of the cross-flow PCM heat exchanger for the energy saving of building HVAC. *Energy Build* 2017;138:468–78. <https://doi.org/10.1016/j.enbuild.2016.12.043>.
- [221] Maccarini A, Hultmark G, Bergsøe NC, Afshari A. Free cooling potential of a PCM-based heat exchanger coupled with a novel HVAC system for simultaneous heating and cooling of buildings. *Sustain Cities Soc* 2018;42:384–95. <https://doi.org/10.1016/j.scs.2018.06.016>.
- [222] Wu W, Dai S, Liu Z, Dou Y, Hua J, Li M, et al. Experimental study on the performance of a novel solar water heating system with and without PCM. *Sol Energy* 2018;171:604–12. <https://doi.org/10.1016/j.solener.2018.07.005>.
- [223] Bouhal T, El Rhafiki T, Kousksou T, Jamil A, Zeraoui Y. PCM addition inside solar water heaters: Numerical comparative approach. *J Energy Storage* 2018;19:232–46. <https://doi.org/10.1016/j.est.2018.08.005>.
- [224] Wu C, Wei Z, Yin H. Virtual and physical experiments of encapsulated phase change material embedded in building envelopes. *Int J Heat Mass Transf* 2021;172:121083. <https://doi.org/10.1016/j.jheatmasstransfer.2021.121083>.
- [225] Amin M, Putra N, Kosasih EA, Prawiro E, Lianto RA, Mahlia TMI. Thermal properties of beeswax/graphene phase change materials as energy storage for building applications. *Appl Therm Eng* 2017;112:273–80.
- [226] Rathore PKS, Shukla SK. Enhanced thermophysical properties of organic PCM through shape stabilization for thermal energy storage in buildings: A state of the art review. *Energy Build* 2021;236:110799. <https://doi.org/10.1016/j.enbuild.2021.110799>.
- [227] Sarrafha H, Kasaiean A, Jahangir MH, Taylor RA. Transient thermal response of multi-walled carbon nanotube phase change materials in building walls. *Energy* 2021;120120.
- [228] Ma Z, Lin W, Sohel MI. Nano-enhanced phase change materials for improved building performance. *Renew Sustain Energy Rev* 2016;58:1256–68.
- [229] Keshteli AN, Sheikholeslami M. Nanoparticle enhanced PCM applications for intensification of thermal performance in building: a review. *J Mol Liq* 2019;274:516–33.
- [230] Bouhal T, Meghari Z, Fertahi S ed-D, El Rhafiki T, Kousksou T, Jamil A, et al. Parametric CFD analysis and impact of PCM intrinsic parameters on melting process inside enclosure integrating fins: Solar building applications. *J Build Eng* 2018;20:634–46. <https://doi.org/10.1016/j.jobe.2018.09.016>.
- [231] Sardari PT, Grant D, Giddings D, Walker GS, Gillott M. Composite metal foam/PCM energy store design for dwelling space air heating. *Energy Convers Manag* 2019;201:112151. <https://doi.org/10.1016/j.enconman.2019.112151>.
- [232] Sheikholeslami M, Jafaryar M, Shafee A, Babazadeh H. Acceleration of discharge process of clean energy storage unit with insertion of porous foam considering nanoparticle enhanced paraffin. *J Clean Prod* 2020;261:121206. <https://doi.org/10.1016/j.jclepro.2020.121206>.
- [233] Li F, Jafaryar M, Hajizadeh MR, Bach Q-V. Performance of ventilation system involving thermal storage unit considering porous media. *J Energy Storage* 2020;31:101709. <https://doi.org/10.1016/j.est.2020.101709>.
- [234] Ye R, Lin W, Yuan K, Fang X, Zhang Z. Experimental and numerical investigations on the thermal performance of building plane containing CaCl₂·6H₂O/expanded graphite composite phase change material. *Appl Energy* 2017;193:325–35.
- [235] Ye R, Zhang C, Sun W, Fang X, Zhang Z. Novel wall panels containing CaCl₂·6H₂O-Mg (NO₃)₂·6H₂O/expanded graphite composites with different phase change temperatures for building energy savings. *Energy Build* 2018;176:407–17.
- [236] Fang Y, Ding Y, Tang Y, Liang X, Jin C, Wang S, et al. Thermal properties enhancement and application of a novel sodium acetate trihydrate-formamide/expanded graphite shape-stabilized composite phase change material for electric radiant floor heating. *Appl Therm Eng* 2019;150:1177–85.
- [237] Liu L, Chen J, Qu Y, Xu T, Wu H, Huang G, et al. A foamed cement blocks with paraffin/expanded graphite composite phase change solar thermal absorption material. *Sol Energy Mater Sol Cells* 2019;200:110038.

- [238] Fraç M, Pichór W, Szoldra P, Szudek W. Cement composites with expanded graphite/paraffin as storage heater. *Constr Build Mater* 2021;275:122126.
- [239] Konuklu Y, Ostry M, Paksoy HO, Charvat P. Review on using microencapsulated phase change materials (PCM) in building applications. *Energy Build* 2015;106:134–55.
- [240] Wijesuriya S, Tabares-Velasco PC. Empirical validation and comparison of methodologies to simulate micro and macro-encapsulated PCMs in the building envelope. *Appl Therm Eng* 2021;116646.
- [241] Soares N, Rosa N, Costa JJ, Lopes AG, Matias T, Simões PN, et al. Validation of different numerical models with benchmark experiments for modelling microencapsulated-PCM-based applications for buildings. *Int J Therm Sci* 2021;159:106565.
- [242] Maleki B, Khadang A, Maddah H, Alizadeh M, Kazemian A, Ali HM. Development and thermal performance of nanoencapsulated PCM/ plaster wallboard for thermal energy storage in buildings. *J Build Eng* 2020;32:101727. <https://doi.org/10.1016/j.jobe.2020.101727>.
- [243] Srinivasaraonai B, Singh LP, Sinha S, Tyagi I, Rawat A. Studies on the mechanical properties and thermal behavior of microencapsulated eutectic mixture in gypsum composite board for thermal regulation in the buildings. *J Build Eng* 2020;31:101400.
- [244] Sari A, Alkan C, Özcan AN. Synthesis and characterization of micro/nano capsules of PMMA/capric-stearic acid eutectic mixture for low temperature-thermal energy storage in buildings. *Energy Build* 2015;90:106–13.
- [245] Philip N, Veerakumar C, Sreekumar A. Lauryl alcohol and stearyl alcohol eutectic for cold thermal energy storage in buildings: Preparation, thermophysical studies and performance analysis. *J Energy Storage* 2020;31:101600.
- [246] Döğüşci DK, Altıntaş A, Sari A, Alkan C. Polystyrene microcapsules with palmitic-capric acid eutectic mixture as building thermal energy storage materials. *Energy Build* 2017;150:376–82.
- [247] Srinivasaraonai B, Sinha S, Singh LP. Studies on microstructural and thermo-physico properties of microencapsulated eutectic phase change material incorporated pure cement system. *J Energy Storage* 2021;35:102318.
- [248] Sarri A, Bechki D, Bouguettaia H, Al-Saadi SN, Boughali S, Farid MM. Effect of using PCMs and shading devices on the thermal performance of buildings in different Algerian climates. A simulation-based optimization. *Sol Energy* 2021;217:375–89. <https://doi.org/10.1016/j.solener.2021.02.024>.
- [249] Kishore RA, Bianchi MVA, Booten C, Vidal J, Jackson R. Parametric and sensitivity analysis of a PCM-integrated wall for optimal thermal load modulation in lightweight buildings. *Appl Therm Eng* 2021;187:116568. <https://doi.org/10.1016/j.applthermaleng.2021.116568>.
- [250] Li C, Yu H, Song Y, Tang Y, Chen P, Hu H, et al. Experimental thermal performance of wallboard with hybrid microencapsulated phase change materials for building application. *J Build Eng* 2020;28:101051. <https://doi.org/10.1016/j.jobe.2019.101051>.
- [251] Boussaba L, Lefebvre G, Makhlof S, Grados A, Royon L. Investigation and properties of a novel composite bio-PCM to reduce summer energy consumptions in buildings of hot and dry climates. *Sol Energy* 2021;214:119–30. <https://doi.org/10.1016/j.solener.2020.11.060>.
- [252] Li C, Zhang B, Xie B, Zhao X, Chen J. Tailored phase change behavior of Na₂SO₄·10H₂O/expanded graphite composite for thermal energy storage. *Energy Convers Manag* 2020;208:112586. <https://doi.org/10.1016/j.enconman.2020.112586>.
- [253] Wi S, Yang S, Park JH, Chang SJ, Kim S. Climatic cycling assessment of red clay/perlite and vermiculite composite PCM for improving thermal inertia in buildings. *Build Environ* 2020;167:106464. <https://doi.org/10.1016/j.buildenv.2019.106464>.
- [254] Hekimoğlu G, Nas M, Ouikhalfan M, Sari A, Tyagi VV, Sharma RK, et al. Silica fume/capric acid-stearic acid PCM included-cementitious composite for thermal controlling of buildings: Thermal energy storage and mechanical properties. *Energy* 2021;219:119588. <https://doi.org/10.1016/j.energy.2020.119588>.
- [255] Yu H, Li C, Zhang K, Tang Y, Song Y, Wang M. Preparation and thermophysical performance of diatomite-based composite PCM wallboard for thermal energy storage in buildings. *J Build Eng* 2020;32:101753. <https://doi.org/10.1016/j.jobe.2020.101753>.
- [256] Kong X, Wang L, Li H, Yuan G, Yao C. Experimental study on a novel hybrid system of active composite PCM wall and solar thermal system for clean heating supply in winter. *Sol Energy* 2020;195:259–70. <https://doi.org/10.1016/j.solener.2019.11.081>.
- [257] Alzoubi HH, Albiss BA, Abu siri SS. Performance of cementitious composites with nano PCMs and cellulose nano fibers. *Constr Build Mater* 2020;236:117483. <https://doi.org/10.1016/j.conbuildmat.2019.117483>.
- [258] Rakkappan SR, Sivan S, Ahmed SN, Naarendharan M, Sai Sudhir P. Preparation, characterisation and energy storage performance study on 1-Decanol-Expanded graphite composite PCM for air-conditioning cold storage system. *Int J Refrig* 2021;123:91–101. <https://doi.org/10.1016/j.jrefrig.2020.11.004>.
- [259] Bondareva NS, Sheikholeslami M, Sheremet MA. The influence of external temperature and convective heat exchange with an environment on heat transfer inside phase change material embedded brick. *J Energy Storage* 2021;33:102087. <https://doi.org/10.1016/j.est.2020.102087>.
- [260] Al-Yasiri Q, Szabó M. Experimental evaluation of the optimal position of a macroencapsulated phase change material incorporated composite roof under hot climate conditions. *Sustain Energy Technol Assessments* 2021;45:101121. <https://doi.org/10.1016/j.seta.2021.101121>.
- [261] Triano-Juárez J, Macías-Melo EV, Hernández-Pérez I, Aguilar-Castro KM, Xamán J. Thermal behavior of a phase change material in a building roof with and without reflective coating in a warm humid zone. *J Build Eng* 2020;32:101648. <https://doi.org/10.1016/j.jobe.2020.101648>.
- [262] Hu Y, Guo R, Heiselberg PK. Performance and control strategy development of a PCM enhanced ventilated window system by a combined experimental and numerical study. *Renew Energy* 2020;155:134–52. <https://doi.org/10.1016/j.renene.2020.03.137>.
- [263] Al-Rashed AAAA, Alnaqi AA, Alsarraf J. Energy-saving of building envelope using passive PCM technique: A case study of Kuwait City climate conditions. *Sustain Energy Technol Assessments* 2021;46:101254. <https://doi.org/10.1016/j.seta.2021.101254>.
- [264] Tafakkori R, Fattahi A. Introducing novel configurations for double-glazed windows with lower energy loss. *Sustain Energy Technol Assessments* 2021;43:100919. <https://doi.org/10.1016/j.seta.2020.100919>.
- [265] Lin Y, Zhong S, Yang W, Hao X, Li C-Q. Multi-objective design optimization on building integrated photovoltaic with Trombe wall and phase change material based on life cycle cost and thermal comfort. *Sustain Energy Technol Assessments* 2021;46:101277. <https://doi.org/10.1016/j.seta.2021.101277>.
- [266] Abu-Hamdeh NH, Melaibari AA, Alquthami TS, Khoshaim A, Oztot HF, Karimipour A. Efficacy of incorporating PCM into the building envelope on the energy saving and AHU power usage in winter. *Sustain Energy Technol Assessments* 2021;43:100969. <https://doi.org/10.1016/j.seta.2020.100969>.
- [267] Tyagi VV, Chopra K, Kalidasan B, Chauhan A, Strithi U, Anand S, et al. Phase change material based advance solar thermal energy storage systems for building heating and cooling applications: A prospective research approach. *Sustain Energy Technol Assessments* 2021;47:101318. <https://doi.org/10.1016/j.seta.2021.101318>.
- [268] Jiang W, Liu B, Zhang X, Zhang T, Li D, Ma L. Energy performance of window with PCM frame. *Sustain Energy Technol Assessments* 2021;45:101109. <https://doi.org/10.1016/j.seta.2021.101109>.
- [269] Wen R, Liu Y, Yang C, Zhu X, Huang Z, Zhang X, et al. Enhanced thermal properties of stearic acid/carbonized maize straw composite phase change material for thermal energy storage in buildings. *J Energy Storage* 2021;36:102420. <https://doi.org/10.1016/j.est.2021.102420>.
- [270] Mahdaoui M, Hamdaoui S, Ait Msaad A, Kouksou T, El Rhafiki T, Jamil A, et al. Building bricks with phase change material (PCM): Thermal performances. *Constr Build Mater* 2021;269:121315. <https://doi.org/10.1016/j.conbuildmat.2020.121315>.
- [271] Bhamare DK, Rathod MK, Banerjee J. Proposal of a unique index for selection of optimum phase change material for effective thermal performance of a building envelope. *Sol Energy* 2021;218:129–41. <https://doi.org/10.1016/j.solener.2021.02.027>.
- [272] Kishore RA, Bianchi MVA, Booten C, Vidal J, Jackson R. Optimizing PCM-integrated walls for potential energy savings in U.S. Buildings. *Energy Build* 2020;226:110355. <https://doi.org/10.1016/j.enbuild.2020.110355>.
- [273] Gholamibozanjani G, Farid M. Application of an active PCM storage system into a building for heating/cooling load reduction. *Energy* 2020;210:118572. <https://doi.org/10.1016/j.energy.2020.118572>.
- [274] Yu J, Yang Q, Ye H, Luo Y, Huang J, Xu X, et al. Thermal performance evaluation and optimal design of building roof with outer-layer shape-stabilized PCM. *Renew Energy* 2020;145:2538–49. <https://doi.org/10.1016/j.renene.2019.08.026>.
- [275] Ručevskis S, Akishin P, Korjakins A. Parametric analysis and design optimisation of PCM thermal energy storage system for space cooling of buildings. *Energy Build* 2020;224:110288. <https://doi.org/10.1016/j.enbuild.2020.110288>.
- [276] Sun W, Zhang Y, Ling Z, Fang X, Zhang Z. Experimental investigation on the thermal performance of double-layer PCM radiant floor system containing two types of inorganic composite PCMs. *Energy Build* 2020;211:109806. <https://doi.org/10.1016/j.enbuild.2020.109806>.
- [277] Mao Q, Yang M. Study on heat transfer performance of a solar double-slope PCM glazed roof with different physical parameters. *Energy Build* 2020;223:110141. <https://doi.org/10.1016/j.enbuild.2020.110141>.
- [278] Wang Z, Qiao Y, Liu Y, Bao J, Gao Q, Chen J, et al. Thermal storage performance of building envelopes for nearly-zero energy buildings during cooling season in Western China: An experimental study. *Build Environ* 2021;194:107709. <https://doi.org/10.1016/j.buildenv.2021.107709>.
- [279] Pirasaci T. Investigation of phase state and heat storage form of the phase change material (PCM) layer integrated into the exterior walls of the residential-apartment during heating season. *Energy* 2020;207:118176. <https://doi.org/10.1016/j.energy.2020.118176>.
- [280] Al-Yasiri Q, Szabó M. Case study on the optimal thickness of phase change material incorporated composite roof under hot climate conditions. *Case Stud Constr Mater* 2021;14:e00522. <https://doi.org/10.1016/j.cscm.2021.e00522>.
- [281] Luo Z, Xu H, Lou Q, Feng L, Ni J. GPU-accelerated lattice Boltzmann simulation of heat transfer characteristics of porous brick roof filled with phase change materials. *Int Commun Heat Mass Transf* 2020;119:104911. <https://doi.org/10.1016/j.icheatmasstransfer.2020.104911>.
- [282] Xamán J, Rodríguez-Ake A, Zavala-Guillén I, Hernández-Pérez I, Arce J, Saucedo D. Thermal performance analysis of a roof with a PCM-layer under Mexican weather conditions. *Renew Energy* 2020;149:773–85. <https://doi.org/10.1016/j.renene.2019.12.084>.
- [283] Sun W, Huang R, Ling Z, Fang X, Zhang Z. Numerical simulation on the thermal performance of a PCM-containing ventilation system with a continuous change in inlet air temperature. *Renew Energy* 2020;145:1608–19. <https://doi.org/10.1016/j.renene.2019.07.089>.
- [284] Drissi S, Ling T-C, Mo KH. Thermal performance of a solar energy storage concrete panel incorporating phase change material aggregates developed for

- thermal regulation in buildings. *Renew Energy* 2020;160:817–29. <https://doi.org/10.1016/j.renene.2020.06.076>.
- [285] Liu L, Li J, Deng Y, Yang Z, Huang K, Zhao S. Optimal design of multi-layer structure composite containing inorganic hydrated salt phase change materials and cement: Lab-scale tests for buildings. *Constr Build Mater* 2021;275:122125. <https://doi.org/10.1016/j.conbuildmat.2020.122125>.
- [286] Amaral C, Silva T, Mohseni F, Amaral JS, Amaral VS, Marques PAAP, et al. Experimental and numerical analysis of the thermal performance of polyurethane foams panels incorporating phase change material. *Energy* 2021;216:119213. <https://doi.org/10.1016/j.energy.2020.119213>.
- [287] Cai R, Sun Z, Yu H, Meng E, Wang J, Dai M. Review on optimization of phase change parameters in phase change material building envelopes. *J Build Eng* 2021;35:101979. <https://doi.org/10.1016/j.jobe.2020.101979>.
- [288] Sevault A, Vullum-Bruer F, Tranås OL. Active PCM-based thermal energy storage in buildings. *Ref. Modul. Earth Syst. Environ. Sci., Elsevier*; 2020. 10.1016/B978-0-12-819723-3.00008-1.
- [289] Bolteya AM, Elsayad MA, Belal AM. Thermal efficiency of PCM filled double glazing units in Egypt. *Ain Shams Eng J* 2021;12:1523–34. <https://doi.org/10.1016/j.asej.2020.12.004>.
- [290] Wierzkowicz A, Heim D. Modelling of thermal processes in a glazing structure with temperature dependent optical properties - An example of PCM-window. *Renew Energy* 2020;160:653–62. <https://doi.org/10.1016/j.renene.2020.06.146>.
- [291] Arranz B, Ruiz-Valero L, González MP, Sánchez SV. Comprehensive experimental assessment of an industrialized modular innovative active glazing and heat recovery system. *Energy* 2020;212:118748. <https://doi.org/10.1016/j.energy.2020.118748>.
- [292] Li D, Wu Y, Wang B, Liu C, Arıcı M. Optical and thermal performance of glazing units containing PCM in buildings: A review. *Constr Build Mater* 2020;233:117327. <https://doi.org/10.1016/j.conbuildmat.2019.117327>.
- [293] Ahmed F, Mahmood M, Waqas A, Ahmad N, Ali M. Thermal analysis of macro-encapsulated phase change material coupled with domestic gas heater for building heating. *Sustain Energy Technol Assessments* 2021;47:101533. <https://doi.org/10.1016/j.seta.2021.101533>.
- [294] Al-Yasiri Q, Szabó M. Incorporation of phase change materials into building envelope for thermal comfort and energy saving: A comprehensive analysis. *J Build Eng* 2021;36:102122. <https://doi.org/10.1016/j.jobe.2020.102122>.
- [295] Jia J, Liu B, Ma L, Wang H, Li D, Wang Y. Energy saving performance optimization and regional adaptability of prefabricated buildings with PCM in different climates. *Case Stud Therm Eng* 2021;26:101164. <https://doi.org/10.1016/j.csste.2021.101164>.
- [296] Dobri A, Tsiantis A, Papatathanasiou TD, Wang Y. Investigation of transient heat transfer in multi-scale PCM composites using a semi-analytical model. *Int J Heat Mass Transf* 2021;175:121389. <https://doi.org/10.1016/j.ijheatmasstransfer.2021.121389>.
- [297] Zhu X, Sheng X, Li J, Chen Y. Thermal comfort and energy saving of novel heat-storage coatings with microencapsulated PCM and their application. *Energy Build* 2021;251:111349. <https://doi.org/10.1016/j.enbuild.2021.111349>.
- [298] Kadrakhmanova M, Memon SA, Saurbayeva A. Implementation of the panel data regression analysis in PCM integrated buildings located in a humid subtropical climate. *Energy* 2021;237:121651. <https://doi.org/10.1016/j.energy.2021.121651>.
- [299] Staszczuk A, Kuczyński T. The impact of wall and roof material on the summer thermal performance of building in a temperate climate. *Energy* 2021;228:120482. <https://doi.org/10.1016/j.energy.2021.120482>.
- [300] Choi SH, Park J, Ko HS, Kang SW. Heat penetration reduction through PCM walls via bubble injections in buildings. *Energy Convers Manag* 2020;221:113187. <https://doi.org/10.1016/j.enconman.2020.113187>.
- [301] Piselli C, Prabhakar M, de Gracia A, Saffari M, Pisello AL, Cabeza LF. Optimal control of natural ventilation as passive cooling strategy for improving the energy performance of building envelope with PCM integration. *Renew Energy* 2020;162:171–81. <https://doi.org/10.1016/j.renene.2020.07.043>.
- [302] Li D, Zhang C, Li Q, Liu C, Arıcı M, Wu Y. Thermal performance evaluation of glass window combining silica aerogels and phase change materials for cold climate of China. *Appl Therm Eng* 2020;165:114547. <https://doi.org/10.1016/j.applthermaleng.2019.114547>.
- [303] Liu C, Zhang G, Arıcı M, Bian J, Li D. Thermal performance of non-ventilated multilayer glazing facades filled with phase change material. *Sol Energy* 2019;177:464–70. <https://doi.org/10.1016/j.solener.2018.11.044>.
- [304] Yang R, Li D, Salazar SL, Rao Z, Arıcı M, Wei W. Photothermal properties and photothermal conversion performance of nano-enhanced paraffin as a phase change thermal energy storage material. *Sol Energy Mater Sol Cells* 2021;219:110792. <https://doi.org/10.1016/j.solmat.2020.110792>.
- [305] Zhang G, Wang Z, Li D, Wu Y, Arıcı M. Seasonal thermal performance analysis of glazed window filled with paraffin including various nanoparticles. *Int J Energy Res* 2020;44:3008–19. <https://doi.org/10.1002/er.5129>.
- [306] Zhang S, Hu W, Li D, Zhang C, Arıcı M, Yıldız Ç, et al. Energy efficiency optimization of PCM and aerogel-filled multiple glazing windows. *Energy* 2021;222:119916. <https://doi.org/10.1016/j.energy.2021.119916>.
- [307] Li Q, Wang Y, Ma L, Arıcı M, Li D, Yıldız Ç, et al. Effect of sunspace and PCM louver combination on the energy saving of rural residences: Case study in a severe cold region of China. *Sustain Energy Technol Assessments* 2021;45:101126. <https://doi.org/10.1016/j.seta.2021.101126>.
- [308] Al Siyabi I, Khanna S, Mallick T, Sundaram S. Experimental and numerical study on the effect of multiple phase change materials thermal energy storage system. *J Energy Storage* 2021;36:102226. <https://doi.org/10.1016/j.est.2020.102226>.
- [309] Hosseini SM, Safaei MR, Goodarzi M, Alrashed AAAA, Nguyen TK. New temperature, interfacial shell dependent dimensionless model for thermal conductivity of nanofluids. *Int J Heat Mass Transf* 2017;114:207–10. <https://doi.org/10.1016/j.ijheatmasstransfer.2017.06.061>.
- [310] Sarafraz MM, Safaei MR, Tian Z, Goodarzi M, Bandarfa Filho EP, Arjomandi M. Thermal assessment of nano-particulate graphene-water/ethylene glycol (WEG 60:40) nano-suspension in a compact heat exchanger. *Energies* 2019;12. <https://doi.org/10.3390/en12101929>.
- [311] Tian Z, Abdollahi A, Shariati M, Amindoust A, Arasteh H, Karimipour A, et al. Turbulent flows in a spiral double-pipe heat exchanger. *Int J Numer Methods Heat Fluid Flow* 2020;30:39–53. <https://doi.org/10.1108/HFF-04-2019-0287>.
- [312] Sardari PT, Babaei-Mahani R, Giddings D, Yasserli S, Moghimi MA, Bahai H. Energy recovery from domestic radiators using a compact composite metal Foam/PCM latent heat storage. *J Clean Prod* 2020;257:120504. <https://doi.org/10.1016/j.jclepro.2020.120504>.
- [313] Zhang C, Li J, Chen Y. Improving the energy discharging performance of a latent heat storage (LHS) unit using fractal-tree-shaped fins. *Appl Energy* 2020;259:114102. <https://doi.org/10.1016/j.apenergy.2019.114102>.
- [314] Maher H, Rocky KA, Bassiouny R, Saha BB. Synthesis and thermal characterization of paraffin-based nanocomposites for thermal energy storage applications. *Therm Sci Eng Prog* 2021;22:100797. <https://doi.org/10.1016/j.tsep.2020.100797>.
- [315] Zhang D, Zhang F, Tang S, Guo C, Qin X. Performance evaluation of cascaded storage system with multiple phase change materials. *Appl Therm Eng* 2021;185:116384. <https://doi.org/10.1016/j.applthermaleng.2020.116384>.
- [316] Khan LA, Khan MM. Role of orientation of fins in performance enhancement of a latent thermal energy storage unit. *Appl Therm Eng* 2020;175:115408. <https://doi.org/10.1016/j.applthermaleng.2020.115408>.
- [317] Mehta DS, Vaghela B, Rathod MK, Banerjee J. Thermal performance augmentation in latent heat storage unit using spiral fin: An experimental analysis. *J Energy Storage* 2020;31:101776. <https://doi.org/10.1016/j.est.2020.101776>.
- [318] Nie C, Deng S, Liu J. Effects of fins arrangement and parameters on the consecutive melting and solidification of PCM in a latent heat storage unit. *J Energy Storage* 2020;29:101319. <https://doi.org/10.1016/j.est.2020.101319>.
- [319] Yu C, Zhang X, Chen X, Zhang C, Chen Y. Melting performance enhancement of a latent heat storage unit using gradient fins. *Int J Heat Mass Transf* 2020;150:119330. <https://doi.org/10.1016/j.ijheatmasstransfer.2020.119330>.
- [320] Zhang S, Pu L, Xu L, Liu R, Li Y. Melting performance analysis of phase change materials in different finned thermal energy storage. *Appl Therm Eng* 2020;176:115425. <https://doi.org/10.1016/j.applthermaleng.2020.115425>.
- [321] Hajizadeh MR, Keshteli AN, Bach Q-V. Solidification of PCM within a tank with longitudinal-Y shape fins and CuO nanoparticle. *J Mol Liq* 2020;317:114188. <https://doi.org/10.1016/j.molliq.2020.114188>.
- [322] Tang S-Z, Tian H-Q, Zhou J-J, Li H. Evaluation and optimization of melting performance in a horizontal thermal energy storage unit with non-uniform fins. *J Energy Storage* 2021;33:102124. <https://doi.org/10.1016/j.est.2020.102124>.
- [323] Shalaby SM, Kabeel AE, Moharram BM, Falefi AH. Experimental study of the solar water heater integrated with shell and finned tube latent heat storage system. *J Energy Storage* 2020;31:101628. <https://doi.org/10.1016/j.est.2020.101628>.
- [324] Chananipoor A, Azizi Z, Raei B, Tahmasebi N. Optimization of the thermal performance of nano-encapsulated phase change material slurry in double pipe heat exchanger: Design of experiments using response surface methodology (RSM). *J Build Eng* 2021;34:101929. <https://doi.org/10.1016/j.jobe.2020.101929>.
- [325] Sarani I, Payan S, Payan A, Nada SA. Enhancement of energy storage capability in RT82 phase change material using strips fins and metal-oxide based nanoparticles. *J Energy Storage* 2020;32:102009. <https://doi.org/10.1016/j.est.2020.102009>.
- [326] Duan J, Xiong Y, Yang D. Study on the effect of multiple spiral fins for improved phase change process. *Appl Therm Eng* 2020;169:114966. <https://doi.org/10.1016/j.applthermaleng.2020.114966>.
- [327] Lu B, Zhang Y, Sun D, Yuan Z, Yang S. Experimental investigation on thermal behavior of paraffin in a vertical shell and spiral fin tube latent heat thermal energy storage unit. *Appl Therm Eng* 2021;187:116575. <https://doi.org/10.1016/j.applthermaleng.2021.116575>.
- [328] Khan Z, Khan ZA. Role of extended fins and graphene nano-platelets in coupled thermal enhancement of latent heat storage system. *Energy Convers Manag* 2020;224:113349. <https://doi.org/10.1016/j.enconman.2020.113349>.
- [329] Alizadeh M, Pahlavanian MH, Tohidi M, Ganji DD. Solidification expedition of Phase Change Material in a triplex-tube storage unit via novel fins and SWCNT nanoparticles. *J Energy Storage* 2020;28:101188. <https://doi.org/10.1016/j.est.2019.101188>.
- [330] Nie C, Liu J, Deng S. Effect of geometry modification on the thermal response of composite metal foam/phase change material for thermal energy storage. *Int J Heat Mass Transf* 2021;165:120652. <https://doi.org/10.1016/j.ijheatmasstransfer.2020.120652>.
- [331] Li F, Almarashi A, Jafaryar M, Hajizadeh MR, Chu Y-M. Melting process of nanoparticle enhanced PCM through storage cylinder incorporating fins. *Powder Technol* 2021;381:551–60. <https://doi.org/10.1016/j.powtec.2020.12.026>.
- [332] Keshteli AN, Sheikholeslami M. Influence of Al₂O₃ nanoparticle and Y-shaped fins on melting and solidification of paraffin. *J Mol Liq* 2020;314:113798. <https://doi.org/10.1016/j.molliq.2020.113798>.
- [333] Mahood HB, Mahdi MS, Monjezi AA, Khadom AA, Campbell AN. Numerical investigation on the effect of fin design on the melting of phase change material in

- a horizontal shell and tube thermal energy storage. *J Energy Storage* 2020;29: 101331. <https://doi.org/10.1016/j.est.2020.101331>.
- [334] Sorour MM, Hassab MA, Zaytoon MM, Alnaakeb MA. The effect of inclination angle on the performance characteristic of a double-pipe latent heat storage unit. *J Energy Storage* 2021;34:102202. <https://doi.org/10.1016/j.est.2020.102202>.
- [335] Lin W, Ling Z, Fang X, Zhang Z. Experimental and numerical research on thermal performance of a novel thermal energy storage unit with phase change material. *Appl Therm Eng* 2021;186:116493. <https://doi.org/10.1016/j.applthermaleng.2020.116493>.
- [336] Li H, Hu C, He Y, Tang D, Wang K. Influence of fin parameters on the melting behavior in a horizontal shell-and-tube latent heat storage unit with longitudinal fins. *J Energy Storage* 2021;34:102230. <https://doi.org/10.1016/j.est.2020.102230>.
- [337] Mazhar AR, Shukla A, Liu S. Numerical analysis of rectangular fins in a PCM for low-grade heat harnessing. *Int J Therm Sci* 2020;152:106306. <https://doi.org/10.1016/j.jthermalsci.2020.106306>.
- [338] Ding P, Liu Z. Numerical investigation of natural convection enhancement in latent heat energy storage units with punched-fin and slit-fin. *Int J Therm Sci* 2021;163:106834. <https://doi.org/10.1016/j.jthermalsci.2021.106834>.
- [339] Wu L, Zhang X, Liu X. Numerical analysis and improvement of the thermal performance in a latent heat thermal energy storage device with spiderweb-like fins. *J Energy Storage* 2020;32:101768. <https://doi.org/10.1016/j.est.2020.101768>.
- [340] Zhang X, Chao X, Lou L, Fan J, Chen Q, Li B, et al. Personal thermal management by thermally conductive composites: A review. *Compos Commun* 2021;23: 100595. <https://doi.org/10.1016/j.coco.2020.100595>.
- [341] Zhu FL, Feng QQ. Recent advances in textile materials for personal radiative thermal management in indoor and outdoor environments. *Int J Therm Sci* 2021; 165:106899. <https://doi.org/10.1016/j.jthermalsci.2021.106899>.
- [342] Peng Y, Cui Y. Advanced textiles for personal thermal management and energy. *Joule* 2020;4:724–42. <https://doi.org/10.1016/j.joule.2020.02.011>.
- [343] Lan X, Wang Y, Peng J, Si Y, Ren J, Ding B, et al. Designing heat transfer pathways for advanced thermoregulatory textiles. *Mater Today Phys* 2021;17:100342. <https://doi.org/10.1016/j.mtphys.2021.100342>.
- [344] Wang F, Ke Y, Udayraj, Yang B, Xu P, Noor N. Effect of cooling strategies on overall performance of a hybrid personal cooling system incorporated with phase change materials (PCMs) and electric fans. *J Therm Biol* 2020;92:102655. <https://doi.org/10.1016/j.jtherbio.2020.102655>.
- [345] Baykara T, Taylan G. Coaxial electrospinning of PVA/Nigella seed oil nanofibers: Processing and morphological characterization. *Mater Sci Eng B* 2021;265: 115012. <https://doi.org/10.1016/j.mseb.2020.115012>.
- [346] Fonseca A, Neves SF, Campos JBLM. Thermal performance of a PCM firefighting suit considering transient periods of fire exposure, post-fire exposure and resting phases. *Appl Therm Eng* 2021;182:115769. <https://doi.org/10.1016/j.applthermaleng.2020.115769>.
- [347] Ruiz-Calleja T, Bonet-Aracil M, Gisbert-Payá J, Bou-Belda E. Analysis of the influence of graphene and phase change microcapsules on thermal behavior of cellulosic fabrics. *Mater Today Commun* 2020;25:101557. <https://doi.org/10.1016/j.mtcomm.2020.101557>.
- [348] Feng W, Zhang Y, Shao Y, Huang T, Zhang N, Yang J, et al. Coaxial electrospun membranes with thermal energy storage and shape memory functions for simultaneous thermal/moisture management in personal cooling textiles. *Eur Polym J* 2021;145:110245. <https://doi.org/10.1016/j.eurpolymj.2020.110245>.
- [349] Dong T, Jiang W, Liu Y, Wu Y, Qi Y, Li J, et al. A phase change material embedded composite consisting of kapok and hollow PET fibers for dynamic thermal comfort regulation. *Ind Crops Prod* 2020;158:112945. <https://doi.org/10.1016/j.indcrop.2020.112945>.
- [350] Fang H, Lin J, Zhang L, Chen A, Wu F, Geng L, et al. Fibrous form-stable phase change materials with high thermal conductivity fabricated by interfacial polyelectrolyte complex spinning. *Carbohydr Polym* 2020;249:116836. <https://doi.org/10.1016/j.carbpol.2020.116836>.
- [351] Hemmatian B, Heidarzadeh N, Fard GC, Maleknia L. Fabrication of phase-change core/shell nanofibers based on a eutectic fatty acid mixture to control body temperature fluctuations. *Mater Chem Phys* 2020;245:122738. <https://doi.org/10.1016/j.matchemphys.2020.122738>.
- [352] Haghighat F, Hosseini Ravandi SA, Esfahany MN, Valipouri A, Zarezade Z. Thermal performance of electrospun core-shell phase change fibrous layers at simulated body conditions. *Appl Therm Eng* 2019;161:113924. <https://doi.org/10.1016/j.applthermaleng.2019.113924>.
- [353] Su Y, Zhu W, Tian M, Wang Y, Zhang X, Li J. Intelligent bidirectional thermal regulation of phase change material incorporated in thermal protective clothing. *Appl Therm Eng* 2020;174:115340. <https://doi.org/10.1016/j.applthermaleng.2020.115340>.
- [354] Hou J, Yang Z, Xu P, Huang G. Design and performance evaluation of novel personal cooling garment. *Appl Therm Eng* 2019;154:131–9. <https://doi.org/10.1016/j.applthermaleng.2019.02.013>.
- [355] Deshpande M, Sundararajan S, Samui AB, Kulkarni PS. Microwave assisted preparation of poly(ethylene) glycol/lignin blends for thermal energy storage. *J Energy Storage* 2021;35:102338. <https://doi.org/10.1016/j.est.2021.102338>.
- [356] Onder E, Sarier N. Fatty acid alkyl ester nanowebbs suitable for renewable thermal energy storage. *Thermochim Acta* 2020;690:178698. <https://doi.org/10.1016/j.tca.2020.178698>.
- [357] Carreira AS, Teixeira RFA, Beirão A, Vaz Vieira R, Figueiredo MM, Gil MH. Preparation of acrylic based microcapsules using different reaction conditions for thermo-regulating textiles production. *Eur Polym J* 2017;93:33–43. <https://doi.org/10.1016/j.eurpolymj.2017.05.027>.
- [358] Karaszewska A, Kamińska I, Nejman A, Gajdzicki B, Fortuniak W, Chojnowski J, et al. Thermal-regulation of nonwoven fabrics by microcapsules of n-eicosane coated with a polysiloxane elastomer. *Mater Chem Phys* 2019;226:204–13. <https://doi.org/10.1016/j.matchemphys.2019.01.029>.
- [359] Kizildag N. Smart composite nanofiber mats with thermal management functionality. *Sci Rep* 2021;11:4256. <https://doi.org/10.1038/s41598-021-83799-5>.
- [360] Hassabo AG, Mohamed AL. Enhancement the thermo-regulating property of cellulosic fabric using encapsulated paraffins in modified pectin. *Carbohydr Polym* 2017;165:421–8. <https://doi.org/10.1016/j.carbpol.2017.02.074>.
- [361] Zhu W, Wang Y, Song S, Ai H, Qiu F, Li D, et al. Environmental-friendly electrospun phase change fiber with exceptional thermal energy storage performance. *Sol Energy Mater Sol Cells* 2021;222:110939. <https://doi.org/10.1016/j.solmat.2020.110939>.
- [362] Qian Y, Han N, Gao X, Gao X, Li W, Zhang X. Cellulose-based phase change fibres for thermal energy storage and management applications. *Chem Eng J* 2021;412: 128596. <https://doi.org/10.1016/j.cej.2021.128596>.
- [363] Zhang J, Narh C, Lv P, Cai Y, Zhou H, Hou X, et al. Preparation of novel form-stable composite phase change materials with porous silica nanofibrous mats for thermal storage/retrieval. *Colloids Surf A Physicochem Eng Asp* 2019; 570:1–10. <https://doi.org/10.1016/j.colsurfa.2019.03.009>.
- [364] Bashiri Rezaei A, Montazer M. Shape-stable thermo-responsive nano Fe₃O₄/fatty acids/PET composite phase-change material for thermal energy management and saving applications. *Appl Energy* 2020;262:114501. <https://doi.org/10.1016/j.apenergy.2020.114501>.
- [365] Luo D, Xiang L, Sun X, Xie L, Zhou D, Qin S. Phase-change smart lines based on paraffin-expanded graphite/polypropylene hollow fiber membrane composite phase change materials for heat storage. *Energy* 2020;197:117252. <https://doi.org/10.1016/j.energy.2020.117252>.
- [366] Noyan ECB, Onder E, Sarier N, Arat R. Development of heat storing poly (acrylonitrile) nanofibers by coaxial electrospinning. *Thermochim Acta* 2018;662: 135–48. <https://doi.org/10.1016/j.tca.2018.02.008>.
- [367] Babapoor A, Karimi G, Golestaneh SI, Mezin MA. Coaxial electro-spun PEG/PA6 composite fibers: Fabrication and characterization. *Appl Therm Eng* 2017;118: 398–407. <https://doi.org/10.1016/j.applthermaleng.2017.02.119>.
- [368] Ke H, Wei Q. Determining influences of silver nanoparticles on morphology and thermal properties of electrospun polyacrylonitrile-based form-stable phase change composite fibrous membranes loading fatty acid ester/eutectics. *Thermochim Acta* 2019;671:10–6. <https://doi.org/10.1016/j.tca.2018.11.002>.
- [369] Zhang W, Zhang X, Xu Y, Xu Y, Qiao J, Shi T, et al. Flexible polyethylene glycol/polyvinylpyrrolidone composite phase change fibres: Preparation, characterization, and thermal conductivity enhancement. *Polymer (Guildf)* 2021; 214:123258. <https://doi.org/10.1016/j.polymer.2020.123258>.
- [370] Song S, Ai H, Zhu W, Qiu F, Wang Y, Zhou J. Eco-friendly electrospun nanofibrous membranes with high thermal energy capacity and improved thermal transfer efficiency. *Renew Energy* 2020;148:504–11. <https://doi.org/10.1016/j.renene.2019.10.056>.
- [371] Lu Y, Xiao X, Fu J, Huan C, Qi S, Zhan Y, et al. Novel smart textile with phase change materials encapsulated core-sheath structure fabricated by coaxial electrospinning. *Chem Eng J* 2019;355:532–9. <https://doi.org/10.1016/j.cej.2018.08.189>.
- [372] Xia W, Fei X, Wang Q, Lu Y, Innocent MT, Zhou J, et al. Nano-hybridized form-stable ester@F-SiO₂ phase change materials for melt-spun PA6 fibers engineered towards smart thermal management fabrics. *Chem Eng J* 2021;403:126369. <https://doi.org/10.1016/j.cej.2020.126369>.
- [373] Lu Y, Xiao X, Liu Y, Wang J, Qi S, Huan C, et al. Achieving multifunctional smart textile with long afterglow and thermal-regulation via coaxial electrospinning. *J Alloys Compd* 2020;812:152144. <https://doi.org/10.1016/j.jallcom.2019.152144>.
- [374] Siah Chehreh Ghadikolaei S. An enviroeconomic review of the solar PV cells cooling technology effect on the CO₂ emission reduction. *Sol Energy* 2021;216: 468–92. <https://doi.org/10.1016/j.solener.2021.01.016>.
- [375] Cui Y, Zhu J, Zoras S, Zhang J. Comprehensive review of the recent advances in PV/T system with loop-pipe configuration and nanofluid. *Renew Sustain Energy Rev* 2021;135:110254. <https://doi.org/10.1016/j.rser.2020.110254>.
- [376] Siah Chehreh Ghadikolaei S. Solar photovoltaic cells performance improvement by cooling technology: An overall review. *Int J Hydrogen Energy* 2021;46: 10939–72. <https://doi.org/10.1016/j.ijhydene.2020.12.164>.
- [377] Anand A, Shukla A, Panchal H, Sharma A. Thermal regulation of photovoltaic system for enhanced power production: A review. *J Energy Storage* 2021;35: 102236. <https://doi.org/10.1016/j.est.2021.102236>.
- [378] Akshayveer, Kumar A, Singh AP, Singh OP. Effect of novel PCM encapsulation designs on electrical and thermal performance of a hybrid photovoltaic solar panel. *Sol Energy* 2020;205:320–33. <https://doi.org/10.1016/j.solener.2020.05.062>.
- [379] Al-Waeli AHA, Sopian K, Chaichan MT, Kazem HA, Ibrahim A, Mat S, et al. Evaluation of the nanofluid and nano-PCM based photovoltaic thermal (PVT) system: an experimental study. *Energy Convers Manag* 2017;151:693–708.
- [380] Al-Waeli AHA, Kazem HA, Chaichan MT, Sopian K. Experimental investigation of using nano-PCM/nanofluid on a photovoltaic thermal system (PVT): technical and economic study. *Therm Sci Eng Prog* 2019;11:213–30.
- [381] Hassan A, Wahab A, Qasim MA, Janjua MM, Ali MA, Ali HM, et al. Thermal management and uniform temperature regulation of photovoltaic modules using hybrid phase change materials-nanofluids system. *Renew Energy* 2020;145: 282–93.

- [382] Islam MM, Hasanuzzaman M, Rahim NA, Pandey AK, Rawa M, Kumar L. Real time experimental performance investigation of a NePCM based photovoltaic thermal system: An energetic and exergetic approach. *Renew Energy* 2021;172: 71–87. <https://doi.org/10.1016/j.renene.2021.02.169>.
- [383] Essa MA, Talaat M, Amer A, Farahat MA. Enhancing the photovoltaic system efficiency using porous metallic media integrated with phase change material. *Energy* 2021;225:120299. <https://doi.org/10.1016/j.energy.2021.120299>.
- [384] Naderi M, Ziapour BM, Gendeshmin MY. Improvement of photocells by the integration of phase change materials and thermoelectric generators (PV-PCM-TEG) and study on the ability to generate electricity around the clock. *J Energy Storage* 2021;36:102384. <https://doi.org/10.1016/j.est.2021.102384>.
- [385] Mahdi JM, Pal Singh R, Taqi Al-Najjar HM, Singh S, Nsofor EC. Efficient thermal management of the photovoltaic/phase change material system with innovative exterior metal-foam layer. *Sol Energy* 2021;216:411–27. <https://doi.org/10.1016/j.solener.2021.01.008>.
- [386] Sarafraz MM, Safaei MR, Leon AS, Tilili I, Alkanhal TA, Tian Z, et al. Experimental investigation on thermal performance of a PV/T-PCM (photovoltaic/thermal) system cooling with a PCM and nanofluid n.d.:1–16.
- [387] Feng J, Huang J, Ling Z, Fang X, Zhang Z. Performance enhancement of a photovoltaic module using phase change material nanoemulsion as a novel cooling fluid. *Sol Energy Mater Sol Cells* 2021;225:111060. <https://doi.org/10.1016/j.solmat.2021.111060>.
- [388] Sakhaei SA, Valipour MS. Thermal behavior of a flat plate solar collector with simultaneous use of helically heat collecting tubes and phase change materials. *Sustain Energy Technol Assessments* 2021;46:101279. <https://doi.org/10.1016/j.seta.2021.101279>.
- [389] Pawar VR, Sobhansarbandi S. Design optimization and heat transfer enhancement of energy storage based solar thermal collector. *Sustain Energy Technol Assessments* 2021;46:101260. <https://doi.org/10.1016/j.seta.2021.101260>.
- [390] Madadi Avargani V, Norton B, Rahimi A, Karimi H. Integrating paraffin phase change material in the storage tank of a solar water heater to maintain a consistent hot water output temperature. *Sustain Energy Technol Assessments* 2021;47:101350. <https://doi.org/10.1016/j.seta.2021.101350>.
- [391] Bhagwat VV, Roy S, Das B, Shah N, Chowdhury A. Performance of finned heat pipe assisted parabolic trough solar collector system under the climatic condition of North East India. *Sustain Energy Technol Assessments* 2021;45:101171. <https://doi.org/10.1016/j.seta.2021.101171>.
- [392] Rahmanian S, Rahmanian-Koushkaki H, Omidvar P, Shahsavari A. Nanofluid-PCM heat sink for building integrated concentrated photovoltaic with thermal energy storage and recovery capability. *Sustain Energy Technol Assessments* 2021;46: 101223. <https://doi.org/10.1016/j.seta.2021.101223>.
- [393] Yadav S, Chandramohan VP. Performance comparison of thermal energy storage system for indirect solar dryer with and without finned copper tube. *Sustain Energy Technol Assessments* 2020;37:100609. <https://doi.org/10.1016/j.seta.2019.100609>.
- [394] Jamil F, Ali HM, Nasir MA, Karahan M, Janjua MM, Naseer A, et al. Evaluation of photovoltaic panels using different nano phase change material and a concise comparison: An experimental study. *Renew Energy* 2021;169:1265–79. <https://doi.org/10.1016/j.renene.2021.01.089>.
- [395] Ahmadi R, Monadinia F, Maleki M. Passive/active photovoltaic-thermal (PVT) system implementing infiltrated phase change material (PCM) in PS-CNT foam. *Sol Energy Mater Sol Cells* 2021;222:110942. <https://doi.org/10.1016/j.solmat.2020.110942>.
- [396] Carmona M, Palacio Bastos A, García JD. Experimental evaluation of a hybrid photovoltaic and thermal solar energy collector with integrated phase change material (PVT-PCM) in comparison with a traditional photovoltaic (PV) module. *Renew Energy* 2021;172:680–96. <https://doi.org/10.1016/j.renene.2021.03.022>.
- [397] Das D, Bordoloi U, Kamble AD, Muigai HH, Pai RK, Kalita P. Performance investigation of a rectangular spiral flow PV/T collector with a novel form-stable composite material. *Appl Therm Eng* 2021;182:116035. <https://doi.org/10.1016/j.applthermaleng.2020.116035>.
- [398] Xu H, Wang N, Zhang C, Qu Z, Karimi F. Energy conversion performance of a PV/T-PCM system under different thermal regulation strategies. *Energy Convers Manag* 2021;229:113660. <https://doi.org/10.1016/j.enconman.2020.113660>.
- [399] Abdumunem AR, Samin PM, Rahman HA, Hussien HA, Mazali II. Enhancing PV Cell's electrical efficiency using phase change material with copper foam matrix and multi-walled carbon nanotubes as passive cooling method. *Renew Energy* 2020;160:663–75. <https://doi.org/10.1016/j.renene.2020.07.037>.
- [400] Karthikeyan V, Sirisamphanwong C, Sukchai S, Sahoo SK, Wongwuttanasatian T. Reducing PV module temperature with radiation based PV module incorporating composite phase change material. *J Energy Storage* 2020;29:101346. <https://doi.org/10.1016/j.est.2020.101346>.
- [401] Vaziri Rad MA, Kasaeian A, Mousavi S, Rajaei F, Kouravand A. Empirical investigation of a photovoltaic-thermal system with phase change materials and aluminum shavings porous media. *Renew Energy* 2021;167:662–75. <https://doi.org/10.1016/j.renene.2020.11.135>.
- [402] Kumar R, Praveen P, Gupta S, Saikiran J, Bharj RS. Performance evaluation of photovoltaic module integrated with phase change material-filled container with external fins for extremely hot climates. *J Energy Storage* 2020;32:101876. <https://doi.org/10.1016/j.est.2020.101876>.
- [403] Shastri DMC, Arunachala UC. Thermal management of photovoltaic module with metal matrix embedded PCM. *J Energy Storage* 2020;28:101312. <https://doi.org/10.1016/j.est.2020.101312>.
- [404] Elsheniti MB, Hemedah MA, Sorour MM, El-Maghlany WM. Novel enhanced conduction model for predicting performance of a PV panel cooled by PCM. *Energy Convers Manag* 2020;205:112456. <https://doi.org/10.1016/j.enconman.2019.112456>.
- [405] Wongwuttanasatian T, Sarikarin T, Sukri A. Performance enhancement of a photovoltaic module by passive cooling using phase change material in a finned container heat sink. *Sol Energy* 2020;195:47–53. <https://doi.org/10.1016/j.solener.2019.11.053>.
- [406] Parhizi M, Jain A. Analytical modeling and optimization of phase change thermal management of a Li-ion battery pack. *Appl Therm Eng* 2019;148:229–37.
- [407] Huang Y-H, Cheng W-L, Zhao R. Thermal management of Li-ion battery pack with the application of flexible form-stable composite phase change materials. *Energy Convers Manag* 2019;182:9–20.
- [408] Zou D, Ma X, Liu X, Zheng P, Hu Y. Thermal performance enhancement of composite phase change materials (PCM) using graphene and carbon nanotubes as additives for the potential application in lithium-ion power battery. *Int J Heat Mass Transf* 2018;120:33–41.
- [409] Heyhat MM, Mousavi S, Siavashi M. Battery thermal management with thermal energy storage composites of PCM, metal foam, fin and nanoparticle. *J Energy Storage* 2020;28:102325.
- [410] Sun Z, Fan R, Yan F, Zhou T, Zheng N. Thermal management of the lithium-ion battery by the composite PCM-Fin structures. *Int J Heat Mass Transf* 2019;145: 118739.
- [411] Safdari M, Ahmadi R, Sadeghzadeh S. Numerical investigation on PCM encapsulation shape used in the passive-active battery thermal management. *Energy* 2020;193:116840.
- [412] Kiani M, Ansari M, Arshadi AA, Houshfar E, Ashjaee M. Hybrid thermal management of lithium-ion batteries using nanofluid, metal foam, and phase change material: an integrated numerical-experimental approach. *J Therm Anal Calorim* 2020:1–13.
- [413] Behi H, Behi M, Karimi D, Jagemont J, Ghanbarpour M, Behnia M, et al. Heat pipe air-cooled thermal management system for lithium-ion batteries: High power applications. *Appl Therm Eng* 2020:116240.
- [414] Wang Y, Peng P, Cao W, Dong T, Zheng Y, Lei B, et al. Experimental study on a novel compact cooling system for cylindrical lithium-ion battery module. *Appl Therm Eng* 2020;180:115772.
- [415] Wu W, Yang X, Zhang G, Chen K, Wang S. Experimental investigation on the thermal performance of heat pipe-assisted phase change material based battery thermal management system. *Energy Convers Manag* 2017;138:486–92.
- [416] Xiao C, Zhang G, Li Z, Yang X. Custom design of solid-solid phase change material with ultra-high thermal stability for battery thermal management. *J Mater Chem A* 2020;8:14624–33. <https://doi.org/10.1039/D0TA05247G>.
- [417] Smith J, Singh R, Hinterberger M, Mochizuki M. Battery thermal management system for electric vehicle using heat pipes. *Int J Therm Sci* 2018;134:517–29.
- [418] Huang Q, Li X, Zhang G, Zhang J, He F, Li Y. Experimental investigation of the thermal performance of heat pipe assisted phase change material for battery thermal management system. *Appl Therm Eng* 2018;141:1092–100.
- [419] B K, Pandey AK, Shahabuddin S, Samykano M, M T, Saidur R. Phase change materials integrated solar thermal energy systems: Global trends and current practices in experimental approaches. *J Energy Storage* 2020;27:101118. <https://doi.org/10.1016/j.est.2019.101118>.
- [420] Omara AAM, Abuelnuor AAA, Mohammed HA, Habibi D, Younis O. Improving solar cooker performance using phase change materials: A comprehensive review. *Sol Energy* 2020;207:539–63. <https://doi.org/10.1016/j.solener.2020.07.015>.
- [421] Teamah HM. Comprehensive review of the application of phase change materials in residential heating applications. *Alexandria Eng J* 2021;60:3829–43. <https://doi.org/10.1016/j.aej.2021.02.053>.
- [422] Xiao Q, Cao J, Zhang Y, Li L, Xu T, Yuan W. The application of solar-to-thermal conversion phase change material in novel solar water heating system. *Sol Energy* 2020;199:484–90. <https://doi.org/10.1016/j.solener.2020.02.054>.
- [423] Liu J, Zou X, Cai Z, Peng Z, Xu Y. Polymer based phase change material for photothermal utilization. *Sol Energy Mater Sol Cells* 2021;220:110852. <https://doi.org/10.1016/j.solmat.2020.110852>.
- [424] Chopra K, Tyagi VV, Pandey AK, Sharma RK, Sari A. PCM integrated glass in glass tube solar collector for low and medium temperature applications: Thermodynamic & techno-economic approach. *Energy* 2020;198:117238. <https://doi.org/10.1016/j.energy.2020.117238>.
- [425] Ding Z, Wu W, Chen Y, Li Y. Dynamic simulation and parametric study of solar water heating system with phase change materials in different climate zones. *Sol Energy* 2020;205:399–408. <https://doi.org/10.1016/j.solener.2020.05.075>.
- [426] Luo Q, Li B, Wang Z, Su S, Xiao H, Zhu C. Thermal modeling of air-type double-pass solar collector with PCM-rod embedded in vacuum tube. *Energy Convers Manag* 2021;235:113952. <https://doi.org/10.1016/j.enconman.2021.113952>.
- [427] Kumar PM, Mulsamy K. A comprehensive study on thermal storage characteristics of nano-CeO₂ embedded phase change material and its influence on the performance of evacuated tube solar water heater. *Renew Energy* 2020;162: 662–76. <https://doi.org/10.1016/j.renene.2020.08.122>.
- [428] Mawire A, Lentswe K, Owusu P, Shobo A, Darkwa J, Calautit J, et al. Performance comparison of two solar cooking storage pots combined with wonderbag slow cookers for off-sunshine cooking. *Sol Energy* 2020;208:1166–80. <https://doi.org/10.1016/j.solener.2020.08.053>.
- [429] Bhawe AG, Kale CK. Development of a thermal storage type solar cooker for high temperature cooking using solar salt. *Sol Energy Mater Sol Cells* 2020;208: 110394. <https://doi.org/10.1016/j.solmat.2020.110394>.
- [430] Saxena A, Cuce E, Tiwari GN, Kumar A. Design and thermal performance investigation of a box cooker with flexible solar collector tubes: An experimental research. *Energy* 2020;206:118144. <https://doi.org/10.1016/j.energy.2020.118144>.

- [431] Coccia G, Aquilanti A, Tomassetti S, Comodi G, Di Nicola G. Design, realization, and tests of a portable solar box cooker coupled with an erythritol-based PCM thermal energy storage. *Sol Energy* 2020;201:530–40. <https://doi.org/10.1016/j.solener.2020.03.031>.
- [432] Jouhara H, Żabnieńska-Góra A, Khordehagh N, Doraghi Q, Ahmad L, Norman L, et al. Thermoelectric generator (TEG) technologies and applications. *Int J Thermofluids* 2021;9:100063. <https://doi.org/10.1016/j.ijft.2021.100063>.
- [433] Shen Z-G, Tian L-L, Liu X. Automotive exhaust thermoelectric generators: Current status, challenges and future prospects. *Energy Convers Manag* 2019;195:1138–73. <https://doi.org/10.1016/j.enconman.2019.05.087>.
- [434] Casi Á, Araiz M, Catalán L, Astrain D. Thermoelectric heat recovery in a real industry: From laboratory optimization to reality. *Appl Therm Eng* 2021;184:116275. <https://doi.org/10.1016/j.applthermaleng.2020.116275>.
- [435] Xia Z, Zhang Z, Meng Z, Yu Z. A 24-hour thermoelectric generator simultaneous using solar heat energy and space cold energy. *J Quant Spectrosc Radiat Transf* 2020;251:107038. <https://doi.org/10.1016/j.jqsrt.2020.107038>.
- [436] Atouei SA, Rezaia A, Ranjbar AA, Rosendahl LA. Protection and thermal management of thermoelectric generator system using phase change materials: An experimental investigation. *Energy* 2018;156:311–8. <https://doi.org/10.1016/j.energy.2018.05.109>.
- [437] Huang K, Yan Y, Wang G, Li B. Improving transient performance of thermoelectric generator by integrating phase change material. *Energy* 2021;219:119648. <https://doi.org/10.1016/j.energy.2020.119648>.
- [438] Borhani SM, Hosseini MJ, Pakrouh R, Ranjbar AA, Nourian A. Performance enhancement of a thermoelectric harvester with a PCM/Metal foam composite. *Renew Energy* 2021;168:1122–40. <https://doi.org/10.1016/j.renene.2021.01.020>.
- [439] Liu A, Zou J, Wu Z, Wang Y, Tian Y, Xie H. Enhancing the performance of TEG system coupled with PCMs by regulating the interfacial thermal conduction. *Energy Rep* 2020;6:1942–9. <https://doi.org/10.1016/j.egy.2020.07.014>.
- [440] Rezaia A, Atouei SA, Rosendahl L. Critical parameters in integration of thermoelectric generators and phase change materials by numerical and Taguchi methods. *Mater Today Energy* 2020;16:100376. <https://doi.org/10.1016/j.mtener.2019.100376>.
- [441] Luo X, Guo Q, Tao Z, Liang Y, Liu Z. Modified phase change materials used for thermal management of a novel solar thermoelectric generator. *Energy Convers Manag* 2020;208:112459. <https://doi.org/10.1016/j.enconman.2019.112459>.
- [442] Tuoi TTK, Van Toan N, Ono T. Theoretical and experimental investigation of a thermoelectric generator (TEG) integrated with a phase change material (PCM) for harvesting energy from ambient temperature changes. *Energy Rep* 2020;6:2022–9. <https://doi.org/10.1016/j.egy.2020.07.023>.
- [443] Montero FJ, Lamba R, Ortega A, Jahn W, Guzmán AM. A novel 24-h day-night operational solar thermoelectric generator using phase change materials. *J Clean Prod* 2021;296:126553. <https://doi.org/10.1016/j.jclepro.2021.126553>.
- [444] Madruga S. Modeling of enhanced micro-energy harvesting of thermal ambient fluctuations with metallic foams embedded in Phase Change Materials. *Renew Energy* 2021;168:424–37. <https://doi.org/10.1016/j.renene.2020.12.041>.
- [445] Liao X, Liu Y, Ren J, Guan L, Sang X, Wang B, et al. Investigation of a double-PCM-based thermoelectric energy-harvesting device using temperature fluctuations in an ambient environment. *Energy* 2020;202:117724. <https://doi.org/10.1016/j.energy.2020.117724>.
- [446] Zhao Y, Zhang X, Xu X, Zhang S. Research progress of phase change cold storage materials used in cold chain transportation and their different cold storage packaging structures. *J Mol Liq* 2020;319:114360. <https://doi.org/10.1016/j.molliq.2020.114360>.
- [447] Alehosseini E, Jafari SM. Micro/nano-encapsulated phase change materials (PCMs) as emerging materials for the food industry. *Trends Food Sci Technol* 2019;91:116–28. <https://doi.org/10.1016/j.tifs.2019.07.003>.
- [448] Arjenaki NO, Soltanizadeh N, Hamdami N. Designing an active phase change material package for thermal and qualitative protection of meat. *Food Packag Shelf Life* 2019;21:100362. <https://doi.org/10.1016/j.fpsl.2019.100362>.
- [449] Rakshamuthu S, Jegan S, Joel Benyameen J, Selvakumar V, Anandeeswaran K, Iyahrja S. Experimental analysis of small size solar dryer with phase change materials for food preservation. *J Energy Storage* 2021;33:102095. <https://doi.org/10.1016/j.est.2020.102095>.
- [450] Tas CE, Unal H. Thermally buffering polyethylene/halloysite/phase change material nanocomposite packaging films for cold storage of foods. *J Food Eng* 2021;292:110351. <https://doi.org/10.1016/j.jfoodeng.2020.110351>.
- [451] Vennapusa JR, Konala A, Dixit P, Chattopadhyay S. Caprylic acid based PCM composite with potential for thermal buffering and packaging applications. *Mater Chem Phys* 2020;253:123453. <https://doi.org/10.1016/j.matchemphys.2020.123453>.
- [452] Chen C, Ding R, Yang S, Wang J, Chen W, Zong L, et al. Development of thermal insulation packaging film based on poly(vinyl alcohol) incorporated with silica aerogel for food packaging application. *LWT* 2020;129:109568. <https://doi.org/10.1016/j.lwt.2020.109568>.
- [453] Kim D, Thanakkasaranee S, Lee K, Sadeghi K, Seo J. Smart packaging with temperature-dependent gas permeability maintains the quality of cherry tomatoes. *Food Biosci* 2021;41:100997. <https://doi.org/10.1016/j.fbio.2021.100997>.
- [454] Chalco-Sandoval W, Fabra MJ, López-Rubio A, Lagaron JM. Use of phase change materials to develop electrospun coatings of interest in food packaging applications. *J Food Eng* 2017;192:122–8. <https://doi.org/10.1016/j.jfoodeng.2015.01.019>.
- [455] Abdolmaleki L, Sadrameli SM, Pirvaram A. Application of environmental friendly and eutectic phase change materials for the efficiency enhancement of household freezers. *Renew Energy* 2020;145:233–41. <https://doi.org/10.1016/j.renene.2019.06.035>.
- [456] Jagadeeswara Reddy V, Akhila K, Dixit P, Singh J, Parvate S, Chattopadhyay S. Thermal buffering performance evaluation of fatty acids blend/fatty alcohol based eutectic phase change material and simulation. *J Energy Storage* 2021;38:102499. <https://doi.org/10.1016/j.est.2021.102499>.